CHAPTER 17

Queueing Theory

Queues (waiting lines) are a part of everyday life. We all wait in queues to buy a movie ticket, make a bank deposit, pay for groceries, mail a package, obtain food in a cafeteria, start a ride in an amusement park, etc. We have become accustomed to considerable amounts of waiting, but still get annoyed by unusually long waits.

However, having to wait is not just a petty personal annoyance. The amount of time that a nation's populace wastes by waiting in queues is a major factor in both the quality of life there and the efficiency of the nation's economy. For example, before its dissolution, the U.S.S.R. was notorious for the tremendously long queues that its citizens frequently had to endure just to purchase basic necessities. Even in the United States, it has been estimated that Americans spend 37,000,000,000 hours per year waiting in queues. If this time could be spent productively instead, it would amount to nearly 20 million person-years of useful work each year!

Even this staggering figure does not tell the whole story of the impact of causing excessive waiting. Great inefficiencies also occur because of other kinds of waiting than people standing in line. For example, making *machines* wait to be repaired may result in lost production. *Vehicles* (including ships and trucks) that need to wait to be unloaded may delay subsequent shipments. *Airplanes* waiting to take off or land may disrupt later travel schedules. Delays in *telecommunication* transmissions due to saturated lines may cause data glitches. Causing *manufacturing jobs* to wait to be performed may disrupt subsequent production. Delaying *service jobs* beyond their due dates may result in lost future business.

Queueing theory is the study of waiting in all these various guises. It uses *queueing models* to represent the various types of *queueing systems* (systems that involve queues of some kind) that arise in practice. Formulas for each model indicate how the corresponding queueing system should perform, including the average amount of waiting that will occur, under a variety of circumstances.

Therefore, these queueing models are very helpful for determining how to operate a queueing system in the most effective way. Providing too much service capacity to operate the system involves excessive costs. But not providing enough service capacity results in excessive waiting and all its unfortunate consequences. The models enable finding an appropriate balance between the cost of service and the amount of waiting.

After some general discussion, this chapter presents most of the more elementary queueing models and their basic results. Section 17.10 discusses how the information provided by queueing theory can be used to design queueing systems that minimize the total

cost of service and waiting, and then Chap. 26 (on the CD-ROM) elaborates considerably further on the application of queueing theory in this way.

17.1 PROTOTYPE EXAMPLE

The emergency room of COUNTY HOSPITAL provides quick medical care for emergency cases brought to the hospital by ambulance or private automobile. At any hour there is always one doctor on duty in the emergency room. However, because of a growing tendency for emergency cases to use these facilities rather than go to a private physician, the hospital has been experiencing a continuing increase in the number of emergency room visits each year. As a result, it has become quite common for patients arriving during peak usage hours (the early evening) to have to wait until it is their turn to be treated by the doctor. Therefore, a proposal has been made that a second doctor should be assigned to the emergency room during these hours, so that two emergency cases can be treated simultaneously. The hospital's management engineer has been assigned to study this question.[1]

The management engineer began by gathering the relevant historical data and then projecting these data into the next year. Recognizing that the emergency room is a queueing system, she applied several alternative queueing theory models to predict the waiting characteristics of the system with one doctor and with two doctors, as you will see in the latter sections of this chapter (see Tables 17.2 and 17.3).

17.2 BASIC STRUCTURE OF QUEUEING MODELS

The Basic Queueing Process

The basic process assumed by most queueing models is the following. *Customers* requiring service are generated over time by an *input source.* These customers enter the *queueing system* and join a *queue.* At certain times, a member of the queue is selected for service by some rule known as the *queue discipline.* The required service is then performed for the customer by the *service mechanism,* after which the customer leaves the queueing system. This process is depicted in Fig. 17.1.

Many alternative assumptions can be made about the various elements of the queueing process; they are discussed next.

[1]For one actual case study of this kind, see W. Blaker Bolling, "Queueing Model of a Hospital Emergency Room," *Industrial Engineering,* September 1972, pp. 26–31.

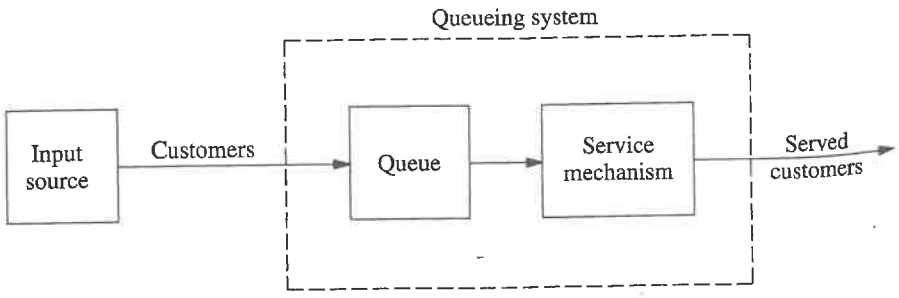


FIGURE 17.1
The basic queueing process.

Input Source (Calling Population)

One characteristic of the input source is its size. The *size* is the total number of customers that might require service from time to time, i.e., the total number of distinct potential customers. This population from which arrivals come is referred to as the **calling population.** The size may be assumed to be either *infinite* or *finite* (so that the input source also is said to be either *unlimited* or *limited*). Because the calculations are far easier for the infinite case, this assumption often is made even when the actual size is some relatively large finite number; and it should be taken to be the implicit assumption for any queueing model that does not state otherwise. The finite case is more difficult analytically because the number of customers in the queueing system affects the number of potential customers outside the system at any time. However, the finite assumption must be made if the rate at which the input source generates new customers is significantly affected by the number of customers in the queueing system.

The statistical pattern by which customers are generated over time must also be specified. The common assumption is that they are generated according to a *Poisson process;* i.e., the number of customers generated until any specific time has a Poisson distribution. As we discuss in Sec. 17.4, this case is the one where arrivals to the queueing system occur randomly but at a certain fixed mean rate, regardless of how many customers already are there (so the *size* of the input source is *infinite*). An equivalent assumption is that the probability distribution of the time between consecutive arrivals is an *exponential* distribution. (The properties of this distribution are described in Sec. 17.4.) The time between consecutive arrivals is referred to as the **interarrival time.**

Any unusual assumptions about the behavior of arriving customers must also be specified. One example is *balking,* where the customer refuses to enter the system and is lost if the queue is too long.

Queue

The queue is where customers wait *before* being served. A queue is characterized by the maximum permissible number of customers that it can contain. Queues are called *infinite* or *finite,* according to whether this number is infinite or finite. The assumption of an *infinite queue* is the standard one for most queueing models, even for situations where there actually is a (relatively large) finite upper bound on the permissible number of customers, because dealing with such an upper bound would be a complicating factor in the analysis. However, for queueing systems where this upper bound is small enough that it actually would be reached with some frequency, it becomes necessary to assume a *finite queue.*

Queue Discipline

The queue discipline refers to the order in which members of the queue are selected for service. For example, it may be first-come-first-served, random, according to some priority procedure, or some other order. First-come-first-served usually is assumed by queueing models, unless it is stated otherwise.

Service Mechanism

The service mechanism consists of one or more *service facilities,* each of which contains one or more *parallel service channels,* called **servers.** If there is more than one service facility, the customer may receive service from a sequence of these (*service channels in series*). At a given facility, the customer enters one of the parallel service channels and is completely serviced by that server. A queueing model must specify the arrangement of the facilities and the number of servers (parallel channels) at each one. Most elementary models assume one service facility with either one server or a finite number of servers.

The time elapsed from the commencement of service to its completion for a customer at a service facility is referred to as the **service time** (or *holding time*). A model of a particular queueing system must specify the probability distribution of service times for each server (and possibly for different types of customers), although it is common to assume the *same* distribution for all servers (all models in this chapter make this assumption). The service-time distribution that is most frequently assumed in practice (largely because it is far more tractable than any other) is the *exponential* distribution discussed in Sec. 17.4, and most of our models will be of this type. Other important service-time distributions are the *degenerate* distribution (constant service time) and the *Erlang* (gamma) distribution, as illustrated by models in Sec. 17.7.

An Elementary Queueing Process

As we have already suggested, queueing theory has been applied to many different types of waiting-line situations. However, the most prevalent type of situation is the following: A single waiting line (which may be empty at times) forms in the front of a single service facility, within which are stationed one or more servers. Each customer generated by an input source is serviced by one of the servers, perhaps after some waiting in the queue (waiting line). The queueing system involved is depicted in Fig. 17.2.

Notice that the queueing process in the illustrative example of Sec. 17.1 is of this type. The input source generates customers in the form of emergency cases requiring medical care. The emergency room is the service facility, and the doctors are the servers.

A server need not be a single individual; it may be a group of persons, e.g., a repair crew that combines forces to perform simultaneously the required service for a customer. Furthermore, servers need not even be people. In many cases, a server can instead be a machine, a vehicle, an electronic device, etc. By the same token, the customers in the waiting line need not be people. For example, they may be items waiting for a certain operation by a given type of machine, or they may be cars waiting in front of a tollbooth.

It is not necessary that there actually be a physical waiting line forming in front of a physical structure that constitutes the service facility. The members of the queue may instead be scattered throughout an area, waiting for a server to come to them, e.g., machines waiting to be repaired. The server or group of servers assigned to a given area constitutes

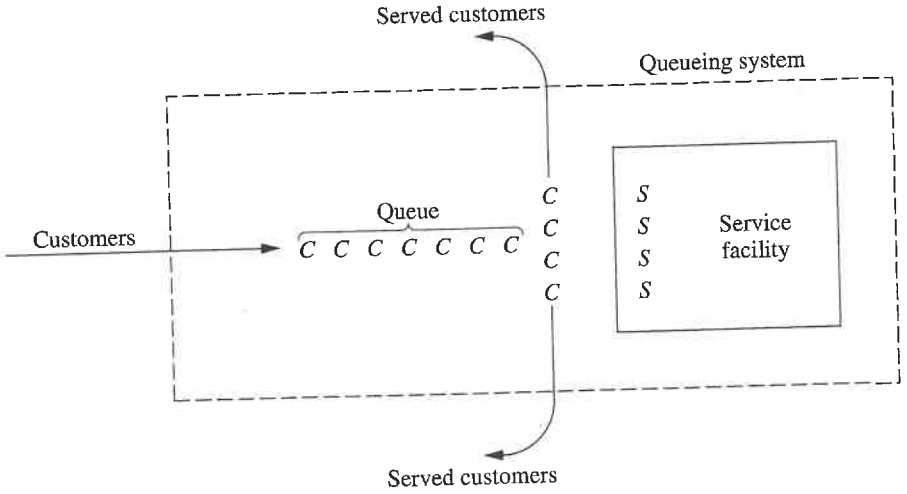
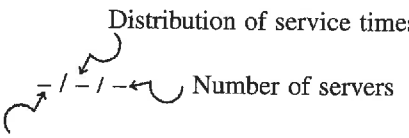


FIGURE 17.2
An elementary queueing system (each customer is indicated by a C and each server by an S).

the service facility for that area. Queueing theory still gives the average number waiting, the average waiting time, and so on, because it is irrelevant whether the customers wait together in a group. The only essential requirement for queueing theory to be applicable is that changes in the number of customers waiting for a given service occur just as though the physical situation described in Fig. 17.2 (or a legitimate counterpart) prevailed.

Except for Sec. 17.9, all the queueing models discussed in this chapter are of the elementary type depicted in Fig. 17.2. Many of these models further assume that all *interarrival times* are independent and identically distributed and that all *service times* are independent and identically distributed. Such models conventionally are labeled as follows:



where M = exponential distribution (Markovian), as described in Sec. 17.4,
D = degenerate distribution (constant times), as discussed in Sec. 17.7,
E_k = Erlang distribution (shape parameter = k), as described in Sec. 17.7,
G = general distribution (any arbitrary distribution allowed),[1] as discussed in Sec. 17.7.

For example, the *M/M/s* model discussed in Sec. 17.6 assumes that both interarrival times and service times have an exponential distribution and that the number of servers is s (any positive integer). The *M/G/*1 model discussed again in Sec. 17.7 assumes that interarrival times have an exponential distribution, but it places no restriction on what the distribution of service times must be, whereas the number of servers is restricted to be exactly 1. Various other models that fit this labeling scheme also are introduced in Sec. 17.7.

Terminology and Notation

Unless otherwise noted, the following standard terminology and notation will be used:

State of system = number of customers in queueing system.
Queue length = number of customers waiting for service to begin
= state of system *minus* number of customers being served.
$N(t)$ = number of customers in queueing system at time t ($t \geq 0$).
$P_n(t)$ = probability of exactly n customers in queueing system at time t, given number at time 0.
s = number of servers (parallel service channels) in queueing system.
λ_n = mean arrival rate (expected number of arrivals per unit time) of new customers when n customers are in system.
μ_n = mean service rate for overall system (expected number of customers completing service per unit time) when n customers are in system. *Note:* μ_n represents *combined* rate at which all *busy* servers (those serving customers) achieve service completions.
λ, μ, ρ = see following paragraph.

[1]When we refer to interarrival times, it is conventional to replace the symbol G by GI = general independent distribution.

When λ_n is a constant for all n, this constant is denoted by λ. When the mean service rate *per busy server* is a constant for all $n \geq 1$, this constant is denoted by μ. (In this case, $\mu_n = s\mu$ when $n \geq s$, that is, when all s servers are busy.) Under these circumstances, $1/\lambda$ and $1/\mu$ are the *expected interarrival time* and the *expected service time,* respectively. Also, $\rho = \lambda/(s\mu)$ is the **utilization factor** for the service facility, i.e., the expected fraction of time the individual servers are busy, because $\lambda/(s\mu)$ represents the fraction of the system's service capacity ($s\mu$) that is being *utilized* on the average by arriving customers (λ).

Certain notation also is required to describe *steady-state* results. When a queueing system has recently begun operation, the state of the system (number of customers in the system) will be greatly affected by the initial state and by the time that has since elapsed. The system is said to be in a **transient condition.** However, after sufficient time has elapsed, the state of the system becomes essentially independent of the initial state and the elapsed time (except under unusual circumstances).[1] The system has now essentially reached a **steady-state condition,** where the probability distribution of the state of the system remains the same (the *steady-state* or *stationary* distribution) over time. Queueing theory has tended to focus largely on the steady-state condition, partially because the transient case is more difficult analytically. (Some transient results exist, but they are generally beyond the technical scope of this book.) The following notation assumes that the system is in a *steady-state condition:*

P_n = probability of exactly n customers in queueing system.

L = expected number of customers in queueing system $= \sum_{n=0}^{\infty} nP_n$.

L_q = expected queue length (excludes customers being served) $= \sum_{n=s}^{\infty} (n - s)P_n$.

$\mathcal{W}$ = waiting time in system (includes service time) for each individual customer.

$W = E(\mathcal{W})$.

$\mathcal{W}_q$ = waiting time in queue (excludes service time) for each individual customer.

$W_q = E(\mathcal{W}_q)$.

Relationships between L, W, L_q, and W_q

Assume that λ_n is a constant λ for all n. It has been proved that in a steady-state queueing process,

$$L = \lambda W.$$

(Because John D. C. Little[2] provided the first rigorous proof, this equation sometimes is referred to as **Little's formula.**) Furthermore, the same proof also shows that

$$L_q = \lambda W_q.$$

If the λ_n are not equal, then λ can be replaced in these equations by $\bar{\lambda}$, the *average* arrival rate over the long run. (We shall show later how $\bar{\lambda}$ can be determined for some basic cases.)

[1] When λ and μ are defined, these unusual circumstances are that $\rho \geq 1$, in which case the state of the system tends to grow continually larger as time goes on.

[2] J. D. C. Little, "A Proof for the Queueing Formula: $L = \lambda W$," *Operations Research,* **9**(3): 383–387, 1961; also see S. Stidham, Jr., "A Last Word on $L = \lambda W$," *Operations Research,* **22**(2): 417–421, 1974.

Now assume that the mean service time is a constant, $1/\mu$ for all $n \geq 1$. It then follows that

$$W = W_q + \frac{1}{\mu}.$$

These relationships are extremely important because they enable all four of the fundamental quantities—L, W, L_q, and W_q—to be immediately determined as soon as one is found analytically. This situation is fortunate because some of these quantities often are much easier to find than others when a queueing model is solved from basic principles.

17.3 EXAMPLES OF REAL QUEUEING SYSTEMS

Our description of queueing systems in Sec. 17.2 may appear relatively abstract and applicable to only rather special practical situations. On the contrary, queueing systems are surprisingly prevalent in a wide variety of contexts. To broaden your horizons on the applicability of queueing theory, we shall briefly mention various examples of real queueing systems that fall into several broad categories. We then will describe queueing systems in several prominent companies (plus one city) and the award-winning studies that were conducted to design these systems.

Some Classes of Queueing Systems

One important class of queueing systems that we all encounter in our daily lives is **commercial service systems,** where outside customers receive service from commercial organizations. Many of these involve person-to-person service at a fixed location, such as a barber shop (the barbers are the servers), bank teller service, checkout stands at a grocery store, and a cafeteria line (service channels in series). However, many others do not, such as home appliance repairs (the server travels to the customers), a vending machine (the server is a machine), and a gas station (the cars are the customers).

Another important class is **transportation service systems.** For some of these systems the vehicles are the customers, such as cars waiting at a tollbooth or traffic light (the server), a truck or ship waiting to be loaded or unloaded by a crew (the server), and airplanes waiting to land or take off from a runway (the server). (An unusual example of this kind is a parking lot, where the cars are the customers and the parking spaces are the servers, but there is no queue because arriving customers go elsewhere to park if the lot is full.) In other cases, the vehicles, such as taxicabs, fire trucks, and elevators, are the servers.

In recent years, queueing theory probably has been applied most to **internal service systems,** where the customers receiving service are *internal* to the organization. Examples include materials-handling systems, where materials-handling units (the servers) move loads (the customers); maintenance systems, where maintenance crews (the servers) repair machines (the customers); and inspection stations, where quality control inspectors (the servers) inspect items (the customers). Employee facilities and departments servicing employees also fit into this category. In addition, machines can be viewed as servers whose customers are the jobs being processed. A related example is a computer laboratory, where each computer is viewed as the server.

There is now growing recognition that queueing theory also is applicable to **social service systems.** For example, a judicial system is a queueing network, where the courts are service facilities, the judges (or panels of judges) are the servers, and the cases waiting to be tried are the customers. A legislative system is a similar queueing network,

where the customers are the bills waiting to be processed. Various health-care systems also are queueing systems. You already have seen one example in Sec. 17.1 (a hospital emergency room), but you can also view ambulances, X-ray machines, and hospital beds as servers in their own queueing systems. Similarly, families waiting for low- and moderate-income housing, or other social services, can be viewed as customers in a queueing system.

Although these are four broad classes of queueing systems, they still do not exhaust the list. In fact, queueing theory first began early in this century with applications to telephone engineering (the founder of queueing theory, A. K. Erlang, was an employee of the Danish Telephone Company in Copenhagen), and telephone engineering still is an important application. Furthermore, we all have our own personal queues—homework assignments, books to be read, and so forth. However, these examples are sufficient to suggest that queueing systems do indeed pervade many areas of society.

Some Award-Winning Studies to Design Queueing Systems

The prestigious *Franz Edelman Awards for Management Science Achievement* are awarded annually by the Institute of Operations Research and Management Sciences (INFORMS) for the year's best applications of OR. A rather substantial number of these awards have been given for innovative applications of queueing theory to the design of queueing systems. We briefly describe some of these applications below.

One of the early first-prize winners (described in the November 1975 issue, Part 2, of *Interfaces*) was the *Xerox Corporation.* The company had recently introduced a major new duplicating system that was proving to be particularly valuable for its owners. Consequently, these customers were demanding that Xerox's tech reps reduce the waiting times to repair the machines. An OR team then applied queueing theory to study how to best meet the new service requirements. This resulted in replacing the previous one-person tech rep territories by larger three-person tech rep territories. This change had the dramatic effect of both substantially reducing the average waiting times of the customers and increasing the utilization of the tech reps by over 50 percent.

In Sec. 3.5, we described an award-winning application by *United Airlines* (January 1986 issue of *Interfaces*) that resulted in annual savings of over $6 million. This application involved scheduling the work assignments of United's 4,000 reservations sales representatives and support personnel at its 11 reservations offices and the 1,000 customer service agents at its 10 largest airports. After determining how many employees are needed at each location during each half hour of the week, we discussed how linear programming was applied to design the work schedules for all the employees to meet these service requirements most efficiently. However, we never mentioned how these service requirements on the number of employees needed each half hour were determined.

We now are in a position to point out that these service requirements were determined by applying *queueing theory.* Each specific location (e.g., the check-in counters at an airport) constitutes a queueing system with the employees as the servers. After forecasting the mean arrival rate during each half hour of the week, queueing models are used to find the minimum number of servers that will provide satisfactory measures of performance for the queueing system.

L.L. Bean, Inc., the large telemarketer and mail-order catalog house, relied mainly on queueing theory for its award-winning study of how to allocate its telecommunications resources. (The article describing this study is in the January 1991 issue of *Interfaces*, and other articles giving additional information are in the November 1989 and March–April 1993 issues of this journal.) The telephone calls coming in to its call center to place

orders are the customers in a large queueing system, with the telephone agents as the servers. The key questions being asked during the study were the following.

1. How many telephone trunk lines should be provided for incoming calls to the call center?
2. How many telephone agents should be scheduled at various times?
3. How many hold positions should be provided for customers waiting for a telephone agent? (Note that the limited number of hold positions causes the system to have a finite queue.)

For each interesting combination of these three quantities, queueing models provide the measures of performance of the queueing system. Given these measures, the OR team carefully assessed the cost of lost sales due to making some customers either incur a busy signal or be placed on hold too long. By adding the cost of the telemarketing resources, the team then was able to find the combination of the three quantities that minimizes the expected total cost. This resulted in cost savings of $9 to $10 million per year.

New York City has a long-standing tradition of using OR techniques in planning and operating many of its complex urban service systems. Starting in the late 1960s, award-winning studies involving queueing theory have been conducted for its Fire Department and its Police Department. (Fires and police emergencies are the customers in these respective queueing systems.) Subsequently, major OR studies (including several more involving queueing theory) have been conducted for its Department of Sanitation, Department of Transportation, Department of Health and Hospitals, Department of Environmental Protection, Office of Management and Budget, and Department of Probation. Because of the success of these studies, many of these departments now have their own in-house OR groups.

The award-winning study in New York City that we will describe here involves its arrest-to-arraignment system. This system consists of the process from when individuals are arrested until they are arraigned (the first court appearance before an arraignment judge, who determines whether there was probable cause for the arrest). Before the study, the city's arrestees (the customers in a queueing system) were in custody waiting to be arraigned for an average of 40 hours (occasionally more than 70 hours). These waiting times were considered excessive, because the arrestees were being held in crowded, noisy conditions that were emotionally stressful, unhealthy, and often physically dangerous. Therefore, a 2-year OR study was conducted to overhaul the system. Both queueing theory and simulation (the subject of Chap. 20) were used. This led to sweeping operational and policy changes that simultaneously reduced average waiting times until arraignment to 24 hours or less and provided annual savings of $9.5 million. (See the January 1993 issue of *Interfaces* for details.)

The first prize in the 1993 competition was won by AT&T for a study that (like the preceding one) also combined the use of queueing theory and simulation (January–February 1994 issue of *Interfaces*). The queueing models are of both AT&T's telecommunication network and the call center environment for the typical business customers of AT&T that have such a center. The purpose of the study was to develop a user-friendly PC-based system that AT&T's business customers can use to guide them in how to design or redesign their call centers. Since call centers comprise one of the United States' fastest-growing industries, this system had been used about 2,000 times by AT&T's business customers by 1992. This resulted in more than $750 million in annual profit for these customers.

KeyCorp is one of the largest bank holding companies in the United States, with more than 1,300 branches and over 6,000 tellers. This company's award-winning OR study (January 1996 issue of *Interfaces*) focused on using queueing theory to improve the performance of each branch's queueing system where the tellers serve the customers. This resulted in developing a companywide service excellence management system (SEMS).

A key part of SEMS is a performance capture system that collects data on a continuous basis for each discrete component of each teller transaction in a completely automated process. This system enables SEMS to measure branch activities and generate reports on customer waiting times, teller proficiency, and productivity levels. These reports help managers schedule tellers to better match customer arrivals. They also identify opportunities for enhancing the productivity and service provided by the tellers by redesigning the service process and providing performance standards. These efforts led to a huge 53 percent reduction in the average service times, a dramatic improvement in customer waiting times, and a major increase in the level of customer satisfaction. At the same time, SEMS is expected to reduce personnel expenses by $98 million over 5 years.

Hewlett-Packard (HP) is a leading multinational manufacturer of electronic equipment. In 1993, the company installed a mechanized assembly-line system for manufacturing ink-jet printers at its plant in Vancouver, Washington, to meet the exploding demand for such printers. It soon became apparent that the system installed would not be fast enough or reliable enough to meet the company's production goals. Therefore, a joint team of management scientists from HP and the Massachusetts Institute of Technology (MIT) was formed to study how to redesign the system to improve its performance.

As described in the January–February 1998 issue of *Interfaces* for this award-winning study, the HP/MIT team quickly realized that the assembly-line system could be modeled as a special kind of queueing system where the customers (the printers to be assembled) go through a series of servers (assembly operations) in a fixed sequence. A special queueing model for this kind of system quickly provided the analytical results that were needed to determine how the system should be redesigned to achieve the required capacity in the most economical way. The changes included adding some buffer storage space at strategic points to better maintain the flow of work to the subsequent stations and to dampen the effect of machine failures. The new design increased productivity about 50 percent and yielded incremental revenues of approximately $280 million in printer sales as well as additional revenue from ancillary products. This innovative application of the special queueing model also provided HP with a new method for creating rapid and effective system designs subsequently in other areas of the company.

There have been many other award-winning applications of queueing theory to the design of queueing systems, as well as numerous additional articles describing other successful applications. However, the several examples presented in this section hopefully have given you a feeling for the kinds of applications that are occurring and for the impact they sometimes have.

17.4 THE ROLE OF THE EXPONENTIAL DISTRIBUTION

The operating characteristics of queueing systems are determined largely by two statistical properties, namely, the probability distribution of *interarrival times* (see "Input Source" in Sec. 17.2) and the probability distribution of *service times* (see "Service Mechanism" in Sec. 17.2). For real queueing systems, these distributions can take on almost any form. (The only restriction is that negative values cannot occur.) However, to formulate a queueing theory *model* as a representation of the real system, it is necessary to specify the assumed form of each of these distributions. To be useful, the assumed form should be *sufficiently realistic* that the model provides *reasonable predictions* while, at the same time, being *sufficiently simple* that the model is *mathematically tractable.* Based on these considerations, the most important probability distribution in queueing theory is the *exponential distribution.*

Suppose that a random variable T represents either interarrival or service times. (We shall refer to the occurrences marking the end of these times—arrivals or service

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completions—as *events*.) This random variable is said to have an *exponential distribution with parameter* α if its probability density function is

$$f_T(t) = \begin{cases} \alpha e^{-\alpha t} & \text{for } t \geq 0 \\ 0 & \text{for } t < 0, \end{cases}$$

as shown in Fig. 17.3. In this case, the cumulative probabilities are

$$P\{T \leq t\} = 1 - e^{-\alpha t}$$
$$P\{T > t\} = e^{-\alpha t} \qquad (t \geq 0),$$

and the expected value and variance of T are, respectively,

$$E(T) = \frac{1}{\alpha},$$

$$\text{var}(T) = \frac{1}{\alpha^2}.$$

What are the implications of assuming that T has an exponential distribution for a queueing model? To explore this question, let us examine six key properties of the exponential distribution.

Property 1: $f_T(t)$ is a strictly *decreasing* function of t ($t \geq 0$).

One consequence of Property 1 is that

$$P\{0 \leq T \leq \Delta t\} > P\{t \leq T \leq t + \Delta t\}$$

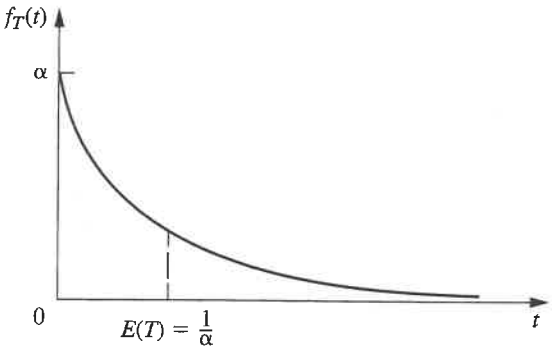
for any strictly positive values of Δt and t. [This consequence follows from the fact that these probabilities are the area under the $f_T(t)$ curve over the indicated interval of length Δt, and the average height of the curve is less for the second probability than for the first.] Therefore, it is not only possible but also relatively likely that T will take on a small value near zero. In fact,

$$P\left\{0 \leq T \leq \frac{1}{2}\frac{1}{\alpha}\right\} = 0.393$$

whereas

$$P\left\{\frac{1}{2}\frac{1}{\alpha} \leq T \leq \frac{3}{2}\frac{1}{\alpha}\right\} = 0.383,$$

so that the value T takes on is more likely to be "small" [i.e., less than half of $E(T)$] than "near" its expected value [i.e., no further away than half of $E(T)$], even though the second interval is twice as wide as the first.



■ **FIGURE 17.3**
Probability density function for the exponential distribution.

Is this really a reasonable property for T in a queueing model? If T represents *service times,* the answer depends upon the general nature of the service involved, as discussed next.

If the service required is essentially identical for each customer, with the server always performing the same sequence of service operations, then the actual service times tend to be near the expected service time. Small deviations from the mean may occur, but usually because of only minor variations in the efficiency of the server. A small service time far below the mean is essentially impossible, because a certain minimum time is needed to perform the required service operations even when the server is working at top speed. The exponential distribution clearly does not provide a close approximation to the service-time distribution for this type of situation.

On the other hand, consider the type of situation where the specific tasks required of the server differ among customers. The broad nature of the service may be the same, but the specific type and amount of service differ. For example, this is the case in the County Hospital emergency room problem discussed in Sec. 17.1. The doctors encounter a wide variety of medical problems. In most cases, they can provide the required treatment rather quickly, but an occasional patient requires extensive care. Similarly, bank tellers and grocery store checkout clerks are other servers of this general type, where the required service is often brief but must occasionally be extensive. An exponential service-time distribution would seem quite plausible for this type of service situation.

If T represents *interarrival times,* Property 1 rules out situations where potential customers approaching the queueing system tend to postpone their entry if they see another customer entering ahead of them. On the other hand, it is entirely consistent with the common phenomenon of arrivals occurring "randomly," described by subsequent properties. Thus, when arrival times are plotted on a time line, they sometimes have the appearance of being clustered with occasional large gaps separating clusters, because of the substantial probability of small interarrival times and the small probability of large interarrival times, but such an irregular pattern is all part of true randomness.

Property 2: Lack of memory.

This property can be stated mathematically as

$$P\{T > t + \Delta t \mid T > \Delta t\} = P\{T > t\}$$

for any positive quantities t and Δt. In other words, the probability distribution of the *remaining* time until the event (arrival or service completion) occurs always is the same, regardless of how much time (Δt) already has passed. In effect, the process "forgets" its history. This surprising phenomenon occurs with the exponential distribution because

$$\begin{aligned} P\{T > t + \Delta t \mid T > \Delta t\} &= \frac{P\{T > \Delta t,\ T > t + \Delta t\}}{P\{T > \Delta t\}} \\ &= \frac{P\{T > t + \Delta t\}}{P\{T > \Delta t\}} \\ &= \frac{e^{-\alpha(t+\Delta t)}}{e^{-\alpha \Delta t}} \\ &= e^{-\alpha t} \\ &= P\{T > t\}. \end{aligned}$$

For *interarrival times,* this property describes the common situation where the time until the next arrival is completely uninfluenced by when the last arrival occurred. For *service times,* the property is more difficult to interpret. We should not expect it to hold in a situation where the server must perform the same fixed sequence of operations for each customer, because then a long elapsed service should imply that probably little

remains to be done. However, in the type of situation where the required service operations differ among customers, the mathematical statement of the property may be quite realistic. For this case, if considerable service has already elapsed for a customer, the only implication may be that this particular customer requires more extensive service than most.

Property 3: The *minimum* of several independent exponential random variables has an exponential distribution.

To state this property mathematically, let $T_1, T_2, \ldots, T_n$ be *independent* exponential random variables with parameters $\alpha_1, \alpha_2, \ldots, \alpha_n$, respectively. Also let U be the random variable that takes on the value equal to the *minimum* of the values actually taken on by $T_1, T_2, \ldots, T_n$; that is,

$$U = \min \{T_1, T_2, \ldots, T_n\}.$$

Thus, if T_i represents the time until a particular kind of event occurs, then U represents the time until the *first* of the n different events occurs. Now note that for any $t \geq 0$,

$$\begin{aligned} P\{U > t\} &= P\{T_1 > t, T_2 > t, \ldots, T_n > t\} \\ &= P\{T_1 > t\}P\{T_2 > t\} \cdots P\{T_n > t\} \\ &= e^{-\alpha_1 t}e^{-\alpha_2 t} \cdots e^{\alpha_n t} \\ &= \exp\left(-\sum_{i=1}^{n} \alpha_i t\right), \end{aligned}$$

so that U indeed has an exponential distribution with parameter

$$\alpha = \sum_{i=1}^{n} \alpha_i.$$

This property has some implications for interarrival times in queueing models. In particular, suppose that there are several (n) *different* types of customers, but the interarrival times for *each* type (type i) have an exponential distribution with parameter α_i ($i = 1, 2, \ldots, n$). By Property 2, the *remaining* time from any specified instant until the next arrival of a customer of type i has this same distribution. Therefore, let T_i be this remaining time, measured from the instant a customer of *any* type arrives. Property 3 then tells us that U, the interarrival times for the queueing system as a whole, has an exponential distribution with parameter α defined by the last equation. As a result, you can choose to ignore the distinction between customers and still have exponential interarrival times for the queueing model.

However, the implications are even more important for *service times* in multiple-server queueing models than for interarrival times. For example, consider the situation where all the servers have the same exponential service-time distribution with parameter μ. For this case, let n be the number of servers *currently* providing service, and let T_i be the *remaining* service time for server i ($i = 1, 2, \ldots, n$), which also has an exponential distribution with parameter $\alpha_i = \mu$. It then follows that U, the time until the *next* service completion from any of these servers, has an exponential distribution with parameter $\alpha = n\mu$. In effect, the queueing system *currently* is performing just like a *single*-server system where service times have an exponential distribution with parameter $n\mu$. We shall make frequent use of this implication for analyzing multiple-server models later in the chapter.

When using this property, it sometimes is useful to also determine the probabilities for *which* of the exponential random variables will turn out to be the one which has the minimum value. For example, you might want to find the probability that a particular server j will finish serving a customer first among n busy exponential servers. It is fairly straightforward (see Prob. 17.4-9) to show that this probability is proportional to the

parameter α_j. In particular, the probability that T_j will turn out to be the smallest of the n random variables is

$$P\{T_j = U\} = \alpha_j / \sum_{i=1}^{n} \alpha_i, \qquad \text{for } j = 1, 2, \ldots, n.$$

Property 4: Relationship to the Poisson distribution.

Suppose that the *time* between consecutive occurrences of some particular kind of event (e.g., arrivals or service completions by a continuously busy server) has an exponential distribution with parameter α. Property 4 then has to do with the resulting implication about the probability distribution of the *number* of times this kind of event occurs over a specified time. In particular, let $X(t)$ be the number of occurrences by time t ($t \geq 0$), where time 0 designates the instant at which the count begins. The implication is that

$$P\{X(t) = n\} = \frac{(\alpha t)^n e^{-\alpha t}}{n!}, \qquad \text{for } n = 0, 1, 2, \ldots;$$

that is, $X(t)$ has a Poisson distribution with parameter αt. For example, with $n = 0$,

$$P\{X(t) = 0\} = e^{-\alpha t},$$

which is just the probability from the exponential distribution that the *first* event occurs after time t. The mean of this Poisson distribution is

$$E\{X(t)\} = \alpha t,$$

so that the expected number of events *per unit time* is α. Thus, α is said to be the *mean rate* at which the events occur. When the events are counted on a continuing basis, the counting process $\{X(t);\ t \geq 0\}$ is said to be a **Poisson process** with parameter α (the mean rate).

This property provides useful information about *service completions* when service times have an exponential distribution with parameter μ. We obtain this information by defining $X(t)$ as the number of service completions achieved by a *continuously busy* server in elapsed time t, where $\alpha = \mu$. For *multiple*-server queueing models, $X(t)$ can also be defined as the number of service completions achieved by n continuously busy servers in elapsed time t, where $\alpha = n\mu$.

The property is particularly useful for describing the probabilistic behavior of *arrivals* when interarrival times have an exponential distribution with parameter λ. In this case, $X(t)$ is the *number* of arrivals in elapsed time t, where $\alpha = \lambda$ is the *mean arrival rate.* Therefore, arrivals occur according to a **Poisson input process** with parameter λ. Such queueing models also are described as assuming a *Poisson input.*

Arrivals sometimes are said to occur *randomly,* meaning that they occur in accordance with a Poisson input process. One intuitive interpretation of this phenomenon is that every time period of fixed length has the *same* chance of having an arrival regardless of when the preceding arrival occurred, as suggested by the following property.

Property 5: For all positive values of t, $P\{T \leq t + \Delta t \mid T > t\} \approx \alpha\, \Delta t$, for small Δt.

Continuing to interpret T as the time from the last event of a certain type (arrival or service completion) until the next such event, we suppose that a time t already has elapsed without the event's occurring. We know from Property 2 that the probability that the event will occur within the next time interval of fixed length Δt is a *constant* (identified in the next paragraph), regardless of how large or small t is. Property 5 goes further to say that when the value of Δt is small, this constant probability can be approximated very closely by $\alpha\, \Delta t$. Furthermore, when considering different small

values of Δt, this probability is essentially *proportional* to Δt, with proportionality factor α. In fact, α is the *mean rate* at which the events occur (see Property 4), so that the *expected number* of events in the interval of length Δt is *exactly* $\alpha\,\Delta t$. The only reason that the probability of an event's occurring differs slightly from this value is the possibility that *more than one* event will occur, which has negligible probability when Δt is small.

To see why Property 5 holds mathematically, note that the constant value of our probability (for a fixed value of $\Delta t > 0$) is just

$$\begin{aligned} P\{T \le t + \Delta t \,|\, T > t\} &= P\{T \le \Delta t\} \\ &= 1 - e^{-\alpha\,\Delta t}, \end{aligned}$$

for any $t \ge 0$. Therefore, because the series expansion of e^x for any exponent x is

$$e^x = 1 + x + \sum_{n=2}^{\infty} \frac{x^n}{n!},$$

it follows that

$$\begin{aligned} P\{T \le t + \Delta t \,|\, T > t\} &= 1 - 1 + \alpha\,\Delta t - \sum_{n=2}^{\infty} \frac{(-\alpha\,\Delta t)^n}{n!} \\ &\approx \alpha\,\Delta t, \qquad \text{for small } \Delta t,^{[1]} \end{aligned}$$

because the summation terms become relatively negligible for sufficiently small values of $\alpha\,\Delta t$.

Because T can represent either interarrival or service times in queueing models, this property provides a convenient approximation of the probability that the event of interest occurs in the next small interval (Δt) of time. An analysis based on this approximation also can be made exact by taking appropriate limits as $\Delta t \to 0$.

Property 6: Unaffected by aggregation or disaggregation.

This property is relevant primarily for verifying that the *input process* is *Poisson.* Therefore, we shall describe it in these terms, although it also applies directly to the exponential distribution (exponential interarrival times) because of Property 4.

We first consider the aggregation (combining) of several Poisson input processes into one overall input process. In particular, suppose that there are several (n) *different* types of customers, where the customers of each type (type i) arrive according to a *Poisson input process* with parameter λ_i ($i = 1, 2, \ldots, n$). Assuming that these are *independent* Poisson processes, the property says that the *aggregate* input process (arrival of all customers without regard to type) also must be Poisson, with parameter (arrival rate) $\lambda = \lambda_1 + \lambda_2 + \cdots + \lambda_n$. In other words, having a Poisson process is *unaffected by aggregation.*

This part of the property follows directly from Properties 3 and 4. The latter property implies that the interarrival times for customers of type i have an exponential distribution with parameter λ_i. For this identical situation, we already discussed for Property 3 that it implies that the interarrival times for all customers also must have an exponential distribution, with parameter $\lambda = \lambda_1 + \lambda_2 + \cdots + \lambda_n$. Using Property 4 again then implies that the aggregate input process is Poisson.

The second part of Property 6 ("unaffected by disaggregation") refers to the reverse case, where the *aggregate* input process (the one obtained by combining the input processes

[1] More precisely,

$$\lim_{\Delta t \to 0} \frac{P\{T \le t + \Delta t \,|\, T > t\}}{\Delta t} = \alpha.$$

for several customer types) is known to be Poisson with parameter λ, but the question now concerns the nature of the *disaggregated* input processes (the individual input processes for the individual customer types). Assuming that each arriving customer has a *fixed* probability p_i of being of type i ($i = 1, 2, \ldots, n$), with

$$\lambda_i = p_i\lambda \qquad \text{and} \qquad \sum_{i=1}^{n} p_i = 1,$$

the property says that the input process for customers of type i also must be Poisson with parameter λ_i. In other words, having a Poisson process is *unaffected by disaggregation.*

As one example of the usefulness of this second part of the property, consider the following situation. Indistinguishable customers arrive according to a Poisson process with parameter λ. Each arriving customer has a fixed probability p of *balking* (leaving without entering the queueing system), so the probability of entering the system is $1 - p$. Thus, there are two types of customers—those who balk and those who enter the system. The property says that each type arrives according to a Poisson process, with parameters $p\lambda$ and $(1 - p)\lambda$, respectively. Therefore, by using the latter Poisson process, queueing models that assume a Poisson input process can still be used to analyze the performance of the queueing system for those customers who enter the system.

One of the examples in the Worked Examples section of the CD-ROM illustrates the application of several of the properties of the exponential distribution presented in this section.

17.5 THE BIRTH-AND-DEATH PROCESS

Most elementary queueing models assume that the inputs (arriving customers) and outputs (leaving customers) of the queueing system occur according to the *birth-and-death process.* This important process in probability theory has applications in various areas. However, in the context of queueing theory, the term **birth** refers to the *arrival* of a new customer into the queueing system, and **death** refers to the *departure* of a served customer. The *state* of the system at time t ($t \geq 0$), denoted by $N(t)$, is the number of customers in the queueing system at time t. The birth-and-death process describes *probabilistically* how $N(t)$ changes as t increases. Broadly speaking, it says that *individual* births and deaths occur *randomly,* where their mean occurrence rates depend only upon the current state of the system. More precisely, the assumptions of the birth-and-death process are the following:

Assumption 1. Given $N(t) = n$, the current probability distribution of the *remaining* time until the next *birth* (arrival) is *exponential* with parameter λ_n ($n = 0, 1, 2, \ldots$).

Assumption 2. Given $N(t) = n$, the current probability distribution of the *remaining* time until the next *death* (service completion) is *exponential* with parameter μ_n ($n = 1, 2, \ldots$).

Assumption 3. The random variable of assumption 1 (the remaining time until the next *birth*) and the random variable of assumption 2 (the remaining time until the next *death*) are mutually independent. The next transition in the state of the process is either

$$n \rightarrow n + 1 \quad \text{(a single birth)}$$

or

$$n \rightarrow n - 1 \quad \text{(a single death),}$$

depending on whether the former or latter random variable is smaller.

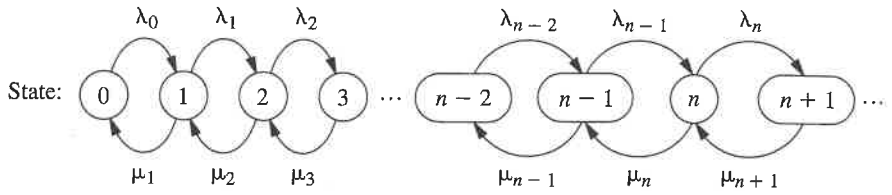


FIGURE 17.4
Rate diagram for the birth-and-death process.

For a queueing system, λ_n and μ_n respectively represent the *mean arrival rate* and the *mean rate of service completions,* when there are n customers in the system. For some queueing systems, the values of the λ_n will be the same for all values of n, and the μ_n also will be the same for all n except for such small n (e.g., $n = 0$) that a server is idle. However, the λ_n and the μ_n also can vary considerably with n for some queueing systems.

For example, one of the ways in which λ_n can be different for different values of n is if potential arriving customers become increasingly likely to *balk* (refuse to enter the system) as n increases. Similarly, μ_n can be different for different n because customers in the queue become increasingly likely to *renege* (leave without being served) as the queue size increases. One of the examples in the Worked Examples section of the CD-ROM illustrates a queueing system where both balking and reneging occur. This example then demonstrates how the general results for the birth-and-death process lead directly to various measures of performance for this queueing system.

Analysis of the Birth-and-Death Process

Because of its assumptions, the birth-and-death process is a special type of *continuous time Markov chain.* (See Sec. 16.8 for a description of continuous time Markov chains and their properties, including an introduction to the general procedure for finding steady-state probabilities that will be applied in the remainder of this section.) Queueing models that can be represented by a continuous time Markov chain are far more tractable analytically than any other.

Because Property 4 for the exponential distribution (see Sec. 17.4) implies that the λ_n and μ_n are mean rates, we can summarize these assumptions by the rate diagram shown in Fig. 17.4. The arrows in this diagram show the only possible *transitions* in the state of the system (as specified by assumption 3), and the entry for each arrow gives the mean rate for that transition (as specified by assumptions 1 and 2) when the system is in the state at the base of the arrow.

Except for a few special cases, analysis of the birth-and-death process is very difficult when the system is in a *transient* condition. Some results about the probability distribution of $N(t)$ have been obtained,[1] but they are too complicated to be of much practical use. On the other hand, it is relatively straightforward to derive this distribution *after* the system has reached a *steady-state* condition (assuming that this condition can be reached). This derivation can be done directly from the rate diagram, as outlined next.

Consider any particular state of the system n ($n = 0, 1, 2, \ldots$). Starting at time 0, suppose that a count is made of the number of times that the process enters this state and the number of times it leaves this state, as denoted below:

$E_n(t)$ = number of times that process enters state n by time t.

$L_n(t)$ = number of times that process leaves state n by time t.

[1] S. Karlin and J. McGregor, "Many Server Queueing Processes with Poisson Input and Exponential Service Times," *Pacific Journal of Mathematics,* **8:** 87–118, 1958.

Because the two types of events (entering and leaving) must alternate, these two numbers must always either be equal or differ by just 1; that is,

$$|E_n(t) - L_n(t)| \leq 1.$$

Dividing through both sides by t and then letting $t \to \infty$ gives

$$\left|\frac{E_n(t)}{t} - \frac{L_n(t)}{t}\right| \leq \frac{1}{t}, \qquad \text{so} \qquad \lim_{t\to\infty}\left|\frac{E_n(t)}{t} - \frac{L_n(t)}{t}\right| = 0.$$

Dividing $E_n(t)$ and $L_n(t)$ by t gives the *actual rate* (number of events per unit time) at which these two kinds of events have occurred, and letting $t \to \infty$ then gives the *mean rate* (expected number of events per unit time):

$$\lim_{t\to\infty}\frac{E_n(t)}{t} = \text{mean rate at which process enters state } n.$$

$$\lim_{t\to\infty}\frac{L_n(t)}{t} = \text{mean rate at which process leaves state } n.$$

These results yield the following key principle:

Rate In = Rate Out Principle. For any state of the system n ($n = 0, 1, 2, \ldots$), mean entering rate = mean leaving rate.

The equation expressing this principle is called the **balance equation** for state n. After constructing the balance equations for all the states in terms of the *unknown* P_n probabilities, we can solve this system of equations (plus an equation stating that the probabilities must sum to 1) to find these probabilities.

To illustrate a balance equation, consider state 0. The process enter this state *only* from state 1. Thus, the steady-state probability of being in state 1 (P_1) represents the proportion of time that it would be *possible* for the process to enter state 0. Given that the process is in state 1, the mean rate of entering state 0 is μ_1. (In other words, for each cumulative unit of time that the process spends in state 1, the expected number of times that it would leave state 1 to enter state 0 is μ_1.) From any *other* state, this mean rate is 0. Therefore, the overall mean rate at which the process leaves its current state to enter state 0 (the *mean entering rate*) is

$$\mu_1 P_1 + 0(1 - P_1) = \mu_1 P_1.$$

By the same reasoning, the *mean leaving rate* must be $\lambda_0 P_0$, so the balance equation for state 0 is

$$\mu_1 P_1 = \lambda_0 P_0.$$

For every other state there are two possible transitions both into and out of the state. Therefore, each side of the balance equations for these states represents the *sum* of the mean rates for the two transitions involved. Otherwise, the reasoning is just the same as for state 0. These balance equations are summarized in Table 17.1.

Notice that the first balance equation contains two variables for which to solve (P_0 and P_1), the first two equations contain three variables (P_0, P_1, and P_2), and so on, so that there always is one "extra" variable. Therefore, the procedure in solving these equations is to solve in terms of one of the variables, the most convenient one being P_0. Thus, the first equation is used to solve for P_1 in terms of P_0; this result and the second equation are then used to solve for P_2 in terms of P_0; and so forth. At the end, the requirement that the sum of all the probabilities equal 1 can be used to evaluate P_0.

TABLE 17.1 Balance equations for the birth-and-death process

State	Rate In = Rate Out
0	$\mu_1 P_1 = \lambda_0 P_0$
1	$\lambda_0 P_0 + \mu_2 P_2 = (\lambda_1 + \mu_1) P_1$
2	$\lambda_1 P_1 + \mu_3 P_3 = (\lambda_2 + \mu_2) P_2$
⋮	⋮
$n-1$	$\lambda_{n-2} P_{n-2} + \mu_n P_n = (\lambda_{n-1} + \mu_{n-1}) P_{n-1}$
n	$\lambda_{n-1} P_{n-1} + \mu_{n+1} P_{n+1} = (\lambda_n + \mu_n) P_n$
⋮	⋮

Results for the Birth-and-Death Process

Applying this procedure yields the following results:

State:

$$0: \quad P_1 = \frac{\lambda_0}{\mu_1} P_0$$

$$1: \quad P_2 = \frac{\lambda_1}{\mu_2} P_1 + \frac{1}{\mu_2}(\mu_1 P_1 - \lambda_0 P_0) = \frac{\lambda_1}{\mu_2} P_1 = \frac{\lambda_1 \lambda_0}{\mu_2 \mu_1} P_0$$

$$2: \quad P_3 = \frac{\lambda_2}{\mu_3} P_2 + \frac{1}{\mu_3}(\mu_2 P_2 - \lambda_1 P_1) = \frac{\lambda_2}{\mu_3} P_2 = \frac{\lambda_2 \lambda_1 \lambda_0}{\mu_3 \mu_2 \mu_1} P_0$$

$$\vdots \qquad \vdots$$

$$n-1: \quad P_n = \frac{\lambda_{n-1}}{\mu_n} P_{n-1} + \frac{1}{\mu_n}(\mu_{n-1} P_{n-1} - \lambda_{n-2} P_{n-2}) = \frac{\lambda_{n-1}}{\mu_n} P_{n-1} = \frac{\lambda_{n-1} \lambda_{n-2} \cdots \lambda_0}{\mu_n \mu_{n-1} \cdots \mu_1} P_0$$

$$n: \quad P_{n+1} = \frac{\lambda_n}{\mu_{n+1}} P_n + \frac{1}{\mu_{n+1}}(\mu_n P_n - \lambda_{n-1} P_{n-1}) = \frac{\lambda_n}{\mu_{n+1}} P_n = \frac{\lambda_n \lambda_{n-1} \cdots \lambda_0}{\mu_{n+1} \mu_n \cdots \mu_1} P_0$$

$$\vdots \qquad \vdots$$

To simplify notation, let

$$C_n = \frac{\lambda_{n-1} \lambda_{n-2} \cdots \lambda_0}{\mu_n \mu_{n-1} \cdots \mu_1}, \qquad \text{for } n = 1, 2, \ldots,$$

and then define $C_n = 1$ for $n = 0$. Thus, the steady-state probabilities are

$$P_n = C_n P_0, \qquad \text{for } n = 0, 1, 2, \ldots.$$

The requirement that

$$\sum_{n=0}^{\infty} P_n = 1$$

implies that

$$\left(\sum_{n=0}^{\infty} C_n \right) P_0 = 1,$$

so that

$$P_0 = \left(\sum_{n=0}^{\infty} C_n \right)^{-1}.$$

When a queueing model is based on the birth-and-death process, so the state of the system n represents the number of customers in the queueing system, the key measures of performance for the queueing system (L, L_q, W, and W_q) can be obtained immediately after calculating the P_n from the above formulas. The definitions of L and L_q given in Sec. 17.2 specify that

$$L = \sum_{n=0}^{\infty} nP_n, \qquad L_q = \sum_{n=s}^{\infty} (n - s)P_n.$$

Furthermore, the relationships given at the end of Sec. 17.2 yield

$$W = \frac{L}{\bar{\lambda}}, \qquad W_q = \frac{L_q}{\bar{\lambda}},$$

where $\bar{\lambda}$ is the *average* arrival rate over the long run. Because λ_n is the mean arrival rate while the system is in state n ($n = 0, 1, 2, \ldots$) and P_n is the proportion of time that the system is in this state,

$$\bar{\lambda} = \sum_{n=0}^{\infty} \lambda_n P_n.$$

Several of the expressions just given involve summations with an infinite number of terms. Fortunately, these summations have analytic solutions for a number of interesting special cases,[1] as seen in the next section. Otherwise, they can be approximated by summing a finite number of terms on a computer.

These steady-state results have been derived under the assumption that the λ_n and μ_n parameters have values such that the process actually can *reach* a steady-state condition. This assumption *always* holds if $\lambda_n = 0$ for some value of n greater than the initial state, so that only a finite number of states (those less than this n) are possible. It also *always* holds when λ and μ are defined (see "Terminology and Notation" in Sec. 17.2) and $\rho = \lambda/(s\mu) < 1$. It does *not* hold if $\sum_{n=1}^{\infty} C_n = \infty$.

Section 17.6 describes several queueing models that are special cases of the birth-and-death process. Therefore, the general steady-state results just given in boxes will be used over and over again to obtain the specific steady-state results for these models.

17.6 QUEUEING MODELS BASED ON THE BIRTH-AND-DEATH PROCESS

Because each of the mean rates $\lambda_0, \lambda_1, \ldots$ and $\mu_1, \mu_2, \ldots$ for the birth-and-death process can be assigned any nonnegative value, we have great flexibility in modeling a queueing system. Probably the most widely used models in queueing theory are based directly upon this process. Because of assumptions 1 and 2 (and Property 4 for the exponential distribution), these models are said to have a **Poisson input** and **exponential service times.** The models differ only in their assumptions about how the λ_n and μ_n change with n. We present three of these models in this section for three important types of queueing systems.

[1]These solutions are based on the following known results for the sum of any geometric series:

$$\sum_{n=0}^{N} x^n = \frac{1 - x^{N+1}}{1 - x}, \qquad \text{for any } x \neq 1,$$

$$\sum_{n=0}^{\infty} x^n = \frac{1}{1 - x}, \qquad \text{if } |x| < 1.$$

FIGU
Rate dia
model.

The *M/M/s* Model

As described in Sec. 17.2, the *M/M/s* model assumes that all *interarrival times* are independently and identically distributed according to an exponential distribution (i.e., the input process is Poisson), that all *service times* are independent and identically distributed according to another exponential distribution, and that the number of servers is s (any positive integer). Consequently, this model is just the special case of the birth-and-death process where the queueing system's *mean arrival rate* and *mean service rate per busy server* are constant (λ and μ, respectively) regardless of the state of the system. When the system has just a *single server* ($s = 1$), the implication is that the parameters for the birth-and-death process are $\lambda_n = \lambda$ ($n = 0, 1, 2, \ldots$) and $\mu_n = \mu$ ($n = 1, 2, \ldots$). The resulting rate diagram is shown in Fig. 17.5*a*.

However, when the system has *multiple servers* ($s > 1$), the μ_n cannot be expressed this simply. Keep in mind that μ_n represents the mean rate of service completions for the *overall* queueing system when there are n customers currently in the system. When the mean service rate per busy server is μ, the overall mean rate of service completions for n busy servers must be $n\mu$. Therefore, $\mu_n = n\mu$ when $n \leq s$, whereas $\mu_n = s\mu$ when $n \geq s$ so that all s servers are busy. The rate diagram for this case is shown in Fig. 17.5*b*.

When $s\mu$ exceeds the mean arrival rate λ, that is, when

$$\rho = \frac{\lambda}{s\mu} < 1,$$

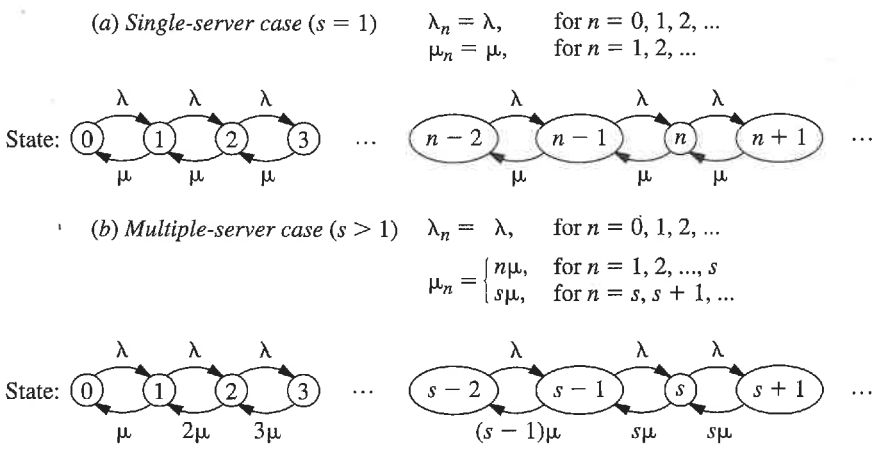
a queueing system fitting this model will eventually reach a steady-state condition. In this situation, the steady-state results derived in Sec. 17.5 for the general birth-and-death process are directly applicable. However, these results simplify considerably for this model and yield closed-form expressions for P_n, L, L_q, and so forth, as shown next.

Results for the Single-Server Case (*M/M/1*). For $s = 1$, the C_n factors for the birth-and-death process reduce to

$$C_n = \left(\frac{\lambda}{\mu}\right)^n = \rho^n, \qquad \text{for } n = 0, 1, 2, \ldots$$

Therefore,

$$P_n = \rho^n P_0, \qquad \text{for } n = 0, 1, 2, \ldots,$$



■ **FIGURE 17.5**
Rate diagrams for the *M/M/s* model.

where

$$P_0 = \left(\sum_{n=0}^{\infty} \rho^n\right)^{-1}$$
$$= \left(\frac{1}{1-\rho}\right)^{-1}$$
$$= 1 - \rho.$$

Thus,

$$P_n = (1-\rho)\rho^n, \qquad \text{for } n = 0, 1, 2, \ldots.$$

Consequently,

$$L = \sum_{n=0}^{\infty} n(1-\rho)\rho^n$$
$$= (1-\rho)\rho \sum_{n=0}^{\infty} \frac{d}{d\rho}(\rho^n)$$
$$= (1-\rho)\rho \frac{d}{d\rho}\left(\sum_{n=0}^{\infty} \rho^n\right)$$
$$= (1-\rho)\rho \frac{d}{d\rho}\left(\frac{1}{1-\rho}\right)$$
$$= \frac{\rho}{1-\rho} = \frac{\lambda}{\mu - \lambda}.$$

Similarly,

$$L_q = \sum_{n=1}^{\infty} (n-1)P_n$$
$$= L - 1(1 - P_0)$$
$$= \frac{\lambda^2}{\mu(\mu - \lambda)}.$$

When $\lambda \geq \mu$, so that the mean arrival rate exceeds the mean service rate, the preceding solution "blows up" (because the summation for computing P_0 diverges). For this case, the queue would "explode" and grow without bound. If the queueing system begins operation with no customers present, the server might succeed in keeping up with arriving customers over a short period of time, but this is impossible in the long run. (Even when $\lambda = \mu$, the *expected* number of customers in the queueing system slowly grows without bound over time because, even though a temporary return to no customers present always is possible, the probabilities of huge numbers of customers present become increasingly significant over time.)

Assuming again that $\lambda < \mu$, we now can derive the probability distribution of the *waiting time in the system* (so *including* service time) W for a random arrival when the queue discipline is first-come-first-served. If this arrival finds n customers already in the system, then the arrival will have to wait through $n + 1$ exponential service times, including his or her own. (For the customer currently being served, recall the lack-of-memory property for the exponential distribution discussed in Sec. 17.4.) Therefore, let $T_1, T_2, \ldots$ be independent service-time random variables having an exponential distribution with parameter μ, and let

$$S_{n+1} = T_1 + T_2 + \cdots + T_{n+1}, \qquad \text{for } n = 0, 1, 2, \ldots,$$

so that S_{n+1} represents the *conditional* waiting time given n customers already in the system. As discussed in Sec. 17.7, S_{n+1} is known to have an *Erlang distribution.*[1] Because the probability that the random arrival will find n customers in the system is P_n, it follows that

$$P\{\mathcal{W} > t\} = \sum_{n=0}^{\infty} P_n P\{S_{n+1} > t\},$$

which reduces after considerable manipulation (see Prob. 17.6-16) to

$$P\{\mathcal{W} > t\} = e^{-\mu(1-\rho)t}, \qquad \text{for } t \geq 0.$$

The surprising conclusion is that $\mathcal{W}$ has an *exponential* distribution with parameter $\mu(1 - \rho)$. Therefore,

$$W = E(\mathcal{W}) = \frac{1}{\mu(1-\rho)} = \frac{1}{\mu - \lambda}.$$

These results *include* service time in the waiting time. In some contexts (e.g., the County Hospital emergency room problem), the more relevant waiting time is just until service begins. Thus, consider the *waiting time in the queue* (so *excluding* service time) $\mathcal{W}_q$ for a random arrival when the queue discipline is first-come-first-served. If this arrival finds no customers already in the system, then the arrival is served immediately, so that

$$P\{\mathcal{W}_q = 0\} = P_0 = 1 - \rho.$$

If this arrival finds $n > 0$ customers already there instead, then the arrival has to wait through n exponential service times until his or her own service begins, so that

$$\begin{aligned} P\{\mathcal{W}_q > t\} &= \sum_{n=1}^{\infty} P_n P\{S_n > t\} \\ &= \sum_{n=1}^{\infty} (1-\rho)\rho^n P\{S_n > t\} \\ &= \rho \sum_{n=0}^{\infty} P_n P\{S_{n+1} > t\} \\ &= \rho P\{\mathcal{W} > t\} \\ &= \rho e^{-\mu(1-\rho)t}, \qquad \text{for } t \geq 0. \end{aligned}$$

Note that Wq does not quite have an exponential distribution, because $P\{\mathcal{W}_q = 0\} > 0$. However, the *conditional* distribution of $\mathcal{W}_q$, given that $\mathcal{W}_q > 0$, does have an exponential distribution with parameter $\mu(1 - \rho)$, just as $\mathcal{W}$ does, because

$$P\{\mathcal{W}_q > t \mid \mathcal{W}_q > 0\} = \frac{P\{\mathcal{W}_q > t\}}{P\{\mathcal{W}_q > 0\}} = e^{-\mu(1-\rho)t}, \qquad \text{for } t \geq 0.$$

By deriving the mean of the (unconditional) distribution of $\mathcal{W}_q$ (or applying either $L_q = \lambda W_q$ or $W_q = W - 1/\mu$),

$$W_q = E(\mathcal{W}_q) = \frac{\lambda}{\mu(\mu - \lambda)}.$$

[1]Outside queueing theory, this distribution is known as the *gamma distribution.*

If you would like to see an example that applies the *M/M/*1 model to determine which type of materials handling equipment a company should purchase, one is provided in the Worked Examples section of the CD-ROM.

Results for the Multiple-Server Case ($s > 1$). When $s > 1$, the C_n factors become

$$C_n = \begin{cases} \dfrac{(\lambda/\mu)^n}{n!} & \text{for } n = 1, 2, \ldots, s \\ \dfrac{(\lambda/\mu)^s}{s!}\left(\dfrac{\lambda}{s\mu}\right)^{n-s} = \dfrac{(\lambda/\mu)^n}{s!\,s^{n-s}} & \text{for } n = s, s+1, \ldots. \end{cases}$$

Consequently, if $\lambda < s\mu$ [so that $\rho = \lambda/(s\mu) < 1$], then

$$P_0 = 1\Big/\left[1 + \sum_{n=1}^{s-1} \frac{(\lambda/\mu)^n}{n!} + \frac{(\lambda/\mu)^s}{s!}\sum_{n=s}^{\infty}\left(\frac{\lambda}{s\mu}\right)^{n-s}\right]$$

$$= 1\Big/\left[\sum_{n=0}^{s-1} \frac{(\lambda/\mu)^n}{n!} + \frac{(\lambda/\mu)^s}{s!}\frac{1}{1-\lambda/(s\mu)}\right],$$

where the $n = 0$ term in the last summation yields the correct value of 1 because of the convention that $n! = 1$ when $n = 0$. These C_n factors also give

$$P_n = \begin{cases} \dfrac{(\lambda/\mu)^n}{n!}P_0 & \text{if } 0 \le n \le s \\ \dfrac{(\lambda/\mu)^n}{s!\,s^{n-s}}P_0 & \text{if } n \ge s. \end{cases}$$

Furthermore,

$$L_q = \sum_{n=s}^{\infty}(n-s)P_n$$

$$= \sum_{j=0}^{\infty} jP_{s+j}$$

$$= \sum_{j=0}^{\infty} j\,\frac{(\lambda/\mu)^s}{s!}\rho^j P_0$$

$$= P_0\frac{(\lambda/\mu)^s}{s!}\rho\sum_{j=0}^{\infty}\frac{d}{d\rho}(\rho^j)$$

$$= P_0\frac{(\lambda/\mu)^s}{s!}\rho\,\frac{d}{d\rho}\left(\sum_{j=0}^{\infty}\rho^j\right)$$

$$= P_0\frac{(\lambda/\mu)^s}{s!}\rho\,\frac{d}{d\rho}\left(\frac{1}{1-\rho}\right)$$

$$= \frac{P_0(\lambda/\mu)^s\rho}{s!(1-\rho)^2},$$

$$W_q = \frac{L_q}{\lambda};$$

$$W = W_q + \frac{1}{\mu};$$

$$L = \lambda\left(W_q + \frac{1}{\mu}\right) = L_q + \frac{\lambda}{\mu}.$$

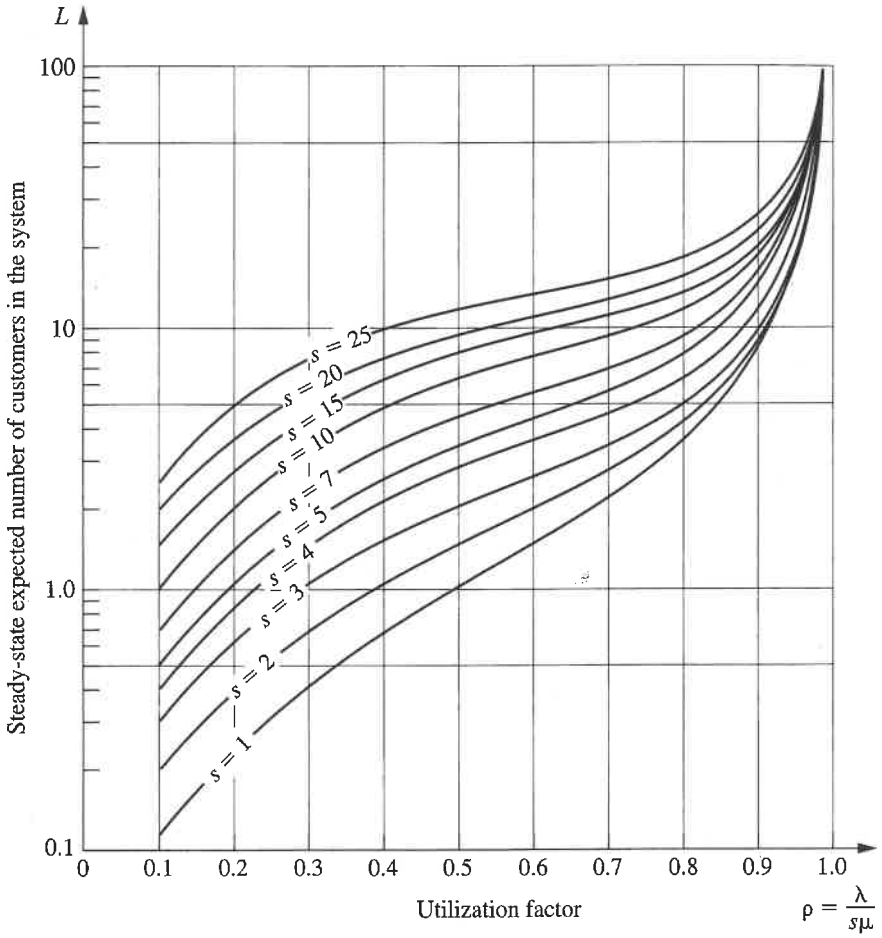
Figure 17.6 shows how L changes with ρ for various values of s.

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■ **FIGURE 17.6**
Values for L for the *M/M/s* model (Sec. 17.6).

The single-server method for finding the probability distribution of waiting times also can be extended to the multiple-server case. This yields[1] (for $t \geq 0$)

$$P\{W > t\} = e^{-\mu t}\left[\frac{1 + P_0(\lambda/\mu)^s}{s!(1-\rho)}\left(\frac{1 - e^{-\mu t(s-1-\lambda/\mu)}}{s - 1 - \lambda/\mu}\right)\right]$$

and

$$P\{W_q > t\} = (1 - P\{W_q = 0\})e^{-s\mu(1-\rho)t},$$

where

$$P\{W_q = 0\} = \sum_{n=0}^{s-1} P_n.$$

The above formulas for the various measures of performance (including the P_n) are relatively imposing for hand calculations. However, this chapter's Excel file in your OR Courseware includes an Excel template that performs all these calculations simultaneously for any values of t, s, λ, and μ you want, provided that $\lambda < s\mu$.

[1]When $s - 1 - \lambda/\mu = 0$, $(1 - e^{-\mu t(s-1-\lambda/\mu)})/(s - 1 - \lambda/\mu)$ should be replaced by μt.

If $\lambda \geq s\mu$, so that the mean arrival rate exceeds the maximum mean rate of service completions, then the queue grows without bound, so the preceding steady-state solutions are not applicable.

The County Hospital Example with the *M/M/s* Model. For the County Hospital emergency room problem (see Sec. 17.1), the management engineer has concluded that the emergency cases arrive pretty much at random (a *Poisson input process*), so that interarrival times have an exponential distribution. She also has concluded that the time spent by a doctor treating the cases approximately follows an *exponential distribution.* Therefore, she has chosen the *M/M/s* model for a preliminary study of this queueing system.

By projecting the available data for the early evening shift into next year, she estimates that patients will arrive at an *average* rate of 1 every $\frac{1}{2}$ hour. A doctor requires an average of 20 minutes to treat each patient. Thus, with one hour as the unit of time,

$$\frac{1}{\lambda} = \frac{1}{2} \text{ hour per customer}$$

and

$$\frac{1}{\mu} = \frac{1}{3} \text{ hour per customer,}$$

so that

$$\lambda = 2 \text{ customers per hour}$$

and

$$\mu = 3 \text{ customers per hour.}$$

The two alternatives being considered are to continue having just one doctor during this shift ($s = 1$) or to add a second doctor ($s = 2$). In both cases,

$$\rho = \frac{\lambda}{s\mu} < 1,$$

so that the system should approach a steady-state condition. (Actually, because λ is somewhat different during other shifts, the system will never truly reach a steady-state condition, but the management engineer feels that steady-state results will provide a good approximation.) Therefore, the preceding equations are used to obtain the results shown in Table 17.2.

On the basis of these results, she tentatively concluded that a single doctor would be inadequate next year for providing the relatively prompt treatment needed in a hospital emergency room. You will see later (Sec. 17.8) how she checked this conclusion by applying another queueing model that provides a better representation of the real queueing system in one crucial way.

You can see another example of an application of the *M/M/*1 model in the Worked Examples section of the CD-ROM, where the issue in this case is whether three employees in a fast-food restaurant should work together as one fast server or separately as three considerably slower servers.

TABLE 17.2 Steady-state results from the $M/M/s$ model for the County Hospital problem

	$s = 1$	$s = 2$
ρ	$\frac{2}{3}$	$\frac{1}{3}$
P_0	$\frac{1}{3}$	$\frac{1}{2}$
P_1	$\frac{2}{9}$	$\frac{1}{3}$
P_n for $n \geq 2$	$\frac{1}{3}\left(\frac{2}{3}\right)^n$	$\left(\frac{1}{3}\right)^n$
L_q	$\frac{4}{3}$	$\frac{1}{12}$
L	2	$\frac{3}{4}$
W_q	$\frac{2}{3}$ hour	$\frac{1}{24}$ hour
W	1 hour	$\frac{3}{8}$ hour
$P\{W_q > 0\}$	0.667	0.167
$P\left\{W_q > \frac{1}{2}\right\}$	0.404	0.022
$P\{W_q > 1\}$	0.245	0.003
$P\{W_q > t\}$	$\frac{2}{3}e^{-t}$	$\frac{1}{6}e^{-4t}$
$P\{W > t\}$	e^{-t}	$\frac{1}{2}e^{-3t}(3 - e^{-t})$

The Finite Queue Variation of the $M/M/s$ Model (Called the $M/M/s/K$ Model)

We mentioned in the discussion of queues in Sec. 17.2 that queueing systems sometimes have a *finite queue;* i.e., the number of customers in the system is not permitted to exceed some specified number (denoted by K) so the queue capacity is $K - s$. Any customer that arrives while the queue is "full" is refused entry into the system and so leaves forever. From the viewpoint of the birth-and-death process, the mean input rate into the system becomes zero at these times. Therefore, the one modification needed in the $M/M/s$ model to introduce a finite queue is to change the λ_n parameters to

$$\lambda_n = \begin{cases} \lambda & \text{for } n = 0, 1, 2, \ldots, K-1 \\ 0 & \text{for } n \geq K. \end{cases}$$

Because $\lambda_n = 0$ for some values of n, a queueing system that fits this model always will eventually reach a steady-state condition, even when $\rho = \lambda/s\mu \geq 1$.

This model commonly is labeled $M/M/s/K$, where the presence of the fourth symbol distinguishes it from the $M/M/s$ model. The single difference in the formulation of these two models is that K is finite for the $M/M/s/K$ model and $K = \infty$ for the $M/M/s$ model.

The usual physical interpretation for the $M/M/s/K$ model is that there is only *limited waiting room* that will accommodate a maximum of K customers in the system. For example, for the County Hospital emergency room problem, this system actually would have

a finite queue if there were only K cots for the patients and if the policy were to send arriving patients to another hospital whenever there were no empty cots.

Another possible interpretation is that arriving customers will leave and "take their business elsewhere" whenever they find too many customers (K) ahead of them in the system because they are not willing to incur a long wait. This balking phenomenon is quite common in commercial service systems. However, there are other models available (e.g., see Prob. 17.5-5) that fit this interpretation even better.

The rate diagram for this model is identical to that shown in Fig. 17.5 for the *M/M/s* model, *except* that it stops with state K.

Results for the Single-Server Case (*M/M/1/K*). For this case,

$$C_n = \begin{cases} \left(\frac{\lambda}{\mu}\right)^n = \rho^n & \text{for } n = 0, 1, 2, \ldots, K \\ 0 & \text{for } n > K. \end{cases}$$

Therefore, for $\rho \neq 1$,[1]

$$\begin{aligned} P_0 &= \frac{1}{\sum_{n=0}^{K} (\lambda/\mu)^n} \\ &= 1 \Big/ \left[\frac{1 - (\lambda/\mu)^{K+1}}{1 - \lambda/\mu} \right] \\ &= \frac{1 - \rho}{1 - \rho^{K+1}}, \end{aligned}$$

so that

$$P_n = \frac{1 - \rho}{1 - \rho^{K+1}} \rho^n, \qquad \text{for } n = 0, 1, 2, \ldots, K.$$

Hence,

$$\begin{aligned} L &= \sum_{n=0}^{K} nP_n \\ &= \frac{1 - \rho}{1 - \rho^{K+1}} \rho \sum_{n=0}^{K} \frac{d}{d\rho}(\rho^n) \\ &= \frac{1 - \rho}{1 - \rho^{K+1}} \rho \frac{d}{d\rho}\left(\sum_{n=0}^{K} \rho^n \right) \\ &= \frac{1 - \rho}{1 - \rho^{K+1}} \rho \frac{d}{d\rho}\left(\frac{1 - \rho^{K+1}}{1 - \rho} \right) \\ &= \rho \frac{-(K+1)\rho^K + K\rho^{K+1} + 1}{(1 - \rho^{K+1})(1 - \rho)} \\ &= \frac{\rho}{1 - \rho} - \frac{(K+1)\rho^{K+1}}{1 - \rho^{K+1}}. \end{aligned}$$

As usual (when $s = 1$),

$$L_q = L - (1 - P_0).$$

Notice that the preceding results do not require that $\lambda < \mu$ (i.e., that $\rho < 1$).

[1] If $\rho = 1$, then $P_n = 1/(K + 1)$ for $n = 0, 1, 2, \ldots, K$, so that $L = K/2$.

When $\rho < 1$, it can be verified that the second term in the final expression for L converges to 0 as $K \to \infty$, so that *all* the preceding results do indeed converge to the corresponding results given earlier for the *M/M/*1 model.

The waiting-time distributions can be derived by using the same reasoning as for the *M/M/*1 model (see Prob. 17.6-27). However, no simple expressions are obtained in this case, so computer calculations are required. Fortunately, even though $L \neq \lambda W$ and $L_q \neq \lambda W_q$ for the current model because the λ_n are not equal for all n (see the end of Sec. 17.2), the *expected* waiting times for customers entering the system still can be obtained directly from the expressions given at the end of Sec. 17.5:

$$W = \frac{L}{\bar{\lambda}}, \qquad W_q = \frac{L_q}{\bar{\lambda}},$$

where

$$\begin{aligned}\bar{\lambda} &= \sum_{n=0}^{\infty} \lambda_n P_n \\ &= \sum_{n=0}^{K-1} \lambda P_n \\ &= \lambda(1 - P_K).\end{aligned}$$

Results for the Multiple-Server Case ($s > 1$). Because this model does not allow more than K customers in the system, K is the maximum number of servers that could ever be used. Therefore, assume that $s \leq K$. In this case, C_n becomes

$$C_n = \begin{cases} \dfrac{(\lambda/\mu)^n}{n!} & \text{for } n = 0, 1, 2, \ldots, s \\ \dfrac{(\lambda/\mu)^s}{s!}\left(\dfrac{\lambda}{s\mu}\right)^{n-s} = \dfrac{(\lambda/\mu)^n}{s!\,s^{n-s}} & \text{for } n = s, s+1, \ldots, K \\ 0 & \text{for } n > K. \end{cases}$$

Hence,

$$P_n = \begin{cases} \dfrac{(\lambda/\mu)^n}{n!} P_0 & \text{for } n = 1, 2, \ldots, s \\ \dfrac{(\lambda/\mu)^n}{s!\,s^{n-s}} P_0 & \text{for } n = s, s+1, \ldots, K \\ 0 & \text{for } n > K, \end{cases}$$

where

$$P_0 = 1 \Big/ \left[\sum_{n=0}^{s} \frac{(\lambda/\mu)^n}{n!} + \frac{(\lambda/\mu)^s}{s!} \sum_{n=s+1}^{K} \left(\frac{\lambda}{s\mu}\right)^{n-s} \right].$$

(These formulas continue to use the convention that $n! = 1$ when $n = 0$.)
Adapting the derivation of L_q for the *M/M/s* model to this case yields

$$L_q = \frac{P_0(\lambda/\mu)^s \rho}{s!(1-\rho)^2}\,[1 - \rho^{K-s} - (K-s)\rho^{K-s}(1-\rho)],$$

where $\rho = \lambda/(s\mu)$.[1] It can then be shown that

$$L = \sum_{n=0}^{s-1} nP_n + L_q + s\left(1 - \sum_{n=0}^{s-1} P_n\right).$$

And W and W_q are obtained from these quantities just as shown for the single-server case.

[1] If $\rho = 1$, it is necessary to apply L'Hôpital's rule twice to this expression for L_q. Otherwise, all these multiple-server results hold for all $\rho > 0$. The reason that this queueing system can reach a steady-state condition even when $\rho \geq 1$ is that $\lambda_n = 0$ for $n \geq K$, so that the number of customers in the system cannot continue to grow indefinitely.

This chapter's Excel file includes an Excel template for calculating the above measures of performance (including the P_n) for this model.

One interesting special case of this model is where $K = s$ so the queue capacity is $K - s = 0$. In this case, customers who arrive when all servers are busy will leave immediately and be lost to the system. This would occur, for example, in a telephone network with s trunk lines so callers get a busy signal and hang up when all the trunk lines are busy. This kind of system (a "queueing system" with no queue) is referred to as *Erlang's loss system* because it was first studied in the early 20th century by A. K. Erlang, a Danish telephone engineer who is considered the founder of queueing theory.

It is common now for the telephone system at a call center to provide some extra trunk lines that place the caller on hold, but additional callers then get a busy signal. Such a system also fits this model, where $(K - s)$ is the number of extra trunk lines that place the caller on hold. One of the examples in the Worked Examples section of the CD-ROM illustrates the application of this model to such a system.

The Finite Calling Population Variation of the *M/M/s* Model

Now assume that the only deviation from the *M/M/s* model is that (as defined in Sec. 17.2) the *input source* is *limited;* i.e., the size of the *calling population* is *finite.* For this case, let N denote the size of the calling population. Thus, when the number of customers in the queueing system is n ($n = 0, 1, 2, \ldots, N$), there are only $N - n$ *potential* customers remaining in the input source.

The most important application of this model has been to the machine repair problem, where one or more maintenance people are assigned the responsibility of maintaining in operational order a certain group of N machines by repairing each one that breaks down. (The example given at the end of Sec. 16.8 illustrates this application when the general procedures for solving any *continuous time Markov chain* are used rather than the specific formulas available for the birth-and-death process.) The maintenance people are considered to be individual servers in the queueing system if they work individually on different machines, whereas the entire crew is considered to be a single server if crew members work together on each machine. The machines constitute the calling population. Each one is considered to be a customer in the queueing system when it is down waiting to be repaired, whereas it is outside the queueing system while it is operational.

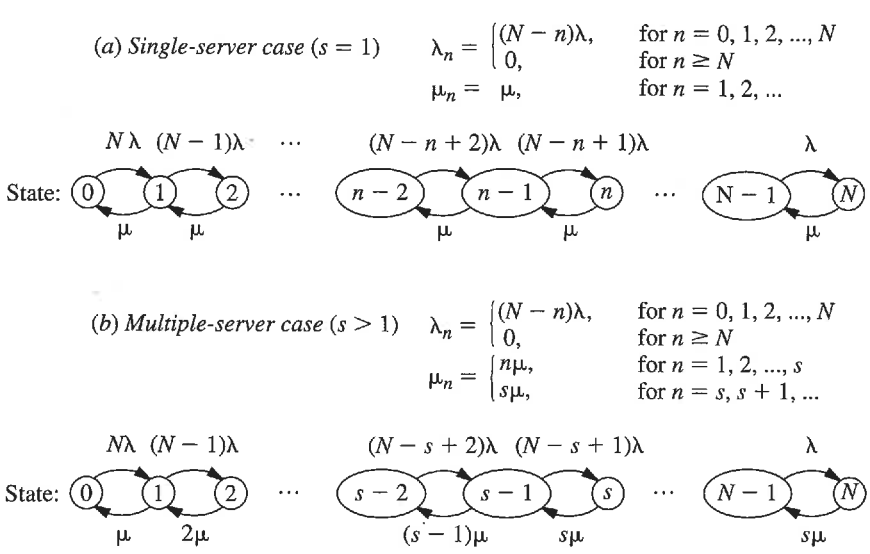
Note that each member of the calling population alternates between being *inside* and *outside* the queueing system. Therefore, the analog of the *M/M/s* model that fits this situation assumes that *each* member's *outside time* (i.e., the elapsed time from leaving the system until returning for the next time) has an *exponential distribution* with parameter λ. When n of the members are *inside,* and so $N - n$ members are *outside,* the current probability distribution of the *remaining* time until the next arrival to the queueing system is the distribution of the *minimum* of the *remaining outside times* for the latter $N - n$ members. Properties 2 and 3 for the exponential distribution imply that this distribution must be exponential with parameter $\lambda_n = (N - n)\lambda$. Hence, this model is just the special case of the birth-and-death process that has the rate diagram shown in Fig. 17.7.

Because $\lambda_n = 0$ for $n = N$, any queueing system that fits this model will eventually reach a steady-state condition. The available steady-state results are summarized as follows:

Results for the Single-Server Case ($s = 1$). When $s = 1$, the C_n factors in Sec. 17.5 reduce to

$$C_n = \begin{cases} N(N-1)\cdots(N-n+1)\left(\frac{\lambda}{\mu}\right)^n = \frac{N!}{(N-n)!}\left(\frac{\lambda}{\mu}\right)^n & \text{for } n \leq N \\ 0 & \text{for } n > N, \end{cases}$$

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■ **FIGURE 17.7**
Rate diagrams for the finite calling population variation of the *M/M/s* model.

for this model. Therefore, again using the convention that $n! = 1$ when $n = 0$,

$$P_0 = 1 \Big/ \sum_{n=0}^{N} \left[\frac{N!}{(N-n)!} \left(\frac{\lambda}{\mu} \right)^n \right];$$

$$P_n = \frac{N!}{(N-n)!} \left(\frac{\lambda}{\mu} \right)^n P_0, \qquad \text{if } n = 1, 2, \ldots, N;$$

$$L_q = \sum_{n=1}^{N} (n-1) P_n,$$

which can be reduced to

$$L_q = N - \frac{\lambda + \mu}{\lambda}(1 - P_0);$$

$$L = \sum_{n=0}^{N} n P_n = L_q + 1 - P_0$$

$$= N - \frac{\mu}{\lambda}(1 - P_0).$$

Finally,

$$W = \frac{L}{\bar{\lambda}} \qquad \text{and} \qquad W_q = \frac{L_q}{\bar{\lambda}},$$

where

$$\bar{\lambda} = \sum_{n=0}^{\infty} \lambda_n P_n = \sum_{n=0}^{N} (N-n)\lambda P_n = \lambda(N - L).$$

At this point, you might find it helpful to refer back to the example at the end of Sec. 16.8, because that example completely fits this model for the single-server case. In particular, $N = 2$, $\lambda = 1$, and $\mu = 2$ for that example, so $P_0 = 0.4$, $P_1 = 0.4$, $P_2 = 0.2$, and so forth.

Results for the Multiple-Server Case ($s > 1$). For $N \geq s > 1$,

$$C_n = \begin{cases} \dfrac{N!}{(N-n)!n!}\left(\dfrac{\lambda}{\mu}\right)^n & \text{for } n = 0, 1, 2, \ldots, s \\ \dfrac{N!}{(N-n)!s!s^{n-s}}\left(\dfrac{\lambda}{\mu}\right)^n & \text{for } n = s, s+1, \ldots, N \\ 0 & \text{for } n > N. \end{cases}$$

Hence,

$$P_n = \begin{cases} \dfrac{N!}{(N-n)!n!}\left(\dfrac{\lambda}{\mu}\right)^n P_0 & \text{if } 0 \leq n \leq s \\ \dfrac{N!}{(N-n)!s!s^{n-s}}\left(\dfrac{\lambda}{\mu}\right)^n P_0 & \text{if } s \leq n \leq N \\ 0 & \text{if } n > N, \end{cases}$$

where

$$P_0 = 1 \Big/ \left[\sum_{n=0}^{s-1} \frac{N!}{(N-n)!n!}\left(\frac{\lambda}{\mu}\right)^n + \sum_{n=s}^{N} \frac{N!}{(N-n)!s!s^{n-s}}\left(\frac{\lambda}{\mu}\right)^n\right].$$

Finally,

$$L_q = \sum_{n=s}^{N} (n-s)P_n$$

and

$$L = \sum_{n=0}^{s-1} nP_n + L_q + s\left(1 - \sum_{n=0}^{s-1} P_n\right),$$

which then yield W and W_q by the same equations as in the single-server case.

This chapter's Excel file includes an Excel template for performing all the above calculations.

Extensive tables of computational results also are available[1] for this model for both the single-server and multiple-server cases.

For both cases, it has been shown[2] that the preceding formulas for P_n and P_0 (and so for L_q, L, W, and W_q) *also* hold for a generalization of this model. In particular, we can *drop* the assumption that the times spent *outside* the queueing system by the members of the calling population have an *exponential distribution,* even though this takes the model outside the realm of the birth-and-death process. As long as these times are identically distributed with mean $1/\lambda$ (and the assumption of exponential service times still holds), these outside times can have *any* probability distribution!

17.7 QUEUEING MODELS INVOLVING NONEXPONENTIAL DISTRIBUTIONS

Because all the queueing theory models in the preceding section (except for one generalization) are based on the birth-and-death process, both their interarrival and service times are required to have *exponential* distributions. As discussed in Sec. 17.4, this type

[1] L. G. Peck and R. N. Hazelwood, *Finite Queueing Tables,* Wiley, New York, 1958.

[2] B. D. Bunday and R. E. Scraton, "The G/M/r Machine Interference Model," *European Journal of Operational Research,* **4:** 399–402, 1980.

of probability distribution has many convenient properties for queueing theory, but it provides a reasonable fit for only certain kinds of queueing systems. In particular, the assumption of exponential interarrival times implies that arrivals occur randomly (a Poisson input process), which is a reasonable approximation in many situations but *not* when the arrivals are carefully scheduled or regulated. Furthermore, the actual service-time distribution frequently deviates greatly from the exponential form, particularly when the service requirements of the customers are quite similar. Therefore, it is important to have available other queueing models that use alternative distributions.

Unfortunately, the mathematical analysis of queueing models with nonexponential distributions is much more difficult. However, it has been possible to obtain some useful results for a few such models. This analysis is beyond the level of this book, but in this section we shall summarize the models and describe their results.

The *M/G/1* Model

As introduced in Sec. 17.2, the *M/G/1* model assumes that the queueing system has a *single server* and a *Poisson input process* (exponential interarrival times) with a *fixed* mean arrival rate λ. As usual, it is assumed that the customers have *independent* service times with the *same* probability distribution. However, no restrictions are imposed on what this service-time distribution can be. In fact, it is only necessary to know (or estimate) the mean $1/\mu$ and variance σ^2 of this distribution.

Any such queueing system can eventually reach a steady-state condition if $\rho = \lambda/\mu < 1$. The readily available steady-state results[1] for this general model are the following:

$$P_0 = 1 - \rho,$$

$$L_q = \frac{\lambda^2\sigma^2 + \rho^2}{2(1 - \rho)},$$

$$L = \rho + L_q,$$

$$W_q = \frac{L_q}{\lambda},$$

$$W = W_q + \frac{1}{\mu}.$$

Considering the complexity involved in analyzing a model that permits *any* service-time distribution, it is remarkable that such a simple formula can be obtained for L_q. This formula is one of the most important results in queueing theory because of its ease of use and the prevalence of *M/G/1* queueing systems in practice. This equation for L_q (or its counterpart for W_q) commonly is referred to as the **Pollaczek-Khintchine formula,** named after two pioneers in the development of queueing theory who derived the formula independently in the early 1930s.

For any fixed expected service time $1/\mu$, notice that L_q, L, W_q, and W all increase as σ^2 is increased. This result is important because it indicates that the consistency of the server has a major bearing on the performance of the service facility—not just the server's average speed. This key point is illustrated in the next subsection.

When the service-time distribution is exponential, $\sigma^2 = 1/\mu^2$, and the preceding results will reduce to the corresponding results for the *M/M/1* model given at the beginning of Sec. 17.6.

[1]A recursion formula also is available for calculating the probability distribution of the number of customers in the system; see A. Hordijk and H. C. Tijms, "A Simple Proof of the Equivalence of the Limiting Distribution of the Continuous-Time and the Embedded Process of the Queue Size in the *M/G/1* Queue," *Statistica Neerlandica,* **36:** 97–100, 1976.

The complete flexibility in the service-time distribution provided by this model is extremely useful, so it is unfortunate that efforts to derive similar results for the multiple-server case have been unsuccessful. However, some multiple-server results have been obtained for the important special cases described by the following two models. (Excel templates are available in this chapter's Excel file for performing the calculations for both the *M*/*G*/1 model and the two models considered below when $s = 1$.)

The *M*/*D*/*s* Model

When the service consists of essentially the same routine task to be performed for all customers, there tends to be little variation in the service time required. The *M*/*D*/*s* model often provides a reasonable representation for this kind of situation, because it assumes that all service times actually equal some fixed *constant* (the *degenerate* service-time distribution) and that we have a *Poisson* input process with a fixed mean arrival rate λ.

When there is just a single server, the *M*/*D*/1 model is just the special case of the *M*/*G*/1 model where $\sigma^2 = 0$, so that the *Pollaczek-Khintchine formula* reduces to

$$L_q = \frac{\rho^2}{2(1-\rho)},$$

where L, W_q, and W are obtained from L_q as just shown. Notice that these L_q and W_q are exactly *half* as large as those for the exponential service-time case of Sec. 17.6 (the *M*/*M*/1 model), where $\sigma^2 = 1/\mu^2$, so decreasing σ^2 can *greatly* improve the measures of performance of a queueing system.

For the multiple-server version of this model (*M*/*D*/*s*), a complicated method is available[1] for deriving the steady-state probability distribution of the number of customers in the system and its mean [assuming $\rho = \lambda/(s\mu) < 1$]. These results have been tabulated for numerous cases,[2] and the means (L) also are given graphically in Fig. 17.8.

The *M*/E_k/*s* Model

The *M*/*D*/*s* model assumes *zero* variation in the service times ($\sigma = 0$), whereas the *exponential* service-time distribution assumes a very large variation ($\sigma = 1/\mu$). Between these two rather extreme cases lies a long middle ground ($0 < \sigma < 1/\mu$), where most *actual* service-time distributions fall. Another kind of theoretical service-time distribution that fills this middle ground is the **Erlang distribution** (named after the founder of queueing theory).

The probability density function for the Erlang distribution is

$$f(t) = \frac{(\mu k)^k}{(k-1)!} t^{k-1} e^{-k\mu t}, \qquad \text{for } t \geq 0,$$

where μ and k are strictly positive parameters of the distribution and k is further restricted to be integer. (Except for this integer restriction and the definition of the parameters, this distribution is *identical* to the *gamma distribution*.) Its mean and standard deviation are

$$\text{Mean} = \frac{1}{\mu}$$

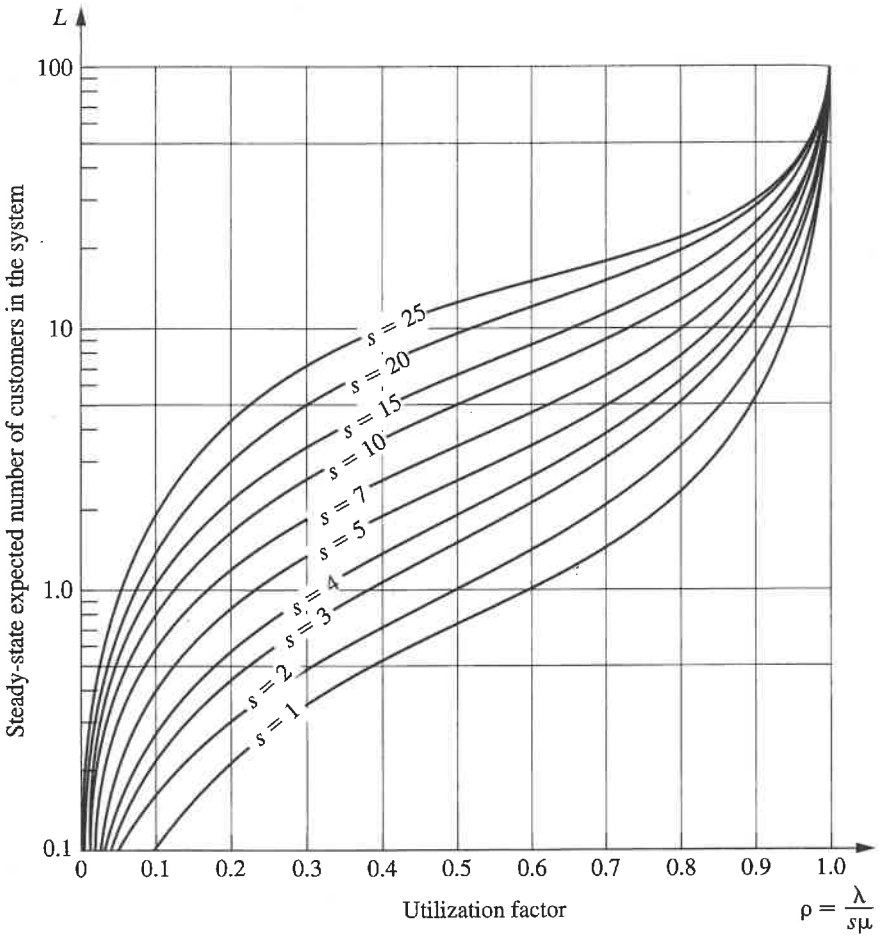
and

$$\text{Standard deviation} = \frac{1}{\sqrt{k}} \frac{1}{\mu}.$$

[1]See N. U. Prabhu: *Queues and Inventories*, Wiley, New York, 1965, pp. 32–34; also see pp. 286–288 in Selected Reference 5.

[2]F. S. Hillier and O. S. Yu, with D. Avis, L. Fossett, F. Lo, and M. Reiman, *Queueing Tables and Graphs*, Elsevier North-Holland, New York, 1981.

■ FIGU
Values
model



■ **FIGURE 17.8**
Values of L for the $M/D/s$ model (Sec. 17.7).

Thus, k is the parameter that specifies the degree of variability of the service times relative to the mean. It usually is referred to as the *shape parameter.*

The Erlang distribution is a very important distribution in queueing theory for two reasons. To describe the first one, suppose that $T_1, T_2, \ldots, T_k$ are k independent random variables with an identical exponential distribution whose mean is $1/(k\mu)$. Then their sum

$$T = T_1 + T_2 + \cdots + T_k$$

has an *Erlang* distribution with parameters μ and k. The discussion of the exponential distribution in Sec. 17.4 suggested that the time required to perform certain kinds of tasks might well have an exponential distribution. However, the total service required by a customer may involve the server's performing not just one specific task but a sequence of k tasks. If the respective tasks have an independent and identical exponential distribution for their duration, the total service time will have an Erlang distribution. This will be the case, e.g., if the server must perform the *same* exponential task k independent times for each customer.

The Erlang distribution also is very useful because it is a large (two-parameter) family of distributions permitting only nonnegative values. Hence, empirical service-time distributions can usually be reasonably approximated by an Erlang distribution. In fact, both

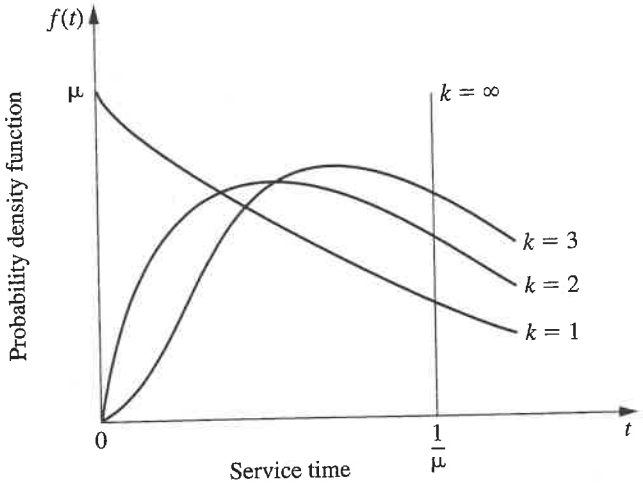


FIGURE 17.9
A family of Erlang distributions with constant mean $1/\mu$.

the *exponential* and the *degenerate* (constant) distributions are special cases of the Erlang distribution, with $k = 1$ and $k = \infty$, respectively. Intermediate values of k provide intermediate distributions with mean $= 1/\mu$, mode $= (k - 1)/(k\mu)$, and variance $= 1/(k\mu^2)$, as suggested by Fig. 17.9. Therefore, after estimating the mean and variance of an empirical service-time distribution, these formulas for the mean and variance can be used to choose the integer value of k that matches the estimates most closely.

Now consider the $M/E_k/1$ model, which is just the special case of the $M/G/1$ model where service times have an Erlang distribution with shape parameter $= k$. Applying the Pollaczek-Khintchine formula with $\sigma^2 = 1/(k\mu^2)$ (and the accompanying results given for $M/G/1$) yields

$$L_q = \frac{\lambda^2/(k\mu^2) + \rho^2}{2(1-\rho)} = \frac{1+k}{2k}\frac{\lambda^2}{\mu(\mu-\lambda)},$$

$$W_q = \frac{1+k}{2k}\frac{\lambda}{\mu(\mu-\lambda)},$$

$$W = W_q + \frac{1}{\mu},$$

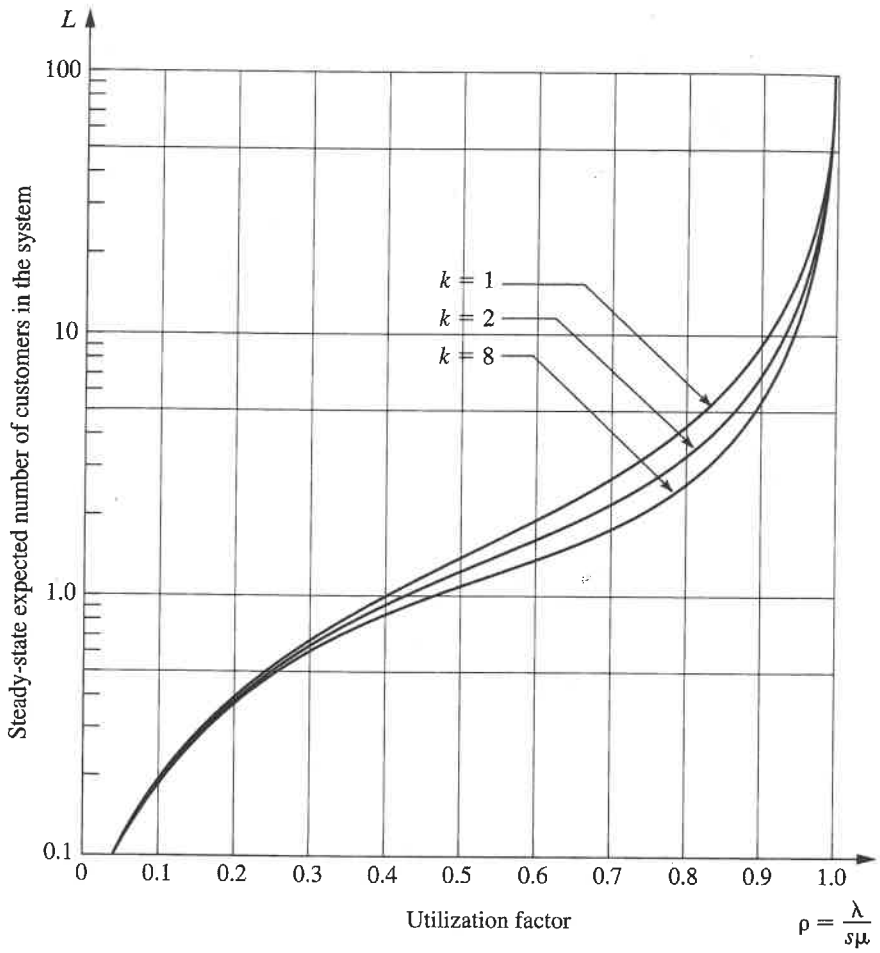
$$L = \lambda W.$$

With multiple servers ($M/E_k/s$), the relationship of the Erlang distribution to the exponential distribution just described can be exploited to formulate a *modified* birth-and-death process (continuous time Markov chain) in terms of individual exponential service phases (k per customer) rather than complete customers. However, it has not been possible to derive a general steady-state solution [when $\rho = \lambda/(s\mu) < 1$] for the probability distribution of the number of customers in the system as we did in Sec. 17.5. Instead, advanced theory is required to solve individual cases numerically. Once again, these results have been obtained and tabulated for numerous cases.[1] The means (L) also are given graphically in Fig. 17.10 for some cases where $s = 2$.

The Worked Examples section of the CD-ROM includes an example that applies the $M/E_k/s$ model for both $s = 1$ and $s = 2$ to choose the less costly alternative.

[1]Ibid.

■ FIGU
Values
model



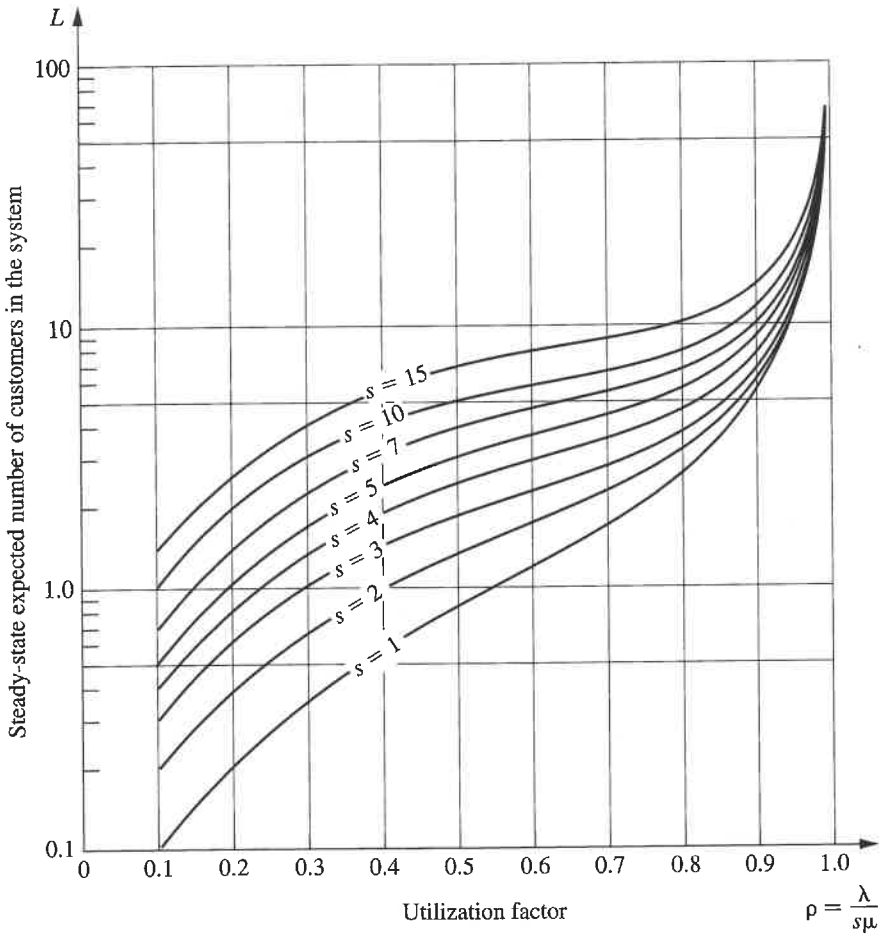
■ **FIGURE 17.10**
Values of L for the $M/E_k/2$ model (Sec. 17.7).

Models without a Poisson Input

All the queueing models presented thus far have assumed a Poisson input process (exponential interarrival times). However, this assumption is violated if the arrivals are scheduled or regulated in some way that prevents them from occurring randomly, in which case another model is needed.

As long as the service times have an exponential distribution with a fixed parameter, three such models are readily available. These models are obtained by merely *reversing* the assumed distributions of the *interarrival* and *service times* in the preceding three models. Thus, the first new model (*GI/M/s*) imposes no restriction on what the *interarrival time* distribution can be. In this case, there are some steady-state results available[1] (particularly in regard to waiting-time distributions) for both the single-server and multiple-server versions of the model, but these results are not nearly as convenient as the simple expressions given for the *M/G/*1 model. The second new model (*D/M/s*) assumes that all interarrival times equal some fixed *constant,* which would represent a queueing system where arrivals are *scheduled* at regular intervals. The third new model ($E_k/M/s$) assumes an *Erlang* interarrival time distribution, which provides a middle ground between

[1]For example, see pp. 248–260 of Selected Reference 5.



■ **FIGURE 17.11**
Values of L for the $D/M/s$ model (Sec. 17.7).

regularly scheduled (constant) and *completely random* (exponential) arrivals. Extensive computational results have been tabulated[1] for these latter two models, including the values of L given graphically in Figs. 17.11 and 17.12.

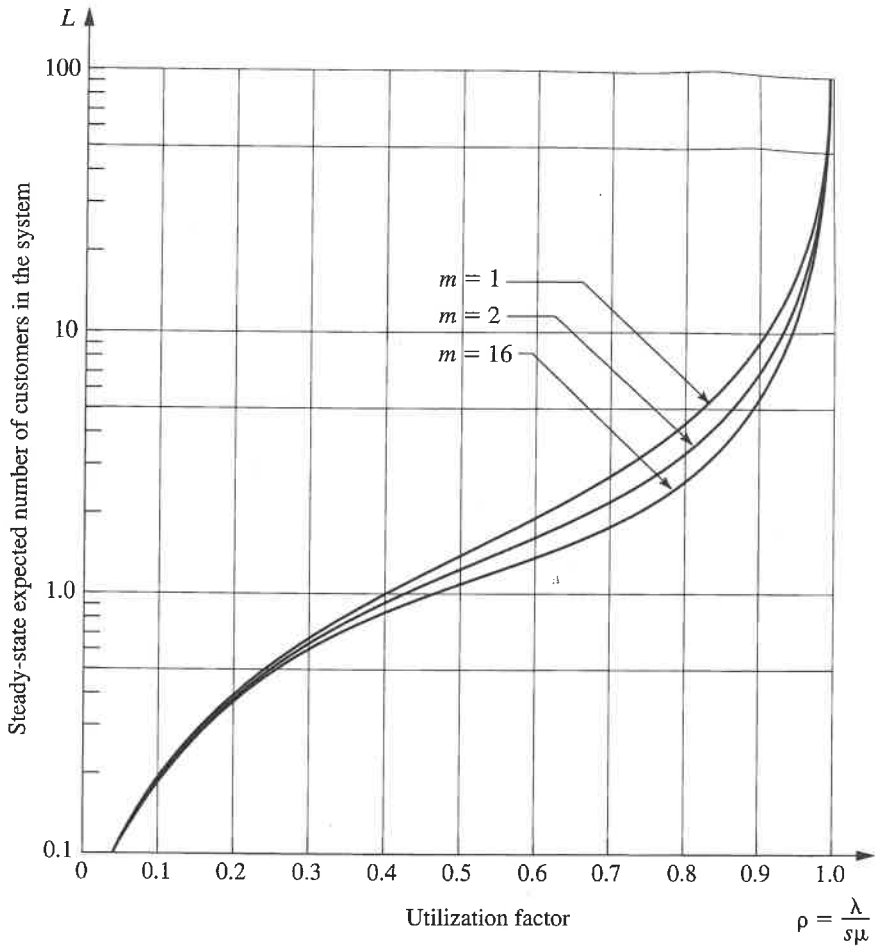
If neither the interarrival times nor the service times for a queueing system have an exponential distribution, then there are three additional queueing models for which computational results also are available.[2] One of these models ($E_m/E_k/s$) assumes an Erlang distribution for both these times. The other two models ($E_k/D/s$ and $D/E_k/s$) assume that one of these times has an Erlang distribution and the other time equals some fixed constant.

Other Models

Although you have seen in this section a large number of queueing models that involve nonexponential distributions, we have far from exhausted the list. For example, another distribution that occasionally is used for either interarrival times or service times is the **hyperexponential distribution.** The key characteristic of this distribution is that even though only nonnegative values are allowed, its standard deviation σ actually is larger than

[1]Hillier and Yu, op. cit.

[2]Ibid.



■ **FIGURE 17.12**
Values of L for the $E_k/M/2$ model (Sec. 17.7).

its mean $1/\mu$. This characteristic is in contrast to the Erlang distribution, where $\sigma < 1/\mu$ in every case except $k = 1$ (exponential distribution), which has $\sigma = 1/\mu$. To illustrate a typical situation where $\sigma > 1/\mu$ can occur, we suppose that the service involved in the queueing system is the repair of some kind of machine or vehicle. If many of the repairs turn out to be routine (small service times) but occasional repairs require an extensive overhaul (very large service times), then the standard deviation of service times will tend to be quite large relative to the mean, in which case the hyperexponential distribution may be used to represent the service-time distribution. Specifically, this distribution would assume that there are fixed probabilities, p and $(1 - p)$, for which kind of repair will occur, that the time required for each kind has an exponential distribution, but that the parameters for these two exponential distributions are different. (In general, the hyperexponential distribution is such a composite of two or more exponential distributions.)

Another family of distributions coming into general use consists of **phase-type distributions** (some of which also are called *generalized Erlangian distributions*). These distributions are obtained by breaking down the total time into a number of phases, each having an exponential distribution, where the parameters of these exponential distributions may be different and the phases may be either in series or in parallel (or both). A group of phases being *in parallel* means that the process randomly selects *one* of the

phases to go through each time according to specified probabilities. This approach is, in fact, how the hyperexponential distribution is derived, so this distribution is a special case of the phase-type distributions. Another special case is the Erlang distribution, which has the restrictions that all its k phases are in series and that these phases have the *same* parameter for their exponential distributions. Removing these restrictions means that phase-type distributions in general can provide considerably more flexibility than the Erlang distribution in fitting the actual distribution of interarrival times or service times observed in a real queueing system. This flexibility is especially valuable when using the actual distribution directly in the model is not analytically tractable, and the ratio of the *mean* to the *standard deviation* for the actual distribution does not closely match the available ratios ($\sqrt{k}$ for $k = 1, 2, \ldots$) for the Erlang distribution.

Since they are built up from combinations of exponential distributions, queueing models using phase-type distributions still can be represented by a *continuous time Markov chain.* This Markov chain generally will have an infinite number of states, so solving for the steady-state distribution of the state of the system requires solving an infinite system of linear equations that has a relatively complicated structure. Solving such a system is far from a routine thing, but recent theoretical advances have enabled us to solve these queueing models numerically in some cases. An extensive tabulation of these results for models with various phase-type distributions (including the hyperexponential distribution) is available.[1]

17.8 PRIORITY-DISCIPLINE QUEUEING MODELS

In priority-discipline queueing models, the queue discipline is based on a *priority system.* Thus, the order in which members of the queue are selected for service is based on their assigned priorities.

Many real queueing systems fit these priority-discipline models much more closely than other available models. Rush jobs are taken ahead of other jobs, and important customers may be given precedence over others. Therefore, the use of priority-discipline models often provides a very welcome refinement over the more usual queueing models.

We present two basic priority-discipline models here. Since both models make the same assumptions, except for the nature of the priorities, we first describe the models together and then summarize their results separately.

The Models

Both models assume that there are N *priority classes* (class 1 has the highest priority and class N has the lowest) and that whenever a server becomes free to begin serving a new customer from the queue, the one customer selected is that member of the *highest* priority class represented in the queue who has waited longest. In other words, customers are selected to begin service in the order of their priority classes, but on a first-come-first-served basis within each priority class. A *Poisson* input process and *exponential* service times are assumed for each priority class. Except for one special case considered later, the models also make the somewhat restrictive assumption that the expected service time is the *same* for all priority classes. However, the models do permit the mean arrival rate to differ among priority classes.

The distinction between the two models is whether the priorities are *nonpreemptive* or *preemptive.* With **nonpreemptive priorities,** a customer being served cannot be ejected back into the queue (preempted) if a higher-priority customer enters the queueing system.

[1] L. P. Seelen, H. C. Tijms, and M. H. Van Hoorn, *Tables for Multi-Server Queues,* North-Holland, Amsterdam, 1985.

Therefore, once a server has begun serving a customer, the service must be completed without interruption. The first model assumes nonpreemptive priorities.

With **preemptive priorities,** the lowest-priority customer being served is *preempted* (ejected back into the queue) whenever a higher-priority customer enters the queueing system. A server is thereby freed to begin serving the new arrival immediately. (When a server does succeed in *finishing* a service, the next customer to begin receiving service is selected just as described at the beginning of this subsection, so a preempted customer normally will get back into service again and, after enough tries, will eventually finish.) Because of the lack-of-memory property of the exponential distribution (see Sec. 17.4), we do not need to worry about defining the point at which service begins when a preempted customer returns to service; the distribution of the *remaining* service time *always* is the same. (For any other service-time distribution, it becomes important to distinguish between *preemptive-resume* systems, where service for a preempted customer resumes at the point of interruption, and *preemptive-repeat* systems, where service must start at the beginning again.) The second model assumes preemptive priorities.

For both models, if the distinction between customers in different priority classes were ignored, Property 6 for the exponential distribution (see Sec. 17.4) implies that *all* customers would arrive according to a Poisson input process. Furthermore, all customers have the *same* exponential distribution for service times. Consequently, the two models actually are identical to the *M/M/s* model studied in Sec. 17.6 *except* for the order in which customers are served. Therefore, when we count just the *total* number of customers in the system, the steady-state distribution for the *M/M/s* model also applies to both models. Consequently, the formulas for L and L_q also carry over, as do the expected waiting-time results (by Little's formula) W and W_q, for a randomly selected customer. What changes is the *distribution* of waiting times, which was derived in Sec. 17.6 under the assumption of a first-come-first-served queue discipline. With a priority discipline, this distribution has a much larger *variance,* because the waiting times of customers in the highest priority classes tend to be much smaller than those under a first-come-first-served discipline, whereas the waiting times in the lowest priority classes tend to be much larger. By the same token, the breakdown of the total number of customers in the system tends to be disproportionately weighted toward the lower-priority classes. But this condition is just the reason for imposing priorities on the queueing system in the first place. We want to *improve* the *measures of performance* for each of the higher-priority classes at the expense of performance for the lower-priority classes. To determine how much improvement is being made, we need to obtain such measures as *expected waiting time in the system* and *expected number of customers in the system* for the individual priority classes. Expressions for these measures are given next for the two models in turn.

Results for the Nonpreemptive Priorities Model

Let W_k be the steady-state expected waiting time in the system (including service time) for a member of priority class k. Then

$$W_k = \frac{1}{AB_{k-1}B_k} + \frac{1}{\mu}, \qquad \text{for } k = 1, 2, \ldots, N,$$

where

$$A = s!\frac{s\mu - \lambda}{r^s}\sum_{j=0}^{s-1}\frac{r^j}{j!} + s\mu,$$

$$B_0 = 1,$$

$$B_k = 1 - \frac{\sum_{i=1}^{k}\lambda_i}{s\mu},$$

s = number of servers,

μ = mean service rate per busy server,

λ_i = mean arrival rate for priority class i,

$$\lambda = \sum_{i=1}^{N} \lambda_i,$$

$$r = \frac{\lambda}{\mu}.$$

(This result assumes that

$$\sum_{i=1}^{k} \lambda_i < s\mu,$$

so that priority class k can reach a steady-state condition.) *Little's formula* still applies to individual priority classes, so L_k, the steady-state expected number of members of priority class k in the queueing system (including those being served), is

$$L_k = \lambda_k W_k, \qquad \text{for } k = 1, 2, \ldots, N.$$

To determine the expected waiting time in the queue (excluding service time) for priority class k, merely subtract $1/\mu$ from W_k; the corresponding expected queue length is again obtained by multiplying by λ_k. For the special case where $s = 1$, the expression for A reduces to $A = \mu^2/\lambda$.

An Excel template is provided in your OR Courseware for performing the above calculations.

The Worked Examples section of the CD-ROM provides an example that illustrates the application of the nonpreemptive priorities model for determining how many turret lathes a factory should have when the jobs fall into three priority classes.

A Single-Server Variation of the Nonpreemptive Priorities Model

The above assumption that the expected service time $1/\mu$ is the same for all priority classes is a fairly restrictive one. In practice, this assumption sometimes is violated because of differences in the service requirements for the different priority classes.

Fortunately, for the special case of a single server, it is possible to allow different expected service times and still obtain useful results. Let $1/\mu_k$ denote the mean of the exponential service-time distribution for priority class k, so

μ_k = mean service rate for priority class k, for $k = 1, 2, \ldots, N$.

Then the steady-state expected waiting time in the system for a member of priority class k is

$$W_k = \frac{a_k}{b_{k-1}b_k} + \frac{1}{\mu_k}, \qquad \text{for } k = 1, 2, \ldots, N,$$

where

$$a_k = \sum_{i=1}^{k} \frac{\lambda_i}{\mu_i^2},$$

$$b_0 = 1,$$

$$b_k = 1 - \sum_{i=1}^{k} \frac{\lambda_i}{\mu_i}.$$

This result holds as long as

$$\sum_{i=1}^{k} \frac{\lambda_i}{\mu_i} < 1,$$

which enables priority class k to reach a steady-state condition. Little's formula can be used as described above to obtain the other main measures of performance for each priority class.

Results for the Preemptive Priorities Model

For the preemptive priorities model, we need to reinstate the assumption that the expected service time is the same for all priority classes. Using the same notation as for the original nonpreemptive priorities model, having the preemption changes the *total* expected waiting time in the system (including the total service time) to

$$W_k = \frac{1/\mu}{B_{k-1}B_k}, \qquad \text{for } k = 1, 2, \ldots, N,$$

for the *single-server* case ($s = 1$). When $s > 1$, W_k can be calculated by an iterative procedure that will be illustrated soon by the County Hospital example. The L_k continue to satisfy the relationship

$$L_k = \lambda_k W_k, \qquad \text{for } k = 1, 2, \ldots, N.$$

The corresponding results for the queue (excluding customers in service) also can be obtained from W_k and L_k as just described for the case of nonpreemptive priorities. Because of the lack-of-memory property of the exponential distribution (see Sec. 17.4), preemptions do not affect the service process (occurrence of service completions) in any way. The expected total service time for any customer still is $1/\mu$.

This chapter's Excel file includes an Excel template for calculating the above measures of performance for the single-server case.

The County Hospital Example with Priorities

For the County Hospital emergency room problem, the management engineer has noticed that the patients are not treated on a first-come-first-served basis. Rather, the admitting nurse seems to divide the patients into roughly three categories: (1) *critical* cases, where prompt treatment is vital for survival; (2) *serious* cases, where early treatment is important to prevent further deterioration; and (3) *stable* cases, where treatment can be delayed without adverse medical consequences. Patients are then treated in this order of priority, where those in the same category are normally taken on a first-come-first-served basis. A doctor will interrupt treatment of a patient if a new case in a higher-priority category arrives. Approximately 10 percent of the patients fall into the first category, 30 percent into the second, and 60 percent into the third. Because the more serious cases will be sent to the hospital for further care after receiving emergency treatment, the average treatment time by a doctor in the emergency room actually does not differ greatly among these categories.

The management engineer has decided to use a priority-discipline queueing model as a reasonable representation of this queueing system, where the three categories of patients constitute the three priority classes in the model. Because treatment is interrupted by the arrival of a higher-priority case, the *preemptive priorities model* is the appropriate one. Given the previously available data ($\mu = 3$ and $\lambda = 2$), the preceding percentages yield $\lambda_1 = 0.2$, $\lambda_2 = 0.6$, and $\lambda_3 = 1.2$. Table 17.3 gives the resulting expected waiting times in the queue (so *excluding* treatment time) for the respective priority classes[1] when there is one ($s = 1$) or two ($s = 2$) doctors on duty. (The corresponding results for the nonpreemptive priorities model also are given in Table 17.3 to show the effect of preempting.)

Deriving the Preemptive Priority Results. These preemptive priority results for $s = 2$ were obtained as follows. Because the waiting times for priority class 1 customers are completely unaffected by the presence of customers in the lower-priority classes, W_1

[1]Note that these expected times can no longer be interpreted as the expected time before treatment begins when $k > 1$, because treatment may be interrupted at least once, causing additional waiting time before service is completed.

TABLE 17.3 Steady-state results from the priority-discipline models for the County Hospital problem

	Preemptive Priorities		Nonpreemptive Priorities	
	$s = 1$	$s = 2$	$s = 1$	$s = 2$
A	—	—	4.5	36
B_1	0.933	—	0.933	0.967
B_2	0.733	—	0.733	0.867
B_3	0.333	—	0.333	0.667
$W_1 - \frac{1}{\mu}$	0.024 hour	0.00037 hour	0.238 hour	0.029 hour
$W_2 - \frac{1}{\mu}$	0.154 hour	0.00793 hour	0.325 hour	0.033 hour
$W_3 - \frac{1}{\mu}$	1.033 hours	0.06542 hour	0.889 hour	0.048 hour

will be the same for any other values of λ_2 and λ_3, including $\lambda_2 = 0$ and $\lambda_3 = 0$. Therefore, W_1 must equal W for the corresponding *one-class* model (the *M/M/s* model in Sec. 17.6) with $s = 2$, $\mu = 3$, and $\lambda = \lambda_1 = 0.2$, which yields

$$W_1 = W = 0.33370 \text{ hour}, \qquad \text{for } \lambda = 0.2$$

so

$$W_1 - \frac{1}{\mu} = 0.33370 - 0.33333 = 0.00037 \text{ hour}.$$

Now consider the first two priority classes. Again note that customers in these classes are completely unaffected by lower-priority classes (just priority class 3 in this case), which can therefore be ignored in the analysis. Let $\overline{W}_{1-2}$ be the expected waiting time in the system (so including service time) of a *random arrival* in *either* of these two classes, so the probability is $\lambda_1/(\lambda_1 + \lambda_2) = \frac{1}{4}$ that this arrival is in class 1 and $\lambda_2/(\lambda_1 + \lambda_2) = \frac{3}{4}$ that it is in class 2. Therefore,

$$\overline{W}_{1-2} = \frac{1}{4}W_1 + \frac{3}{4}W_2.$$

Furthermore, because the *expected* waiting time is the same for *any* queue discipline, $\overline{W}_{1-2}$ must also equal W for the *M/M/s* model in Sec. 17.6, with $s = 2$, $\mu = 3$, and $\lambda = \lambda_1 + \lambda_2 = 0.8$, which yields

$$\overline{W}_{1-2} = W = 0.33937 \text{ hour}, \qquad \text{for } \lambda = 0.8.$$

Combining these facts gives

$$W_2 = \frac{4}{3}\left[0.33937 - \frac{1}{4}(0.33370)\right] = 0.34126 \text{ hour}.$$

$$\left(W_2 - \frac{1}{\mu} = 0.00793 \text{ hour}.\right)$$

Finally, let $\overline{W}_{1-3}$ be the expected waiting time in the system (so including service time) for a *random arrival* in *any* of the three priority classes, so the probabilities are 0.1, 0.3, and 0.6 that it is in classes 1, 2, and 3, respectively. Therefore,

$$\overline{W}_{1-3} = 0.1W_1 + 0.3W_2 + 0.6W_3.$$

Furthermore, $\overline{W}_{1-3}$ must also equal W for the *M/M/s* model in Sec. 17.6, with $s = 2$, $\mu = 3$, and $\lambda = \lambda_1 + \lambda_2 + \lambda_3 = 2$, so that (from Table 17.2)

$$\overline{W}_{1-3} = W = 0.375 \text{ hour}, \qquad \text{for } \lambda = 2.$$

Consequently,

$$W_3 = \frac{1}{0.6}[0.375 - 0.1(0.33370) - 0.3(0.34126)]$$

$$= 0.39875 \text{ hour.}$$

$$\left(W_3 - \frac{1}{\mu} = 0.06542 \text{ hour.}\right)$$

The corresponding W_q results for the *M/M/s* model in Sec. 17.6 also could have been used in exactly the same way to derive the $W_k - 1/\mu$ quantities directly.

Conclusions. When $s = 1$, the $W_k - 1/\mu$ values in Table 17.3 for the preemptive priorities case indicate that providing just a single doctor would cause critical cases to wait about $1\frac{1}{2}$ minutes (0.024 hour) on the average, serious cases to wait more than 9 minutes, and stable cases to wait more than 1 hour. (Contrast these results with the average wait of $W_q = \frac{2}{3}$ hour for all patients that was obtained in Table 17.2 under the first-come-first-served queue discipline.) However, these values represent *statistical expectations,* so some patients have to wait considerably longer than the average for their priority class. This wait would not be tolerable for the critical and serious cases, where a few minutes can be vital. By contrast, the $s = 2$ results in Table 17.3 (preemptive priorities case) indicate that adding a second doctor would virtually eliminate waiting for all but the stable cases. Therefore, the management engineer recommended that there be two doctors on duty in the emergency room during the early evening hours next year. The board of directors for County Hospital adopted this recommendation and simultaneously raised the charge for using the emergency room!

17.9 QUEUEING NETWORKS

Thus far we have considered only queueing systems that have a *single* service facility with one or more servers. However, queueing systems encountered in OR studies are sometimes actually *queueing networks,* i.e., networks of service facilities where customers must receive service at some of or all these facilities. For example, orders being processed through a job shop must be routed through a sequence of machine groups (service facilities). It is therefore necessary to study the entire network to obtain such information as the expected total waiting time, expected number of customers in the entire system, and so forth.

Because of the importance of queueing networks, research into this area has been very active. However, this is a difficult area, so we limit ourselves to a brief introduction.

One result is of such fundamental importance for queueing networks that this finding and its implications warrant special attention here. This fundamental result is the following *equivalence property* for the *input process* of arriving customers and the *output process* of departing customers for certain queueing systems.

> **Equivalence property:** Assume that a service facility with s servers and an infinite queue has a Poisson input with parameter λ and the same exponential service-time distribution with parameter μ for each server (the *M/M/s* model), where $s\mu > \lambda$. Then the steady-state *output* of this service facility is also a Poisson process[1] with parameter λ.

[1]For a proof, see P. J. Burke, "The Output of a Queueing System," *Operations Research,* 4(6): 699–704, 1956.

Notice that this property makes no assumption about the type of queue discipline used. Whether it is first-come-first-served, random, or even a priority discipline as in Sec. 17.8, the served customers will leave the service facility according to a Poisson process. The crucial implication of this fact for queueing networks is that if these customers must then go to another service facility for further service, this second facility *also* will have a Poisson input. With an exponential service-time distribution, the equivalence property will hold for this facility as well, which can then provide a Poisson input for a third facility, etc. We discuss the consequences for two basic kinds of networks next.

Infinite Queues in Series

Suppose that customers must all receive service at a *series* of m service facilities in a fixed sequence. Assume that each facility has an infinite queue (no limitation on the number of customers allowed in the queue), so that the series of facilities form a system of *infinite queues in series*. Assume further that the customers arrive at the first facility according to a Poisson process with parameter λ and that each facility i ($i = 1, 2, \ldots, m$) has an exponential service-time distribution with parameter μ_i for its s_i servers, where $s_i\mu_i > \lambda$. It then follows from the equivalence property that (under steady-state conditions) each service facility has a Poisson input with parameter λ. Therefore, the elementary *M/M/s* model of Sec. 17.6 (or its priority-discipline counterparts in Sec. 17.8) can be used to analyze each service facility independently of the others!

Being able to use the *M/M/s* model to obtain all measures of performance for each facility independently, rather than analyzing interactions between facilities, is a tremendous simplification. For example, the probability of having n customers at a given facility is given by the formula for P_n in Sec. 17.6 for the *M/M/s* model. The *joint probability* of n_1 customers at facility 1, n_2 customers at facility 2, . . . , then, is the *product* of the individual probabilities obtained in this simple way. In particular, this joint probability can be expressed as

$$P\{(N_1, N_2, \ldots, N_m) = (n_1, n_2, \ldots, n_m)\} = P_{n_1}P_{n_2}\cdots P_{n_m}.$$

(This simple form for the solution is called the **product form solution.**) Similarly, the expected total waiting time and the expected number of customers in the entire system can be obtained by merely summing the corresponding quantities obtained at the respective facilities.

Unfortunately, the equivalence property and its implications do not hold for the case of *finite* queues discussed in Sec. 17.6. This case is actually quite important in practice, because there is often a definite limitation on the queue length in front of service facilities in networks. For example, only a small amount of buffer storage space is typically provided in front of each facility (station) in a production-line system. For such systems of finite queues in series, no simple product form solution is available. The facilities must be analyzed jointly instead, and only limited results have been obtained.

Jackson Networks

Systems of infinite queues in series are not the only queueing networks where the *M/M/s* model can be used to analyze each service facility independently of the others. Another prominent kind of network with this property (a product form solution) is the *Jackson network*, named after the individual who first characterized the network and showed that this property holds.[1]

[1]See J. R. Jackson, "Jobshop-Like Queueing Systems," *Management Science*, **10**(1): 131–142, 1963.

The characteristics of a Jackson network are the same as assumed above for the system of infinite queues in series, except now the customers visit the facilities in different orders (and may not visit them all). For each facility, its arriving customers come from *both* outside the system (according to a Poisson process) and the other facilities. These characteristics are summarized below.

A **Jackson network** is a system of m service facilities where facility i ($i = 1, 2, \ldots, m$) has

1. An infinite queue
2. Customers arriving from outside the system according to a Poisson input process with parameter a_i
3. s_i servers with an exponential service-time distribution with parameter μ_i.

A customer leaving facility i is routed next to facility j ($j = 1, 2, \ldots, m$) with probability p_{ij} or departs the system with probability

$$q_i = 1 - \sum_{j=1}^{m} p_{ij}.$$

Any such network has the following key property.

Under steady-state conditions, each facility j ($j = 1, 2, \ldots, m$) in a Jackson network behaves as if it were an *independent M/M/s* queueing system with arrival rate

$$\lambda_j = a_j + \sum_{i=1}^{m} \lambda_i p_{ij},$$

where $s_j\mu_j > \lambda_j$.

This key property cannot be *proved* directly from the equivalence property this time (the reasoning would become circular), but its *intuitive underpinning* is still provided by the latter property. The intuitive viewpoint (not quite technically correct) is that, for each facility i, its input processes from the various sources (outside and other facilities) are *independent Poisson processes,* so the *aggregate* input process is Poisson with parameter λ_i (Property 6 in Sec. 17.4). The equivalence property then says that the *aggregate output* process for facility i must be Poisson with parameter λ_i. By disaggregating this output process (Property 6 again), the process for customers going from facility i to facility j must be Poisson with parameter $\lambda_i p_{ij}$. This process becomes one of the Poisson *input* processes for facility j, thereby helping to maintain the series of Poisson processes in the overall system.

The equation given for obtaining λ_j is based on the fact that λ_i is the *departure rate* as well as the arrival rate for all customers using facility i. Because p_{ij} is the proportion of customers departing from facility i who go next to facility j, the rate at which customers from facility i arrive at facility j is $\lambda_i p_{ij}$. Summing this product over all i, and then adding this sum to a_j, gives the *total arrival rate* to facility j from all sources.

To calculate λ_j from this equation requires knowing the λ_i for $i \neq j$, but these λ_i also are unknowns given by the corresponding equations. Therefore, the procedure is to solve *simultaneously* for $\lambda_1, \lambda_2, \ldots, \lambda_m$ by obtaining the simultaneous solution of the entire system of linear equations for λ_j for $j = 1, 2, \ldots, m$. Your IOR Tutorial includes an interactive procedure for solving for the λ_j in this way.

To illustrate these calculations, consider a Jackson network with three service facilities that have the parameters shown in Table 17.4. Plugging into the formula for λ_j for $j = 1, 2, 3$, we obtain

$$\begin{aligned} \lambda_1 &= 1 + 0.1\lambda_2 + 0.4\lambda_3 \\ \lambda_2 &= 4 + 0.6\lambda_1 + 0.4\lambda_3 \\ \lambda_3 &= 3 + 0.3\lambda_1 + 0.3\lambda_2. \end{aligned}$$

TABLE 17.4 Data for the example of a Jackson network

Facility j	s_j	μ_j	a_j	p_{ij}		
				$i = 1$	$i = 2$	$i = 3$
$j = 1$	1	10	1	0	0.1	0.4
$j = 2$	2	10	4	0.6	0	0.4
$j = 3$	1	10	3	0.3	0.3	0

(Reason through each equation to see why it gives the total arrival rate to the corresponding facility.) The simultaneous solution for this system is

$$\lambda_1 = 5, \qquad \lambda_2 = 10, \qquad \lambda_3 = 7\frac{1}{2}.$$

Given this simultaneous solution, each of the three service facilities now can be analyzed *independently* by using the formulas for the *M/M/s* model given in Sec. 17.6. For example, to obtain the distribution of the number of customers $N_i = n_i$ at facility i, note that

$$\rho_i = \frac{\lambda_i}{s_i \mu_i} = \begin{cases} \frac{1}{2} & \text{for } i = 1 \\ \frac{1}{2} & \text{for } i = 2 \\ \frac{3}{4} & \text{for } i = 3. \end{cases}$$

Plugging these values (and the parameters in Table 17.4) into the formula for P_n gives

$$P_{n_1} = \frac{1}{2}\left(\frac{1}{2}\right)^{n_1} \qquad \text{for facility 1,}$$

$$P_{n_2} = \begin{cases} \frac{1}{3} & \text{for } n_2 = 0 \\ \frac{1}{3} & \text{for } n_2 = 1 \\ \frac{1}{3}\left(\frac{1}{2}\right)^{n_2 - 1} & \text{for } n_2 \geq 2 \end{cases} \qquad \text{for facility 2,}$$

$$P_{n_3} = \frac{1}{4}\left(\frac{3}{4}\right)^{n_3} \qquad \text{for facility 3.}$$

The *joint probability* of (n_1, n_2, n_3) then is given simply by the product form solution

$$P\{(N_1, N_2, N_3) = (n_1, n_2, n_3)\} = P_{n_1} P_{n_2} P_{n_3}.$$

In a similar manner, the expected number of customers L_i at facility i can be calculated from Sec. 17.6 as

$$L_1 = 1, \qquad L_2 = \frac{4}{3}, \qquad L_3 = 3.$$

The expected *total* number of customers in the entire system then is

$$L = L_1 + L_2 + L_3 = 5\frac{1}{3}.$$

Obtaining W, the expected *total* waiting time in the system (including service times) for a customer, is a little trickier. You cannot simply add the expected waiting times at the respective facilities, because a customer does not necessarily visit each facility exactly

once. However, Little's formula can still be used, where the system arrival rate λ is the sum of the arrival rates *from outside* to the facilities, $\lambda = a_1 + a_2 + a_3 = 8$. Thus,

$$W = \frac{L}{a_1 + a_2 + a_3} = \frac{2}{3}.$$

In conclusion, we should point out that there do exist other (more complicated) kinds of queueing networks where the individual service facilities can be analyzed independently from the others. In fact, finding queueing networks with a product form solution has been the Holy Grail for research on queueing networks. Some sources of additional information are Selected References 1, 2, 12, and 14.

17.10 THE APPLICATION OF QUEUEING THEORY

Because of the wealth of information provided by queueing theory, it is widely used to guide the design (or redesign) of queueing systems. We now turn our focus to how queueing theory is applied in this way.

The most common decision that needs to be made when designing a queueing system is how many servers to provide. However, a number of other decisions also may be needed. The possible decisions include

1. Number of servers at a service facility.
2. Efficiency of the servers.
3. Number of service facilities.
4. Amount of waiting space in the queue.
5. Any priorities for different categories of customers.

The two primary considerations in making these kinds of decisions typically are (1) the cost of the service capacity provided by the queueing system and (2) the consequences of making the customers wait in the queueing system. Providing too much service capacity causes excessive costs. Providing too little causes excessive waiting. Therefore, the goal is to find an appropriate trade-off between the service cost and the amount of waiting.

Two basic approaches are available for seeking this trade-off. One is to establish one or more criteria for a satisfactory level of service in terms of how much waiting would be acceptable. For example, one possible criterion might be that the expected waiting time in the system should not exceed a certain number of minutes. Another might be that at least 95 percent of the customers should wait no longer than a certain number of minutes in the system. Similar criteria in terms of the expected number of customers in the system (or the probability distribution of this number) also could be used. The criteria also might be stated in terms of the waiting time or the number of customers in the *queue* instead of in the system. Once the criterion or criteria have been selected, it then is usually straightforward to use trial and error to find the least costly design of the queueing system that satisfies all the criteria.

The other basic approach for seeking the best trade-off involves assessing the costs associated with the consequences of making customers wait. For example, suppose that the queueing system is an *internal service system* (as described in Sec. 17.3), where the customers are the employees of a for-profit company. Making these employees wait at the queueing system causes *lost productivity,* which results in *lost profit.* This lost profit is the **waiting cost** associated with the queueing system. By expressing this waiting cost as a function of the amount of waiting, the problem of determining the best design of the queueing system can now be posed as minimizing the expected *total cost* (service cost plus waiting cost) per unit time.

We spell out this latter approach below for the problem of determining the optimal number of servers to provide.

How Many Servers Should Be Provided?

To formulate the objective function when the decision variable is the number of servers s, let

$E(\text{TC})$ = expected total cost per unit time,

$E(\text{SC})$ = expected service cost per unit time,

$E(\text{WC})$ = expected waiting cost per unit time.

Then the objective is to choose the number of servers so as to

$$\text{Minimize } E(\text{TC}) = E(\text{SC}) + E(\text{WC}).$$

When each server costs the same, the **service cost** is

$$E(\text{SC}) = C_s s,$$

where C_s is the marginal cost of a server per unit time. To evaluate WC for any value of s, note that $L = \lambda W$ gives the expected total amount of waiting in the queueing system per unit time. Therefore, when the waiting cost is proportional to the amount of waiting, this cost can be expressed as

$$E(\text{WC}) = C_w L,$$

where C_w is the waiting cost per unit time for each customer in the queueing system. Therefore, after estimating the constants, C_s and C_w, the goal is to choose the value of s so as to

$$\text{Minimize } E(\text{TC}) = C_s s + C_w L.$$

By choosing the queueing model that fits the queueing system, the value of L can be obtained for various values of s. Increasing s decreases L, at first rapidly and then gradually more slowly.

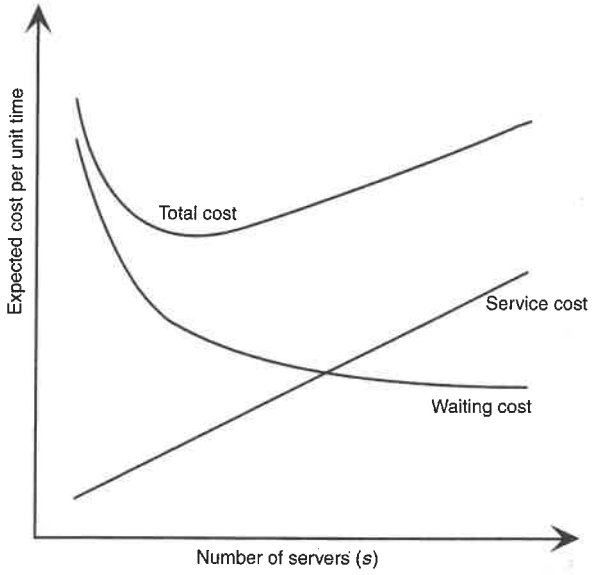
Figure 17.13 shows the general shape of the $E(\text{SC})$, $E(\text{WC})$, and $E(\text{TC})$ curves versus the number of servers s. (For better conceptualization, we have drawn these as smooth curves even though the only feasible values of s are $s = 1, 2, \ldots$.) By calculating $E(\text{TC})$ for consecutive values of s until $E(\text{TC})$ stops decreasing and starts increasing instead, it is straightforward to find the number of servers that minimizes total cost. The following example illustrates this process.

An Example

The Acme Machine Shop has a tool crib to store tools required by the shop mechanics. Two clerks run the tool crib. The clerks hand out the tools as the mechanics arrive and request them. The tools then are returned to the clerks when they are no longer needed. There have been complaints from supervisors that their mechanics have had to waste too much time waiting to be served at the tool crib, so it appears as if there should be *more* clerks. On the other hand, management is exerting pressure to reduce overhead in the plant, and this reduction would lead to *fewer* clerks. To resolve these conflicting pressures, an OR study is being conducted to determine just how many clerks the tool crib should have.

The tool crib constitutes a queueing system, with the clerks as its servers and the mechanics as its customers. After gathering some data on interarrival times and service times, the OR team has concluded that the queueing model that fits this queueing system best is

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■ **FIGURE 17.13**
The shape of the expected cost curves for determining the number of servers to provide.

the *M/M/s* model. The estimates of the mean arrival rate λ and the mean service rate (per server) μ are

$\lambda = 120$ customers per hour,
$\mu = 80$ customers per hour,

so the utilization factor for the two clerks is

$$\rho = \frac{\lambda}{s\mu} = \frac{120}{2(80)} = 0.75.$$

The total cost to the company of each tool crib clerk is about \$20 per hour, so $C_s = \$20$. While a mechanic is busy, the value to the company of his or her output averages about \$48 per hour, so $C_w = \$48$. Therefore, the OR team now needs to find the number of servers (tool crib clerks) s that will

Minimize $E(\text{TC}) = \$20\ s + \$48\ L$.

An Excel template has been provided in your OR Courseware for calculating these costs with the *M/M/s* model. All you need to do is enter the data for the model along with the unit service cost C_s, the unit waiting cost C_w, and the number of servers s you want to try. The template then calculates $E(\text{SC})$, $E(\text{WC})$, and $E(\text{TC})$. This is illustrated in Fig. 17.14 with $s = 3$ for this example. By repeatedly entering alternative values of s, the template then can reveal which value minimizes $E(\text{TC})$ in a matter of seconds.

Table 17.5 shows the data that would be generated from this template by repeating these calculations for $s = 1, 2, 3, 4$, and 5. Since the utilization factor for $s = 1$ is $\rho = 1.5$, a single clerk would be unable to keep up with the customers, so this option is ruled out. All larger values of s are feasible, but $s = 3$ has the smallest expected total cost. Furthermore, $s = 3$ would decrease the current expected total cost for $s = 2$ by \$61 per hour. Therefore, despite management's current drive to reduce overhead (which includes the cost of tool crib clerks), the OR team recommends that a third clerk be added to the tool crib. Note that this recommendation would decrease the utilization factor for the clerks from an already modest 0.75 all the way down to 0.5. However, because of the large improvement in the productivity of the mechanics (who are much more expensive than the clerks) through decreasing their time wasted waiting at the tool crib, management adopts the recommendation.

	A	B	C	D	E	F	G
1	Economic Analysis of Acme Machine Shop Example						
2							
3			Data				Results
4		λ =	120	(mean arrival rate)		L =	1.736842105
5		μ =	80	(mean service rate)		L_q =	0.236842105
6		s =	3	(# servers)			
7						W =	0.014473684
8		Pr(W > t) =	0.02581732			W_q =	0.001973684
9		when t =	0.05				
10						ρ =	0.5
11		Prob(W_q > t) =	0.00058707				
12		when t =	0.05			n	P_n
13						0	0.210526316
14		Economic Analysis:				1	0.315789474
15		C_s =	$20.00	(cost / server / unit time)		2	0.236842105
16		C_w =	$48.00	(waiting cost / unit time)		3	0.118421053
17						4	0.059210526
18		Cost of Service	$60.00			5	0.029605263
19		Cost of Waiting	$83.37			6	0.014802632
20		Total Cost	$143.37			7	0.007401316

	B	C
18	Cost of Service	=C_s*s
19	Cost of Waiting	=C_w*L
20	Total Cost	=CostOfService+CostOfWaiting

Range Name	Cell
CostOfService	C18
CostOfWaiting	C19
C_s*s	C15
C_w*L	C16
L	G4
s	C6
TotalCost	C20

FIGURE 17.14
This Excel template for using economic analysis to choose the number of servers with the *M/M/s* model is applied here to the Acme Machine Shop example with $s = 3$.

TABLE 17.5 Calculation of $E(TC)$ for alternative s in the Acme Machine Shop example

s	ρ	L	$E(SC) = C_s s$	$E(WC) = C_w L$	$E(TC) = E(SC) + E(WC)$
1	1.50	∞	$20	∞	∞
2	0.75	3.43	$40	$164.57	$204.57
3	0.50	1.74	$60	$83.37	$143.37
4	0.375	1.54	$80	$74.15	$154.15
5	0.30	1.51	$100	$72.41	$172.41

Other Issues

Chapter 26 on the CD-ROM expands considerably further on the application of queueing theory, including how to deal with some other issues not considered above.

For example, the analysis on p. 816 assumed that the waiting cost is proportional to the amount of waiting, but this sometimes is not the case. If a company has one or two of its employees in a queueing system, this may not be very serious in terms of their lost productivity because others may be able to handle all of the available productive work. However, having additional employees in the queueing system may result in a sharp increase in lost productivity and the resulting lost profit, so the waiting cost becomes a nonlinear function of the number in the system. Similarly, the consequences to a commercial service system for making its customers wait may be minimal for short waits but much more serious for long waits. In this case, the waiting cost becomes a nonlinear function of the waiting time. Section 26.3 describes the formulation of nonlinear waiting-cost functions and then the calculation of $E(\text{WC})$ with such functions.

Section 26.4 discusses a decision model where the decision variables are *both* the number of servers and the mean service rate for the servers. An interesting issue that arises here is whether it is better have *one fast server* (several people working together to serve each customer rapidly) or *several slow servers* (several people working separately to serve different customers).

Section 26.4 also presents a decision model where the decision variables are the number of service facilities and the number of servers per facility to provide service to a calling population of potential customers. Given the mean arrival rate for the entire calling population, increasing the number of facilities enables decreasing the mean arrival rate (workload) at each facility. The number of service facilities also affects how much time each customer will need to spend in traveling to and from the nearest facility. The waiting cost now needs to be a function of the total time lost by a customer by either waiting at a service facility or traveling to and from the facility. Therefore, Sec. 26.5 presents some travel-time models for determining the expected round-trip travel time for each customer.

17.11 CONCLUSIONS

Queueing systems are prevalent throughout society. The adequacy of these systems can have an important effect on the quality of life and productivity.

Queueing theory studies queueing systems by formulating mathematical models of their operation and then using these models to derive measures of performance. This analysis provides vital information for effectively designing queueing systems that achieve an appropriate balance between the cost of providing a service and the cost associated with waiting for that service.

This chapter presented the most basic models of queueing theory for which particularly useful results are available. However, many other interesting models could be considered if space permitted. In fact, several thousand research papers formulating and/or analyzing queueing models have already appeared in the technical literature, and many more are being published each year!

The *exponential distribution* plays a fundamental role in queueing theory for representing the distribution of interarrival and service times, because this assumption enables us to represent the queueing system as a *continuous time Markov chain.* For the same reason, *phase-type distributions* such as the *Erlang distribution,* where the total time is broken down into individual phases having an exponential distribution, are very useful. Useful analytical results have been obtained for only a relatively few queueing models making other assumptions.

Priority-discipline queueing models are useful for the common situation where some categories of customers are given priority over others for receiving service.

In another common situation, customers must receive service at several different service facilities. Models for queueing networks are gaining widespread use for such situations. This is an area of especially active ongoing research.

When no tractable model that provides a reasonable representation of the queueing system under study is available, a common approach is to obtain relevant performance data by developing a computer program for simulating the operation of the system. This technique is discussed in Chap. 20.

Section 17.10 briefly describes how queueing theory can be used to help design effective queueing systems and then Chap. 26 (on the CD-ROM) expands considerably further on this subject.

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LEARNING AIDS FOR THIS CHAPTER ON CD-ROM

Worked Examples:

Examples for Chapter 17

An Interactive Procedure in IOR Tutorial:

Jackson Network

"Ch. 17—Queueing Theory" Excel Files:

Template for *M/M/s* Model
Template for Finite Queue Variation of *M/M/s* Model
Template for Finite Calling Population Variation of *M/M/s* Model
Template for *M/G/*1 Model
Template for *M/D/*1 Model
Template for $M/E_k/1$ Model
Template for Nonpreemptive Priorities Model
Template for Preemptive Priorities Model
Template for *M/M/s* Economic Analysis of Number of Servers

"Ch. 17—Queueing Theory" LINGO File for Selected Examples

Glossary for Chapter 17

See Appendix 1 for documentation of the software.

PROBLEMS[1]

To the left of each of the following problems (or their parts), we have inserted a T whenever one of the templates listed above can be helpful. An asterisk on the problem number indicates that at least a partial answer is given in the back of the book.

17.2-1.* Consider a typical barber shop. Demonstrate that it is a queueing system by describing its components.

17.2-2.* Newell and Jeff are the two barbers in a barber shop they own and operate. They provide two chairs for customers who are waiting to begin a haircut, so the number of customers in the shop varies between 0 and 4. For $n = 0, 1, 2, 3, 4$, the probability P_n that exactly n customers are in the shop is $P_0 = \frac{1}{16}$, $P_1 = \frac{4}{16}$, $P_2 = \frac{6}{16}$, $P_3 = \frac{4}{16}$, $P_4 = \frac{1}{16}$.

(a) Calculate L. How would you describe the meaning of L to Newell and Jeff?

(b) For each of the possible values of the number of customers in the queueing system, specify how many customers are in the queue. Then calculate L_q. How would you describe the meaning of L_q to Newell and Jeff?

(c) Determine the expected number of customers being served.

(d) Given that an average of 4 customers per hour arrive and stay to receive a haircut, determine W and W_q. Describe these two quantities in terms meaningful to Newell and Jeff.

(e) Given that Newell and Jeff are equally fast in giving haircuts, what is the average duration of a haircut?

17.2-3. Mom-and-Pop's Grocery Store has a small adjacent parking lot with three parking spaces reserved for the store's customers. During store hours, cars enter the lot and use one of the spaces at a mean rate of 2 per hour. For $n = 0, 1, 2, 3$, the probability P_n that exactly n spaces currently are being used is $P_0 = 0.2$, $P_1 = 0.3$, $P_2 = 0.3$, $P_3 = 0.2$.

(a) Describe how this parking lot can be interpreted as being a queueing system. In particular, identify the customers and the servers. What is the service being provided? What constitutes a service time? What is the queue capacity?

(b) Determine the basic measures of performance—L, L_q, W, and W_q—for this queueing system.

(c) Use the results from part (*b*) to determine the average length of time that a car remains in a parking space.

17.2-4. For each of the following statements about the queue in a queueing system, label the statement as true or false and then justify your answer by referring to a specific statement in the chapter.

(a) The queue is where customers wait in the queueing system until their service is completed.

(b) Queueing models conventionally assume that the queue can hold only a limited number of customers.

(c) The most common queue discipline is first-come-first-served.

17.2-5. Midtown Bank always has two tellers on duty. Customers arrive to receive service from a teller at a mean rate of 40 per hour. A teller requires an average of 2 minutes to serve a customer. When both tellers are busy, an arriving customer joins a single line to wait for service. Experience has shown that customers wait in line an average of 1 minute before service begins.

(a) Describe why this is a queueing system.

(b) Determine the basic measures of performance—W_q, W, L_q, and L—for this queueing system. (*Hint*: We don't know the probability distributions of interarrival times and service times for this queueing system, so you will need to use the relationships between these measures of performance to help answer the question.)

[1]See also the end of Chap. 26 (on the CD-ROM) for additional problems involving the application of queueing theory.

17.2-6. Explain why the utilization factor ρ for the server in a single-server queueing system must equal $1 - P_0$, where P_0 is the probability of having 0 customers in the system.

17.2-7. You are given two queueing systems, Q_1 and Q_2. The mean arrival rate, the mean service rate per busy server, and the steady-state expected number of customers for Q_2 are twice the corresponding values for Q_1. Let W_i = the steady-state expected waiting time in the system for Q_i, for $i = 1, 2$. Determine W_2/W_1.

17.2-8. Consider a single-server queueing system with *any* service-time distribution and *any* distribution of interarrival times (the *GI/G*/1 model). Use only basic definitions and the relationships given in Sec. 17.2 to verify the following general relationships:
(a) $L = L_q + (1 - P_0)$.
(b) $L = L_q + \rho$.
(c) $P_0 = 1 - \rho$.

17.2-9. Show that

$$L = \sum_{n=0}^{s-1} nP_n + L_q + s\left(1 - \sum_{n=0}^{s-1} P_n\right)$$

by using the statistical definitions of L and L_q in terms of the P_n.

17.3-1. Identify the customers and the servers in the queueing system in each of the following situations:
(a) The checkout stand in a grocery store.
(b) A fire station.
(c) The toll booth for a bridge.
(d) A bicycle repair shop.
(e) A shipping dock.
(f) A group of semiautomatic machines assigned to one operator.
(g) The materials-handling equipment in a factory area.
(h) A plumbing shop.
(i) A job shop producing custom orders.
(j) A secretarial typing pool.

17.4-1. Suppose that a queueing system has two servers, an exponential interarrival time distribution with a mean of 2 hours, and an exponential service-time distribution with a mean of 2 hours for each server. Furthermore, a customer has just arrived at 12:00 noon.
(a) What is the probability that the next arrival will come (i) before 1:00 P.M., (ii) between 1:00 and 2:00 P.M., and (iii) after 2:00 P.M.?
(b) Suppose that no additional customers arrive before 1:00 P.M. Now what is the probability that the next arrival will come between 1:00 and 2:00 P.M.?
(c) What is the probability that the number of arrivals between 1:00 and 2:00 P.M. will be (i) 0, (ii) 1, and (iii) 2 or more?
(d) Suppose that both servers are serving customers at 1:00 P.M. What is the probability that *neither* customer will have service completed (i) before 2:00 P.M., (ii) before 1:10 P.M., and (iii) before 1:01 P.M.?

17.4-2.* The jobs to be performed on a particular machine arrive according to a *Poisson* input process with a mean rate of two per hour. Suppose that the machine breaks down and will require 1 hour to be repaired. What is the probability that the number of new jobs that will arrive during this time is (*a*) 0, (*b*) 2, and (*c*) 5 or more?

17.4-3. The time required by a mechanic to repair a machine has an exponential distribution with a mean of 4 hours. However, a special tool would reduce this mean to 2 hours. If the mechanic repairs a machine in less than 2 hours, he is paid \$100; otherwise, he is paid \$80. Determine the mechanic's expected increase in pay per machine repaired if he uses the special tool.

17.4-4. A three-server queueing system has a controlled arrival process that provides customers in time to keep the servers continuously busy. Service times have an exponential distribution with mean 0.5.

You observe the queueing system starting up with all three servers beginning service at time $t = 0$. You then note that the first completion occurs at time $t = 1$. Given this information, determine the expected amount of time after $t = 1$ until the next service completion occurs.

17.4-5. A queueing system has three servers with expected service times of 20 minutes, 15 minutes, and 10 minutes. The service times have an exponential distribution. Each server has been busy with a current customer for 5 minutes. Determine the expected remaining time until the next service completion.

17.4-6. Consider a queueing system with two types of customers. Type 1 customers arrive according to a Poisson process with a mean rate of 5 per hour. Type 2 customers also arrive according to a Poisson process with a mean rate of 5 per hour. The system has two servers, both of which serve both types of customers. For both types, service times have an exponential distribution with a mean of 10 minutes. Service is provided on a first-come-first-served basis.
(a) What is the probability distribution (including its mean) of the time between consecutive arrivals of customers of any type?
(b) When a particular type 2 customer arrives, she finds two type 1 customers there in the process of being served but no other customers in the system. What is the probability distribution (including its mean) of this type 2 customer's waiting time in the queue?

17.4-7. Consider a two-server queueing system where all service times are independent and identically distributed according to an exponential distribution with a mean of 10 minutes. Service is provided on a first-come-first-served basis. When a particular customer arrives, he finds that both servers are busy and no one is waiting in the queue.
(a) What is the probability distribution (including its mean and standard deviation) of this customer's waiting time in the queue?
(b) Determine the expected value and standard deviation of this customer's waiting time in the system.
(c) Suppose that this customer still is waiting in the queue 5 minutes after its arrival. Given this information, how does this change the expected value and the standard deviation of this customer's total waiting time in the system from the answers obtained in part (*b*)?

17.4-8. For each of the following statements regarding service times modeled by the exponential distribution, label the statement as true or false and then justify your answer by referring to specific statements (with page citations) in the chapter.

(a) The expected value and variance of the service times are always equal.

(b) The exponential distribution always provides a good approximation of the actual service-time distribution when each customer requires the same service operations.

(c) At an s-server facility, $s > 1$, with exactly s customers already in the system, a new arrival would have an expected waiting time before entering service of $1/\mu$ time units, where μ is the mean service rate for each busy server.

17.4-9. As for Property 3 of the exponential distribution, let T_1, T_2, . . . , T_n be independent exponential random variables with parameters α_1, α_2, . . . , α_n, respectively, and let $U = \min\{T_1, T_2, \ldots, T_n\}$. Show that the probability that a particular random variable T_j will turn out to be smallest of the n random variables is

$$P\{T_j = U\} = \alpha_j \Big/ \sum_{i=1}^{n} \alpha_i, \qquad \text{for } j = 1, 2, \ldots, n.$$

(*Hint*: $P\{T_j = U\} = \int_0^\infty P\{T_i > T_j \text{ for all } i \neq j \,|\, T_j = t\}\alpha_j e^{-\alpha_j t} dt$.)

17.5-1. Consider the birth-and-death process with all $\mu_n = 2$ ($n = 1, 2, \ldots$), $\lambda_0 = 3$, $\lambda_1 = 2$, $\lambda_2 = 1$, and $\lambda_n = 0$ for $n = 3, 4, \ldots$.

(a) Display the rate diagram.

(b) Calculate P_0, P_1, P_2, P_3, and P_n for $n = 4, 5, \ldots$.

(c) Calculate L, L_q, W, and W_q.

17.5-2. Consider a birth-and-death process with just three attainable states (0, 1, and 2), for which the steady-state probabilities are P_0, P_1, and P_2, respectively. The birth-and-death rates are summarized in the following table:

State	Birth Rate	Death Rate
0	1	—
1	1	2
2	0	2

(a) Construct the rate diagram for this birth-and-death process.

(b) Develop the balance equations.

(c) Solve these equations to find P_0, P_1, and P_2.

(d) Use the general formulas for the birth-and-death process to calculate P_0, P_1, and P_2. Also calculate L, L_q, W, and W_q.

17.5-3. Consider the birth-and-death process with the following mean rates. The birth rates are $\lambda_0 = 2$, $\lambda_1 = 3$, $\lambda_2 = 2$, $\lambda_3 = 1$, and $\lambda_n = 0$ for $n > 3$. The death rates are $\mu_1 = 3$, $\mu_2 = 4$, $\mu_3 = 1$, and $\mu_n = 2$ for $n > 4$.

(a) Construct the rate diagram for this birth-and-death process.

(b) Develop the balance equations.

(c) Solve these equations to find the steady-state probability distribution P_0, P_1, ...

(d) Use the general formulas for the birth-and-death process to calculate P_0, P_1, ... Also calculate L, L_q, W, and W_q.

17.5-4. Consider the birth-and-death process with all $\lambda_n = 2$ ($n = 0, 1, \ldots$), $\mu_1 = 2$, and $\mu_n = 4$ for $n = 2, 3, \ldots$.

(a) Display the rate diagram.

(b) Calculate P_0 and P_1. Then give a general expression for P_n in terms of P_0 for $n = 2, 3, \ldots$.

(c) Consider a queueing system with two servers that fits this process. What is the mean arrival rate for this queueing system? What is the mean service rate for each server when it is busy serving customers?

17.5-5.* A service station has one gasoline pump. Cars wanting gasoline arrive according to a Poisson process at a mean rate of 15 per hour. However, if the pump already is being used, these potential customers may *balk* (drive on to another service station). In particular, if there are n cars already at the service station, the probability that an arriving potential customer will balk is $n/3$ for $n = 1, 2, 3$. The time required to service a car has an exponential distribution with a mean of 4 minutes.

(a) Construct the rate diagram for this queueing system.

(b) Develop the balance equations.

(c) Solve these equations to find the steady-state probability distribution of the number of cars at the station. Verify that this solution is the same as that given by the general solution for the birth-and-death process.

(d) Find the expected waiting time (including service) for those cars that stay.

17.5-6. A maintenance person has the job of keeping two machines in working order. The amount of time that a machine works before breaking down has an exponential distribution with a mean of 10 hours. The time then spent by the maintenance person to repair the machine has an exponential distribution with a mean of 8 hours.

(a) Show that this process fits the birth-and-death process by defining the states, specifying the values of the λ_n and μ_n, and then constructing the rate diagram.

(b) Calculate the P_n.

(c) Calculate L, L_q, W, and W_q.

(d) Determine the proportion of time that the maintenance person is busy.

(e) Determine the proportion of time that any given machine is working.

(f) Refer to the nearly identical example of a *continuous time Markov chain* given at the end of Sec. 16.8. Describe the relationship between continuous time Markov chains and the birth-and-death process that enables both to be applied to this same problem.

17.5-7. Consider a single-server queueing system where interarrival times have an exponential distribution with parameter λ and service times have an exponential distribution with parameter μ. In addition, customers *renege* (leave the queueing system without being served) if their waiting time in the queue grows too large. In particular, assume that the time each customer is willing to wait in

the queue before reneging has an exponential distribution with a mean of $1/\theta$.

(a) Construct the rate diagram for this queueing system.

(b) Develop the balance equations.

17.5-8.* A certain small grocery store has a single checkout stand with a full-time cashier. Customers arrive at the stand "randomly" (i.e., a Poisson input process) at a mean rate of 30 per hour. When there is only one customer at the stand, she is processed by the cashier alone, with an expected service time of 1.5 minutes. However, the stock boy has been given standard instructions that whenever there is more than one customer at the stand, he is to help the cashier by bagging the groceries. This help reduces the expected time required to process a customer to 1 minute. In both cases, the service-time distribution is exponential.

(a) Construct the rate diagram for this queueing system.

(b) What is the steady-state probability distribution of the number of customers at the checkout stand?

(c) Derive L for this system. (*Hint:* Refer to the derivation of L for the $M/M/1$ model at the beginning of Sec. 17.6.) Use this information to determine L_q, W, and W_q.

17.5-9. A department has one word-processing operator. Documents produced in the department are delivered for word processing according to a Poisson process with an expected interarrival time of 20 minutes. When the operator has just one document to process, the expected processing time is 15 minutes. When she has more than one document, then editing assistance that is available reduces the expected processing time for each document to 10 minutes. In both cases, the processing times have an exponential distribution.

(a) Construct the rate diagram for this queueing system.

(b) Find the steady-state distribution of the number of documents that the operator has received but not yet completed.

(c) Derive L for this system. (*Hint:* Refer to the derivation of L for the $M/M/1$ model at the beginning of Sec. 17.6.) Use this information to determine L_q, W, and W_q.

17.5-10. Customers arrive at a queueing system according to a Poisson process with a mean arrival rate of 2 customers per minute. The service time has an exponential distribution with a mean of 1 minute. An unlimited number of servers are available as needed so customers never wait for service to begin. Calculate the steady-state probability that exactly 1 customer is in the system.

17.5-11. Suppose that a single-server queueing system fits all the assumptions of the birth-and-death process *except* that customers always arrive in *pairs*. The mean arrival rate is 2 pairs per hour (4 customers per hour) and the mean service rate (when the server is busy) is 5 customers per hour.

(a) Construct the rate diagram for this queueing system.

(b) Develop the balance equations.

(c) For comparison purposes, display the rate diagram for the corresponding queueing system that completely fits the birth-and-death process, i.e., where customers arrive *individually* at a mean rate of 4 per hour.

17.5-12. Consider a single-server queueing system with a finite queue that can hold a maximum of 2 customers *excluding* any being served. The server can provide *batch service* to 2 customers simultaneously, where the service time has an exponential distribution with a mean of 1 unit of time regardless of the number being served. Whenever the queue is not full, customers arrive individually according to a Poisson process at a mean rate of 1 per unit of time.

(a) Assume that the server *must* serve 2 customers simultaneously. Thus, if the server is idle when only 1 customer is in the system, the server must wait for another arrival before beginning service. Formulate the queueing model as a continuous time Markov chain by defining the states and then constructing the rate diagram. Give the balance equations, but do not solve further.

(b) Now assume that the batch size for a service is 2 only if 2 customers are in the queue when the server finishes the preceding service. Thus, if the server is idle when only 1 customer is in the system, the server must serve this single customer, and any subsequent arrivals must wait in the queue until service is completed for this customer. Formulate the resulting queueing model as a continuous time Markov chain by defining the states and then constructing the rate diagram. Give the balance equations, but do not solve further.

17.5-13. Consider a queueing system that has two classes of customers, two clerks providing service, and *no queue*. Potential customers from each class arrive according to a Poisson process, with a mean arrival rate of 10 customers per hour for class 1 and 5 customers per hour for class 2, but these arrivals are lost to the system if they cannot immediately enter service.

Each customer of class 1 that enters the system will receive service from either one of the clerks that is free, where the service times have an exponential distribution with a mean of 5 minutes.

Each customer of class 2 that enters the system requires the *simultaneous use of both clerks* (the two clerks work together as a single server), where the service times have an exponential distribution with a mean of 5 minutes. Thus, an arriving customer of this kind would be lost to the system unless both clerks are free to begin service immediately.

(a) Formulate the queueing model as a continuous time Markov chain by defining the states and constructing the rate diagram.

(b) Now describe how the formulation in part (*a*) can be fitted into the format of the birth-and-death process.

(c) Use the results for the birth-and-death process to calculate the steady-state joint distribution of the number of customers of each class in the system.

(d) For each of the two classes of customers, what is the expected fraction of arrivals who are unable to enter the system?

17.6-1.* The 4M Company has a single turret lathe as a key work center on its factory floor. Jobs arrive at this work center according to a Poisson process at a mean rate of 2 per day. The processing time to perform each job has an exponential distribution with a mean of $\frac{1}{4}$ day. Because the jobs are bulky, those not being worked on are currently being stored in a room some distance from the

machine. However, to save time in fetching the jobs, the production manager is proposing to add enough in-process storage space next to the turret lathe to accommodate 3 jobs in addition to the one being processed. (Excess jobs will continue to be stored temporarily in the distant room.) Under this proposal, what proportion of the time will this storage space next to the turret lathe be adequate to accommodate all waiting jobs?

(a) Use available formulas to calculate your answer.

T **(b)** Use the corresponding Excel template to obtain the probabilities needed to answer the question.

17.6-2. Customers arrive at a single-server queueing system according to a Poisson process at a mean rate of 10 per hour. If the server works continuously, the number of customers that can be served in an hour has a Poisson distribution with a mean of 15. Determine the proportion of time during which no one is waiting to be served.

17.6-3. Consider the *M*/*M*/1 model, with $\lambda < \mu$.

(a) Determine the steady-state probability that a customer's actual waiting time in the system is longer than the expected waiting time in the system, i.e., $P\{W > W\}$.

(b) Determine the steady-state probability that a customer's actual waiting time in the queue is longer than the expected waiting time in the queue, i.e., $P\{W_q > W_q\}$.

17.6-4. Verify the following relationships for an *M*/*M*/1 queueing system:

$$\lambda = \frac{(1 - P_0)^2}{W_q P_0}, \qquad \mu = \frac{1 - P_0}{W_q P_0}.$$

17.6-5. It is necessary to determine how much in-process storage space to allocate to a particular work center in a new factory. Jobs arrive at this work center according to a Poisson process with a mean rate of 3 per hour, and the time required to perform the necessary work has an exponential distribution with a mean of 0.25 hour. Whenever the waiting jobs require more in-process storage space than has been allocated, the excess jobs are stored temporarily in a less convenient location. If each job requires 1 square foot of floor space while it is in in-process storage at the work center, how much space must be provided to accommodate all waiting jobs (*a*) 50 percent of the time, (*b*) 90 percent of the time, and (*c*) 99 percent of the time? Derive an analytical expression to answer these three questions. *Hint:* The sum of a geometric series is

$$\sum_{n=0}^{N} x^n = \frac{1 - x^{N+1}}{1 - x}.$$

17.6-6. Consider the following statements about an *M*/*M*/1 queueing system and its utilization factor ρ. Label each of the statements as true or false, and then justify your answer.

(a) The probability that a customer has to wait before service begins is proportional to ρ.

(b) The expected number of customers in the system is proportional to ρ.

(c) If ρ has been increased from $\rho = 0.9$ to $\rho = 0.99$, the effect of any further increase in ρ on L, L_q, W, and W_q will be relatively small as long as $\rho < 1$.

17.6-7. Customers arrive at a single-server queueing system in accordance with a Poisson process with an expected interarrival time of 25 minutes. Service times have an exponential distribution with a mean of 30 minutes.

Label each of the following statements about this system as true or false, and then justify your answer.

(a) The server definitely will be busy forever after the first customer arrives.

(b) The queue will grow without bound.

(c) If a second server with the same service-time distribution is added, the system can reach a steady-state condition.

17.6-8. For each of the following statements about an *M*/*M*/1 queueing system, label the statement as true or false and then justify your answer by referring to specific statements (with page citations) in the chapter.

(a) The waiting time in the system has an exponential distribution.

(b) The waiting time in the queue has an exponential distribution.

(c) The conditional waiting time in the system, given the number of customers already in the system, has an Erlang (gamma) distribution.

17.6-9. The Friendly Neighbor Grocery Store has a single checkout stand with a full-time cashier. Customers arrive randomly at the stand at a mean rate of 30 per hour. The service-time distribution is exponential, with a mean of 1.5 minutes. This situation has resulted in occasional long lines and complaints from customers. Therefore, because there is no room for a second checkout stand, the manager is considering the alternative of hiring another person to help the cashier by bagging the groceries. This help would reduce the expected time required to process a customer to 1 minute, but the distribution still would be exponential.

The manager would like to have the percentage of time that there are more than two customers at the checkout stand down below 25 percent. She also would like to have no more than 5 percent of the customers needing to wait at least 5 minutes before beginning service, or at least 7 minutes before finishing service.

(a) Use the formulas for the *M*/*M*/1 model to calculate L, W, W_q, L_q, P_0, P_1, and P_2 for the current mode of operation. What is the probability of having more than two customers at the checkout stand?

T **(b)** Use the Excel template for this model to check your answers in part (*a*). Also find the probability that the waiting time before beginning service exceeds 5 minutes, and the probability that the waiting time before finishing service exceeds 7 minutes.

(c) Repeat part (*a*) for the alternative being considered by the manager.

(d) Repeat part (*b*) for this alternative.

(e) Which approach should the manager use to satisfy her criteria as closely as possible?

T **17.6-10.** The Centerville International Airport has two runways, one used exclusively for takeoffs and the other exclusively for landings. Airplanes arrive in the Centerville air space to request landing instructions according to a Poisson process at a mean rate of

10 per hour. The time required for an airplane to land after receiving clearance to do so has an exponential distribution with a mean of 3 minutes, and this process must be completed before giving clearance to do so to another airplane. Airplanes awaiting clearance must circle the airport.

The Federal Aviation Administration has a number of criteria regarding the safe level of congestion of airplanes waiting to land. These criteria depend on a number of factors regarding the airport involved, such as the number of runways available for landing. For Centerville, the criteria are (1) the average number of airplanes waiting to receive clearance to land should not exceed 1, (2) 95 percent of the time, the actual number of airplanes waiting to receive clearance to land should not exceed 4, (3) for 99 percent of the airplanes, the amount of time spent circling the airport before receiving clearance to land should not exceed 30 minutes (since exceeding this amount of time often would require rerouting the plane to another airport for an emergency landing before its fuel runs out).

(a) Evaluate how well these criteria are currently being satisfied.

(b) A major airline is considering adding this airport as one of its hubs. This would increase the mean arrival rate to 15 airplanes per hour. Evaluate how well the above criteria would be satisfied if this happens.

(c) To attract additional business [including the major airline mentioned in part (*b*)], airport management is considering adding a second runway for landings. It is estimated that this eventually would increase the mean arrival rate to 25 airplanes per hour. Evaluate how well the above criteria would be satisfied if this happens.

T **17.6-11.** The Security & Trust Bank employs 4 tellers to serve its customers. Customers arrive according to a Poisson process at a mean rate of 2 per minute. However, business is growing and management projects that the mean arrival rate will be 3 per minute a year from now. The transaction time between the teller and customer has an exponential distribution with a mean of 1 minute.

Management has established the following guidelines for a satisfactory level of service to customers. The average number of customers waiting in line to begin service should not exceed 1. At least 95 percent of the time, the number of customers waiting in line should not exceed 5. For at least 95 percent of the customers, the time spent in line waiting to begin service should not exceed 5 minutes.

(a) Use the *M/M/s* model to determine how well these guidelines are currently being satisfied.

(b) Evaluate how well the guidelines will be satisfied a year from now if no change is made in the number of tellers.

(c) Determine how many tellers will be needed a year from now to completely satisfy these guidelines.

17.6-12. Consider the *M/M/s* model.

T **(a)** Suppose there is one server and the expected service time is exactly 1 minute. Compare L for the cases where the mean arrival rate is 0.5, 0.9, and 0.99 customers per minute, respectively. Do the same for L_q, W, W_q, and $P\{W > 5\}$. What conclusions do you draw about the impact of increasing the utilization factor ρ from small values (e.g., $\rho = 0.5$) to fairly large values (e.g., $\rho = 0.9$) and then to even larger values very close to 1 (e.g., $\rho = 0.99$)?

(b) Now suppose there are two servers and the expected service time is exactly 2 minutes. Follow the instructions for part (*a*).

T **17.6-13.** Consider the *M/M/s* model with a mean arrival rate of 10 customers per hour and an expected service time of 5 minutes. Use the Excel template for this model to obtain and print out the various measures of performance (with $t = 10$ and $t = 0$, respectively, for the two waiting time probabilities) when the number of servers is 1, 2, 3, 4, and 5. Then, for each of the following possible criteria for a satisfactory level of service (where the unit of time is 1 minute), use the printed results to determine how many servers are needed to satisfy this criterion.

(a) $L_q \leq 0.25$

(b) $L \leq 0.9$

(c) $W_q \leq 0.1$

(d) $W \leq 6$

(e) $P\{W_q > 0\} \leq 0.01$

(f) $P\{W > 10\} \leq 0.2$

(g) $\sum_{n=0}^{s} P_n \geq 0.95$

17.6-14. A gas station with only one gas pump employs the following policy: If a customer has to wait, the price is \$1 per gallon; if she does not have to wait, the price is \$1.20 per gallon. Customers arrive according to a Poisson process with a mean rate of 15 per hour. Service times at the pump have an exponential distribution with a mean of 3 minutes. Arriving customers always wait until they can eventually buy gasoline. Determine the expected price of gasoline per gallon.

17.6-15. You are given an *M/M/1* queueing system with mean arrival rate λ and mean service rate μ. An arriving customer receives n dollars if n customers are already in the system. Determine the expected cost in dollars per customer.

17.6-16. Section 17.6 gives the following equations for the *M/M/1* model:

$$(1) \quad P\{W > t\} = \sum_{n=0}^{\infty} P_n P\{S_{n+1} > t\}.$$

$$(2) \quad P\{W > t\} = e^{-\mu(1-\rho)t}.$$

Show that Eq. (1) reduces algebraically to Eq. (2). (*Hint:* Use differentiation, algebra, and integration.)

17.6-17. Derive W_q directly for the following cases by developing and reducing an expression analogous to Eq. (1) in Prob. 17.6-16. (*Hint:* Use the *conditional* expected waiting time in the queue given that a random arrival finds n customers already in the system.)

(a) The *M/M/1* model

(b) The *M/M/s* model

T **17.6-18.** Consider an *M/M/2* queueing system with $\lambda = 4$ and $\mu = 3$. Determine the mean rate at which service completions occur during the periods when no customers are waiting in the queue.

T **17.6-19.** You are given an *M/M/*2 queueing system with $\lambda = 4$ per hour and $\mu = 6$ per hour. Determine the probability that an arriving customer will wait more than 30 minutes in the queue, given that at least 2 customers are already in the system.

17.6-20.* In the Blue Chip Life Insurance Company, the deposit and withdrawal functions associated with a certain investment product are separated between two clerks, Clara and Clarence. Deposit slips arrive randomly (a Poisson process) at Clara's desk at a mean rate of 16 per hour. Withdrawal slips arrive randomly (a Poisson process) at Clarence's desk at a mean rate of 14 per hour. The time required to process either transaction has an exponential distribution with a mean of 3 minutes. To reduce the expected waiting time in the system for both deposit slips and withdrawal slips, the actuarial department has made the following recommendations: (1) Train each clerk to handle both deposits and withdrawals, and (2) put both deposit and withdrawal slips into a single queue that is accessed by both clerks.

(a) Determine the expected waiting time in the system under current procedures for each type of slip. Then combine these results to calculate the expected waiting time in the system for a random arrival of either type of slip.

T (b) If the recommendations are adopted, determine the expected waiting time in the system for arriving slips.

T (c) Now suppose that adopting the recommendations would result in a slight increase in the expected processing time. Use the Excel template for the *M/M/s* model to determine by trial and error the expected processing time (within 0.001 hour) that would cause the expected waiting time in the system for a random arrival to be essentially the same under current procedures and under the recommendations.

17.6-21. People's Software Company has just set up a call center to provide technical assistance on its new software package. Two technical representatives are taking the calls, where the time required by either representative to answer a customer's questions has an exponential distribution with a mean of 8 minutes. Calls are arriving according to a Poisson process at a mean rate of 10 per hour.

By next year, the mean arrival rate of calls is expected to decline to 5 per hour, so the plan is to reduce the number of technical representatives to one then.

T (a) Assuming that μ will continue to be 7.5 calls per hour for next year's queueing system, determine L, L_q, W, and W_q for both the current system and next year's system. For each of these four measures of performance, which system yields the smaller value?

(b) Now assume that μ will be adjustable when the number of technical representatives is reduced to one. Solve algebraically for the value of μ that would yield the same value of W as for the current system.

(c) Repeat part (*b*) with W_q instead of W.

17.6-22. Consider a generalization of the *M/M/*1 model where the server needs to "warm up" at the beginning of a busy period, and so serves the first customer of a busy period at a slower rate than other customers. In particular, if an arriving customer finds the server idle, the customer experiences a service time that has an exponential distribution with parameter μ_1. However, if an arriving customer finds the server busy, that customer joins the queue and subsequently experiences a service time that has an exponential distribution with parameter μ_2, where $\mu_1 < \mu_2$. Customers arrive according to a Poisson process with mean rate λ.

(a) Formulate this model as a continuous time Markov chain by defining the states and constructing the rate diagram accordingly.

(b) Develop the balance equations.

(c) Suppose that numerical values are specified for μ_1, μ_2, and λ, and that $\lambda < \mu_2$ (so that a steady-state distribution exists). Since this model has an infinite number of states, the steady-state distribution is the simultaneous solution of an infinite number of balance equations (plus the equation specifying that the sum of the probabilities equals 1). Suppose that you are unable to obtain this solution analytically, so you wish to use a computer to solve the model numerically. Considering that it is impossible to solve an infinite number of equations numerically, briefly describe what still can be done with these equations to obtain an approximation of the steady-state distribution. Under what circumstances will this approximation be essentially exact?

(d) Given that the steady-state distribution has been obtained, give explicit expressions for calculating L, L_q, W, and W_q.

(e) Given this steady-state distribution, develop an expression for $P\{W > t\}$ that is analogous to Eq. (1) in Prob. 17.6-16.

17.6-23. For each of the following models, write the balance equations and show that they are satisfied by the solution given in Sec. 17.6 for the steady-state distribution of the number of customers in the system.

(a) The *M/M/*1 model.

(b) The finite queue variation of the *M/M/*1 model, with $K = 2$.

(c) The finite calling population variation of the *M/M/*1 model, with $N = 2$.

T **17.6-24.** Consider a telephone system with three lines. Calls arrive according to a Poisson process at a mean rate of 6 per hour. The duration of each call has an exponential distribution with a mean of 15 minutes. If all lines are busy, calls will be put on hold until a line becomes available.

(a) Print out the measures of performance provided by the Excel template for this queueing system (with $t = 1$ hour and $t = 0$, respectively, for the two waiting time probabilities).

(b) Use the printed result giving $P\{W_q > 0\}$ to identify the steady-state probability that a call will be answered immediately (not put on hold). Then verify this probability by using the printed results for the P_n.

(c) Use the printed results to identify the steady-state probability distribution of the number of calls on hold.

(d) Print out the new measures of performance if arriving calls are lost whenever all lines are busy. Use these results to identify the steady-state probability that an arriving call is lost.

17.6-25.* Janet is planning to open a small car-wash operation, and she must decide how much space to provide for waiting cars.

Janet estimates that customers would arrive randomly (i.e., a Poisson input process) with a mean rate of 1 every 4 minutes, unless the waiting area is full, in which case the arriving customers would take their cars elsewhere. The time that can be attributed to washing one car has an exponential distribution with a mean of 3 minutes. Compare the expected fraction of potential customers that will be *lost* because of inadequate waiting space if (*a*) 0 spaces (not including the car being washed), (*b*) 2 spaces, and (*c*) 4 spaces were provided.

17.6-26. Consider the finite queue variation of the *M*/*M*/*s* model. Derive the expression for L_q given in Sec. 17.6 for this model.

17.6-27. For the finite queue variation of the *M*/*M*/1 model, develop an expression analogous to Eq. (1) in Prob. 17.6-16 for the following probabilities:

(a) $P\{W > t\}$.

(b) $P\{W_q > t\}$.

[*Hint:* Arrivals can occur only when the system is not full, so the probability that a random arrival finds n customers already there is $P_n/(1 - P_K)$.]

17.6-28. George is planning to open a drive-through photo-developing booth with a single service window that will be open approximately 200 hours per month in a busy commercial area. Space for a drive-through lane is available for a rental of $200 per month per car length. George needs to decide how many car lengths of space to provide for his customers.

Excluding this rental cost for the drive-through lane, George believes that he will average a profit of $4 per customer served (nothing for a drop off of film and $8 when the photographs are picked up). He also estimates that customers will arrive randomly (a Poisson process) at a mean rate of 20 per hour, although those who find the drive-through lane full will be forced to leave. Half of the customers who find the drive-through lane full wanted to drop off film, and the other half wanted to pick up their photographs. The half who wanted to drop off film will take their business elsewhere instead. The other half of the customers who find the drive-through lane full will not be lost because they will keep trying later until they can get in and pick up their photographs. George assumes that the time required to serve a customer will have an exponential distribution with a mean of 2 minutes.

T **(a)** Find L and the mean rate at which customers are lost when the number of car lengths of space provided is 2, 3, 4, and 5.

(b) Calculate W from L for the cases considered in part (*a*).

(c) Use the results from part (*a*) to calculate the decrease in the mean rate at which customers are lost when the number of car lengths of space provided is increased from 2 to 3, from 3 to 4, and from 4 to 5. Then calculate the increase in expected profit per hour (excluding space rental costs) for each of these three cases.

(d) Compare the increases in expected profit found in part (*c*) with the cost per hour of renting each car length of space. What conclusion do you draw about the number of car lengths of space that George should provide?

17.6-29. At the Forrester Manufacturing Company, one repair technician has been assigned the responsibility of maintaining three machines. For each machine, the probability distribution of the running time before a breakdown is exponential, with a mean of 9 hours. The repair time also has an exponential distribution, with a mean of 2 hours.

(a) Which queueing model fits this queueing system?

T **(b)** Use this queueing model to find the probability distribution of the number of machines not running, and the mean of this distribution.

(c) Use this mean to calculate the expected time between a machine breakdown and the completion of the repair of that machine.

(d) What is the expected fraction of time that the repair technician will be busy?

T **(e)** As a crude approximation, assume that the calling population is infinite and that machine breakdowns occur randomly at a mean rate of 3 every 9 hours. Compare the result from part (*b*) with that obtained by making this approximation while using (i) the *M*/*M*/*s* model and (ii) the finite queue variation of the *M*/*M*/*s* model with $K = 3$.

T **(f)** Repeat part (*b*) when a second repair technician is made available to repair a second machine whenever more than one of these three machines require repair.

17.6-30. Reconsider the specific birth-and-death process described in Prob. 17.5-1.

(a) Identify a queueing model (and its parameter values) in Sec. 17.6 that fits this process.

T **(b)** Use the corresponding Excel template to obtain the answers for parts (*b*) and (*c*) of Prob. 17.5-1.

T **17.6-31.*** The Dolomite Corporation is making plans for a new factory. One department has been allocated 12 semiautomatic machines. A small number (yet to be determined) of operators will be hired to provide the machines the needed occasional servicing (loading, unloading, adjusting, setup, and so on). A decision now needs to be made on how to organize the operators to do this. Alternative 1 is to assign each operator to her own machines. Alternative 2 is to pool the operators so that any idle operator can take the next machine needing servicing. Alternative 3 is to combine the operators into a single crew that will work together on any machine needing servicing.

The running time (time between completing service and the machine's requiring service again) of each machine is expected to have an exponential distribution, with a mean of 150 minutes. The service time is assumed to have an exponential distribution, with a mean of 15 minutes (for Alternatives 1 and 2) or 15 minutes divided by the number of operators in the crew (for Alternative 3). For the department to achieve the required production rate, the machines must be running at least 89 percent of the time on average.

(a) For Alternative 1, what is the maximum number of machines that can be assigned to an operator while still achieving the required production rate? What is the resulting utilization of each operator?

(b) For Alternative 2, what is the minimum number of operators needed to achieve the required production rate? What is the resulting utilization of the operators?

(c) For Alternative 3, what is the minimum size of the crew needed to achieve the required production rate? What is the resulting utilization of the crew?

17.6-32. A shop contains three identical machines that are subject to a failure of a certain kind. Therefore, a maintenance system is provided to perform the maintenance operation (recharging) required by a failed machine. The time required by each operation has an exponential distribution with a mean of 30 minutes. However, with probability $\frac{1}{3}$, the operation must be performed a second time (with the same distribution of time) in order to bring the failed machine back to a satisfactory operational state. The maintenance system works on only one failed machine at a time, performing all the operations (one or two) required by that machine, on a first-come-first-served basis. After a machine is repaired, the time until its next failure has an exponential distribution with a mean of 3 hours.

(a) How should the states of the system be defined in order to formulate this queueing system as a continuous time Markov chain? (*Hint:* Given that a first operation is being performed on a failed machine, completing this operation *successfully* and completing it *unsuccessfully* are two separate events of interest. Then use Property 6 regarding disaggregation for the exponential distribution.)

(b) Construct the corresponding rate diagram.

(c) Develop the balance equations.

17.7-1.* Consider the *M*/*G*/1 model.

(a) Compare the expected waiting time in the queue if the service-time distribution is (i) exponential, (ii) constant, (iii) Erlang with the amount of variation (i.e., the standard deviation) halfway between the constant and exponential cases.

(b) What is the effect on the expected waiting time in the queue and on the expected queue length if both λ and μ are doubled and the scale of the service-time distribution is changed accordingly?

17.7-2. Consider the *M*/*G*/1 model with $\lambda = 0.2$ and $\mu = 0.25$.

T **(a)** Use the Excel template for this model (or hand calculations) to find the main measures of performance—L, L_q, W, W_q—for each of the following values of σ: 4, 3, 2, 1, 0.

(b) What is the ratio of L_q with $\sigma = 4$ to L_q with $\sigma = 0$? What does this say about the importance of reducing the variability of the service times?

(c) Calculate the reduction in L_q when σ is reduced from 4 to 3, from 3 to 2, from 2 to 1, and from 1 to 0. Which is the largest reduction? Which is the smallest?

(d) Use trial and error with the template to see approximately how much μ would need to be increased with $\sigma = 4$ to achieve the same L_q as with $\mu = 0.25$ and $\sigma = 0$.

17.7-3. Consider the following statements about an *M*/*G*/1 queueing system, where σ^2 is the variance of service times. Label each statement as true or false, and then justify your answer.

(a) Increasing σ^2 (with fixed λ and μ) will increase L_q and L, but will not change W_q and W.

(b) When choosing between a tortoise (small μ and σ^2) and a hare (large μ and σ^2) to be the server, the tortoise always wins by providing a smaller L_q.

(c) With λ and μ fixed, the value of L_q with an exponential service-time distribution is twice as large as with constant service times.

(d) Among all possible service-time distributions (with λ and μ fixed), the exponential distribution yields the largest value of L_q.

17.7-4. Marsha operates an expresso stand. Customers arrive according to a Poisson process at a mean rate of 30 per hour. The time needed by Marsha to serve a customer has an exponential distribution with a mean of 75 seconds.

(a) Use the *M*/*G*/1 model to find L, L_q, W, and W_q.

(b) Suppose Marsha is replaced by an expresso vending machine that requires exactly 75 seconds for each customer to operate. Find L, L_q, W, and W_q.

(c) What is the ratio of L_q in part (*b*) to L_q in part (*a*)?

T **(d)** Use trial and error with the Excel template for the *M*/*G*/1 model to see approximately how much Marsha would need to reduce her expected service time to achieve the same L_q as with the expresso vending machine.

17.7-5. Antonio runs a shoe repair store by himself. Customers arrive to bring a pair of shoes to be repaired according to a Poisson process at a mean rate of $\frac{1}{4}$ per hour. The time Antonio requires to repair each individual shoe has an exponential distribution with a mean of 15 minutes.

(a) Consider the formulation of this queueing system where the individual shoes (not pairs of shoes) are considered to be the customers. For this formulation, construct the rate diagram and develop the balance equations, but do not solve further.

(b) Now consider the formulation of this queueing system where the pairs of shoes are considered to be the customers. Identify the specific queueing model that fits this formulation.

(c) Calculate the expected number of pairs of shoes in the shop.

(d) Calculate the expected amount of time from when a customer drops off a pair of shoes until they are repaired and ready to be picked up.

T **(e)** Use the corresponding Excel template to check your answers in parts (*c*) and (*d*).

17.7-6.* The maintenance base for Friendly Skies Airline has facilities for overhauling only one airplane engine at a time. Therefore, to return the airplanes to use as soon as possible, the policy has been to stagger the overhauling of the four engines of each airplane. In other words, only one engine is overhauled each time an airplane comes into the shop. Under this policy, airplanes have arrived according to a Poisson process at a mean rate of 1 per day. The time required for an engine overhaul (once work has begun) has an exponential distribution with a mean of $\frac{1}{2}$ day.

A proposal has been made to change the policy so that all four engines are overhauled consecutively each time an airplane comes into the shop. Although this would quadruple the expected service time, each plane would need to come to the maintenance base only one-fourth as often.

Management now needs to decide whether to continue the status quo or adopt the proposal. The objective is to minimize the average amount of flying time lost by the entire fleet per day due to engine overhauls.

(a) Compare the two alternatives with respect to the average amount of flying time lost by an airplane each time it comes to the maintenance base.

(b) Compare the two alternatives with respect to the average number of airplanes losing flying time due to being at the maintenance base.
(c) Which of these two comparisons is the appropriate one for making management's decision? Explain.

17.7-7. Reconsider Prob. 17.7-6. Management has adopted the proposal but now wants further analysis conducted of this new queueing system.
(a) How should the state of the system be defined in order to formulate the queueing model as a continuous time Markov chain?
(b) Construct the corresponding rate diagram.

17.7-8. The McAllister Company factory currently has *two* tool cribs, each with a *single* clerk, in its manufacturing area. One tool crib handles only the tools for the heavy machinery; the second one handles all other tools. However, for each crib the mechanics arrive to obtain tools at a mean rate of 24 per hour, and the expected service time is 2 minutes.

Because of complaints that the mechanics coming to the tool crib have to wait too long, it has been proposed that the two tool cribs be combined so that either clerk can handle either kind of tool as the demand arises. It is believed that the mean arrival rate to the combined two-clerk tool crib would double to 48 per hour and that the expected service time would continue to be 2 minutes. However, information is not available on the *form* of the probability distributions for interarrival and service times, so it is not clear which queueing model would be most appropriate.

Compare the status quo and the proposal with respect to the total expected number of mechanics at the tool crib(s) and the expected waiting time (including service) for each mechanic. Do this by tabulating these data for the four queueing models considered in Figs. 17.6, 17.8, 17.10, and 17.11 (use $k = 2$ when an Erlang distribution is appropriate).

17.7-9.* Consider a single-server queueing system with a Poisson input, Erlang service times, and a finite queue. In particular, suppose that $k = 2$, the mean arrival rate is 2 customers per hour, the expected service time is 0.25 hour, and the maximum permissible number of customers in the system is 2. This system can be formulated as a continuous time Markov chain by dividing each service time into two consecutive phases, each having an exponential distribution with a mean of 0.125 hour, and then defining the state of the system as (n, p), where n is the number of customers in the system ($n = 0, 1, 2$), and p indicates the phase of the customer being served ($p = 0, 1, 2$, where $p = 0$ means that no customer is being served).
(a) Construct the corresponding rate diagram. Write the balance equations, and then use these equations to solve for the steady-state distribution of the state of this Markov chain.
(b) Use the steady-state distribution obtained in part (*a*) to identify the steady-state distribution of the number of customers in the system (P_0, P_1, P_2) and the steady-state expected number of customers in the system (L).
(c) Compare the results from part (*b*) with the corresponding results when the service-time distribution is exponential.

17.7-10. Consider the $E_2/M/1$ model with $\lambda = 4$ and $\mu = 5$. This model can be formulated as a continuous time Markov chain by dividing each interarrival time into two consecutive phases, each having an exponential distribution with a mean of $1/(2\lambda) = 0.125$, and then defining the state of the system as (n, p), where n is the number of customers in the system ($n = 0, 1, 2, \ldots$) and p indicates the phase of the *next* arrival (not yet in the system) ($p = 1, 2$).

Construct the corresponding rate diagram (but do not solve further).

17.7-11. A company has one repair technician to keep a large group of machines in running order. Treating this group as an infinite calling population, individual breakdowns occur according to a Poisson process at a mean rate of 1 per hour. For each breakdown, the probability is 0.9 that only a minor repair is needed, in which case the repair time has an exponential distribution with a mean of $\frac{1}{2}$ hour. Otherwise, a major repair is needed, in which case the repair time has an exponential distribution with a mean of 5 hours. Because both of these *conditional* distributions are exponential, the *unconditional* (combined) distribution of repair times is *hyperexponential*.
(a) Compute the mean and standard deviation of this hyperexponential distribution. [*Hint:* Use the general relationships from probability theory that, for any random variable X and any pair of mutually exclusive events E_1 and E_2, $E(X) = E(X|E_1)P(E_1) + E(X|E_2)P(E_2)$ and $\text{var}(X) = E(X^2) - E(X)^2$.] Compare this standard deviation with that for an exponential distribution having this mean.
(b) What are P_0, L_q, L, W_q, and W for this queueing system?
(c) What is the conditional value of W, given that the machine involved requires major repair? A minor repair? What is the division of L between machines requiring the two types of repairs? (*Hint:* Little's formula still applies for the individual categories of machines.)
(d) How should the states of the system be defined in order to formulate this queueing system as a continuous time Markov chain? (*Hint:* Consider what additional information must be given, besides the number of machines down, for the conditional distribution of the time remaining until the next event of each kind to be exponential.)
(e) Construct the corresponding rate diagram.

17.7-12. Consider the finite queue variation of the $M/G/1$ model, where K is the maximum number of customers allowed in the system. For $n = 1, 2, \ldots$, let the random variable X_n be the number of customers in the system at the moment t_n when the nth customer has just finished being served. (Do not count the departing customer.) The times $\{t_1, t_2, \ldots\}$ are called *regeneration points*. Furthermore, $\{X_n\}$ ($n = 1, 2, \ldots$) is a discrete time Markov chain and is known as an *embedded Markov chain*. Embedded Markov chains are useful for studying the properties of continuous time stochastic processes such as for an $M/G/1$ model.

Now consider the particular special case where $K = 4$, the service time of successive customers is a fixed constant, say, 10 minutes, and the mean arrival rate is 1 every 50 minutes. Therefore, $\{X_n\}$ is an embedded Markov chain with states 0, 1, 2, 3. (Because

there are never more than 4 customers in the system, there can never be more than 3 in the system at a regeneration point.) Because the system is observed at successive departures, X_n can never decrease by more than 1. Furthermore, the probabilities of transitions that result in increases in X_n are obtained directly from the Poisson distribution.

(a) Find the one-step transition matrix for the embedded Markov chain. (*Hint:* In obtaining the transition probability from state 3 to state 3, use the probability of 1 or more arrivals rather than just 1 arrival, and similarly for other transitions to state 3.)

(b) Use the corresponding routine in the Markov chains area of your OR Courseware to find the steady-state probabilities for the number of customers in the system at regeneration points.

(c) Compute the expected number of customers in the system at regeneration points, and compare it to the value of L for the $M/D/1$ model (with $K = \infty$) in Sec. 17.7.

17.8-1.* Southeast Airlines is a small commuter airline serving primarily the state of Florida. Their ticket counter at a certain airport is staffed by a single ticket agent. There are two separate lines—one for first-class passengers and one for coach-class passengers. When the ticket agent is ready for another customer, the next first-class passenger is served if there are any in line. If not, the next coach-class passenger is served. Service times have an exponential distribution with a mean of 3 minutes for both types of customers. During the 12 hours per day that the ticket counter is open, passengers arrive randomly at a mean rate of 2 per hour for first-class passengers and 10 per hour for coach-class passengers.

(a) What kind of queueing model fits this queueing system?

T **(b)** Find the main measures of performance—L, L_q, W, and W_q—for both first-class passengers and coach-class passengers.

(c) What is the expected waiting time before service begins for first-class customers as a fraction of this waiting time for coach-class customers?

(d) Determine the average number of hours per day that the ticket agent is busy.

T **17.8-2.** Consider the model with nonpreemptive priorities presented in Sec. 17.8. Suppose there are two priority classes, with $\lambda_1 = 4$ and $\lambda_2 = 4$. In designing this queueing system, you are offered the choice between the following alternatives: (1) one fast server ($\mu = 10$) and (2) two slow servers ($\mu = 5$).

Compare these alternatives with the usual four mean measures of performance (W, L, W_q, L_q) for the individual priority classes (W_1, W_2, L_1, L_2, and so forth). Which alternative is preferred if your primary concern is expected waiting time in the *system* for priority class 1 (W_1)? Which is preferred if your primary concern is expected waiting time in the *queue* for priority class 1?

17.8-3. Consider the single-server variation of the nonpreemptive priorities model presented in Sec. 17.8. Suppose there are three priority classes, with $\lambda_1 = 1$, $\lambda_2 = 1$, and $\lambda_3 = 1$. The expected service times for priority classes 1, 2, and 3 are 0.4, 0.3, and 0.2, respectively, so $\mu_1 = 2.5$, $\mu_2 = 3\frac{1}{3}$, and $\mu_3 = 5$.

(a) Calculate W_1, W_2, and W_3.

(b) Repeat part (*a*) when using the approximation of applying the general model for nonpreemptive priorities presented in Sec. 17.8 instead. Since this general model assumes that the expected service time is the same for all priority classes, use an expected service time of 0.3 so $\mu = 3\frac{1}{3}$. Compare the results with those obtained in part (*a*) and evaluate how good an approximation is provided by making this assumption.

T **17.8-4.*** A particular work center in a job shop can be represented as a single-server queueing system, where jobs arrive according to a Poisson process, with a mean rate of 8 per day. Although the arriving jobs are of three distinct types, the time required to perform any of these jobs has the same exponential distribution, with a mean of 0.1 working day. The practice has been to work on arriving jobs on a first-come-first-served basis. However, it is important that jobs of type 1 not wait very long, whereas the wait is only moderately important for jobs of type 2 and is relatively unimportant for jobs of type 3. These three types arrive with a mean rate of 2, 4, and 2 per day, respectively. Because all three types have experienced rather long delays on average, it has been proposed that the jobs be selected according to an appropriate priority discipline instead.

Compare the expected waiting time (including service) for each of the three types of jobs if the queue discipline is (*a*) first-come-first-served, (*b*) nonpreemptive priority, and (*c*) preemptive priority.

T **17.8-5.** Reconsider the *County Hospital* emergency room problem as analyzed in Sec. 17.8. Suppose that the definitions of the three categories of patients are tightened somewhat in order to move marginal cases into a lower category. Consequently, only 5 percent of the patients will qualify as critical cases, 20 percent as serious cases, and 75 percent as stable cases. Develop a table showing the data presented in Table 17.3 for this revised problem.

17.8-6. Reconsider the queueing system described in Prob. 17.4-6. Suppose now that type 1 customers are more important than type 2 customers. If the queue discipline were changed from first-come-first-served to a priority system with type 1 customers being given nonpreemptive priority over type 2 customers, would this increase, decrease, or keep unchanged the expected total number of customers in the system?

(a) Determine the answer without any calculations, and then present the reasoning that led to your conclusion.

T **(b)** Verify your conclusion in part (*a*) by finding the expected total number of customers in the system under each of these two queue disciplines.

17.8-7. Consider the queueing model with a preemptive priority queue discipline presented in Sec. 17.8. Suppose that $s = 1$, $N = 2$, and $(\lambda_1 + \lambda_2) < \mu$; and let P_{ij} be the steady-state probability that there are i members of the higher-priority class and j members of the lower-priority class in the queueing system ($i = 0, 1, 2, \ldots$; $j = 0, 1, 2, \ldots$). Use a method analogous to that presented in Sec. 17.5 to derive a system of linear equations whose simultaneous solution is the P_{ij}. Do not actually obtain this solution.

17.9-1. Consider a queueing system with two servers, where the customers arrive from two different sources. From source 1, the customers always arrive 2 at a time, where the time between consecutive arrivals of pairs of customers has an exponential distribution with a mean of 20 minutes. Source 2 is itself a two-server queueing system, which has a Poisson input process with a mean rate of 7 customers per hour, and the service time from each of these two servers has an exponential distribution with a mean of 15 minutes. When a customer completes service at source 2, he or she immediately enters the queueing system under consideration for another type of service. In the latter queueing system, the queue discipline is preemptive priority where customers from source 1 always have preemptive priority over customers from source 2. However, service times are independent and identically distributed for both types of customers according to an exponential distribution with a mean of 6 minutes.

(a) First focus on the problem of deriving the steady-state distribution of *only* the number of source 1 customers in the queueing system under consideration. Using a continuous time Markov chain formulation, define the states and construct the rate diagram for most efficiently deriving this distribution (but do not actually derive it).

(b) Now focus on the problem of deriving the steady-state distribution of the *total* number of customers of both types in the queueing system under consideration. Using a continuous time Markov chain formulation, define the states and construct the rate diagram for most efficiently deriving this distribution (but do not actually derive it).

(c) Now focus on the problem of deriving the steady-state *joint* distribution of the number of customers of each type in the queueing system under consideration. Using a continuous time Markov chain formulation, define the states and construct the rate diagram for deriving this distribution (but do not actually derive it).

17.9-2. Consider a system of two infinite queues in series, where each of the two service facilities has a single server. All service times are independent and have an exponential distribution, with a mean of 3 minutes at facility 1 and 4 minutes at facility 2. Facility 1 has a Poisson input process with a mean rate of 10 per hour.

(a) Find the steady-state distribution of the number of customers at facility 1 and then at facility 2. Then show the product form solution for the *joint* distribution of the number at the respective facilities.

(b) What is the probability that both servers are idle?

(c) Find the expected *total* number of customers in the system and the expected *total* waiting time (including service times) for a customer.

17.9-3. Under the assumptions specified in Sec. 17.9 for a system of infinite queues in series, this kind of queueing network actually is a special case of a Jackson network. Demonstrate that this is true by describing this system as a Jackson network, including specifying the values of the a_j and the p_{ij}, given λ for this system.

17.9-4. Consider a Jackson network with three service facilities having the parameter values shown below.

Facility j	s_j	μ_j	a_j	p_{ij}		
				$i = 1$	$i = 2$	$i = 3$
$j = 1$	1	40	10	0	0.3	0.4
$j = 2$	1	50	15	0.5	0	0.5
$j = 3$	1	30	3	0.3	0.2	0

T (a) Find the total arrival rate at each of the facilities.

(b) Find the steady-state distribution of the number of customers at facility 1, facility 2, and facility 3. Then show the product form solution for the joint distribution of the number at the respective facilities.

(c) What is the probability that all the facilities have empty queues (no customers waiting to begin service)?

(d) Find the expected total number of customers in the system.

(e) Find the expected total waiting time (including service times) for a customer.

T **17.10-1.** When describing economic analysis of the number of servers to provide in a queueing system, Sec. 17.10 introduces a basic cost model where the objective is to minimize $E(\text{TC}) = C_s s + C_w L$. The purpose of this problem is to enable you to explore the effect that the relative sizes of C_s and C_w have on the optimal number of servers.

Suppose that the queueing system under consideration fits the *M/M/s* model with $\lambda = 8$ customers per hour and $\mu = 10$ customers per hour. Use the Excel template in your OR Courseware for economic analysis with the *M/M/s* model to find the optimal number of servers for each of the following cases.

(a) $C_s = \$100$ and $C_w = \$10$.

(b) $C_s = \$100$ and $C_w = \$100$.

(c) $C_s = \$10$ and $C_w = \$100$.

T **17.10-2.*** Jim McDonald, manager of the fast-food hamburger restaurant McBurger, realizes that providing fast service is a key to the success of the restaurant. Customers who have to wait very long are likely to go to one of the other fast-food restaurants in town next time. He estimates that each minute a customer has to wait in line before completing service costs him an average of 30 cents in lost future business. Therefore, he wants to be sure that enough cash registers always are open to keep waiting to a minimum. Each cash register is operated by a part-time employee who obtains the food ordered by each customer and collects the payment. The total cost for each such employee is $9 per hour.

During lunch time, customers arrive according to a Poisson process at a mean rate of 66 per hour. The time needed to serve a customer is estimated to have an exponential distribution with a mean of 2 minutes.

Determine how many cash registers Jim should have open during lunch time to minimize his expected total cost per hour.

T **17.10-3.** The Garrett-Tompkins Company provides three copy machines in its copying room for the use of its employees. However, due to recent complaints about considerable time being wasted waiting for a copier to become free, management is considering adding one or more additional copy machines.

During the 2,000 working hours per year, employees arrive at the copying room according to a Poisson process at a mean rate of 30 per hour. The time each employee needs with a copy machine is believed to have an exponential distribution with a mean of 5 minutes. The lost productivity due to an employee spending time in the copying room is estimated to cost the company an average of $25 per hour. Each copy machine is leased for $3,000 per year.

Determine how many copy machines the company should have to minimize its expected total cost per hour.

CASES

CASE 17.1 Reducing In-Process Inventory

Jim Wells, vice-president for manufacturing of the Northern Airplane Company, is exasperated. His walk through the company's most important plant this morning has left him in a foul mood. However, he now can vent his temper at Jerry Carstairs, the plant's production manager, who has just been summoned to Jim's office.

"Jerry, I just got back from walking through the plant, and I am very upset." "What is the problem, Jim?" "Well, you know how much I have been emphasizing the need to cut down on our in-process inventory." "Yes, we've been working hard on that," responds Jerry. "Well, not hard enough!" Jim raises his voice even higher. "Do you know what I found by the presses?" "No." "Five metal sheets still waiting to be formed into wing sections. And then, right next door at the inspection station, 13 wing sections! The inspector was inspecting one of them, but the other 12 were just sitting there. You know we have a couple hundred thousand dollars tied up in each of those wing sections. So between the presses and the inspection station, we have a few million bucks worth of terribly expensive metal just sitting there. We can't have that!"

The chagrined Jerry Carstairs tries to respond. "Yes, Jim, I am well aware that that inspection station is a bottleneck. It usually isn't nearly as bad as you found it this morning, but it is a bottleneck. Much less so for the presses. You really caught us on a bad morning." "I sure hope so," retorts Jim, "but you need to prevent anything nearly this bad happening even occasionally. What do you propose to do about it?" Jerry now brightens noticeably in his response. "Well actually, I've already been working on this problem. I have a couple proposals on the table and I have asked an operations research analyst on my staff to analyze these proposals and report back with recommendations." "Great," responds Jim, "glad to see you are on top of the problem. Give this your highest priority and report back to me as soon as possible." "Will do," promises Jerry.

Here is the problem that Jerry and his OR analyst are addressing. Each of 10 identical presses is being used to form wing sections out of large sheets of specially processed metal. The sheets arrive randomly to the group of presses at a mean rate of 7 per hour. The time required by a press to form a wing section out of a metal sheet has an exponential distribution with a mean of 1 hour. When finished, the wing sections arrive randomly at an inspection station at the same mean rate as the metal sheets arrived at the presses (7 per hour). A single inspector has the full-time job of inspecting these wing sections to make sure they meet specifications. Each inspection takes her $7\frac{1}{2}$ minutes, so she can inspect 8 wing sections per hour. This inspection rate has resulted in a substantial average amount of in-process inventory at the inspection station (i.e., the average number of wing sheets waiting to complete inspection is fairly large), in addition to that already found at the group of machines.

The cost of this in-process inventory is estimated to be $8 per hour for each metal sheet at the presses or each wing section at the inspection station. Therefore, Jerry Carstairs has made two alternative proposals to reduce the average level of in-process inventory.

Proposal 1 is to use slightly less power for the presses (which would increase their average time to form a wing section to 1.2 hours), so that the inspector can keep up with their output better. This also would reduce the cost of the power for running each machine from $7.00 to $6.50 per hour. (By contrast, increasing to maximum power would increase this cost to $7.50 per hour while decreasing the average time to form a wing section to 0.8 hour.)

Proposal 2 is to substitute a certain younger inspector for this task. He is somewhat faster (albeit with some variability in his inspection times because of less experience), so he should keep up better. (His inspection time would have an Erlang distribution with a mean of 7.2 minutes and a shape parameter $k = 2$.) This inspector is in a job classification that calls for a total compensation (including benefits) of $19 per hour, whereas the current inspector is in a lower job classification where the compensation is $17 per hour. (The inspection

times for each of these inspectors are typical of those in the same job classification.)

You are the OR analyst on Jerry Carstair's staff who has been asked to analyze this problem. He wants you to "use the latest OR techniques to see how much each proposal would cut down on in-process inventory and then make your recommendations."

(a) To provide a basis of comparison, begin by evaluating the status quo. Determine the expected amount of in-process inventory at the presses and at the inspection station. Then calculate the expected total cost per hour when considering all of the following: the cost of the in-process inventory, the cost of the power for runnng the presses, and the cost of the inspector.

(b) What would be the effect of proposal 1? Why? Make specific comparisons to the results from part (*a*). Explain this outcome to Jerry Carstairs.

(c) Determine the effect of proposal 2. Make specific comparisons to the results from part (*a*). Explain this outcome to Jerry Carstairs.

(d) Make your recommendations for reducing the average level of in-process inventory at the inspection station and at the group of machines. Be specific in your recommendations, and support them with quantitative analysis like that done in part (*a*). Make specific comparisons to the results from part (*a*), and cite the improvements that your recommendations would yield.

PREVIEW OF AN ADDED CASE ON CD-ROM

CASE 17.2 Queueing Quandary

Many angry customers are complaining about the long waits needed to get through to a call center. It appears that more service representatives are needed to answer the calls. Another option is to train the service representatives further to enable them to answer calls more efficiently. Some possible criteria for satisfactory levels of service have been proposed. Queueing theory needs to be applied to determine how the operation of the call center should be redesigned.